

transcribed DNA (e.g., ribosomal RNA, histones), and diverged copies of ancestral genes (e.g., globin family genes).

19. The most direct method of determining the direction of transcription of the histone genes would be to clone and sequence the region, from which transcriptional information can be ascertained.
21. Cloning and then sequencing the region would provide the answer. Analysis would show genes of similar sequence to the active genes but lacking the sequences for transcription.
23. It is very rich in A-T sequences. Density is proportional to G-C content. Since the molecule has a low density, it has low G-C, and therefore high A-T.
25. Spaces between the nucleosomes must contain many promoter sequences. For the DNA to be digested, it must be unprotected. Since we see little transcription, the promoters must be missing and must have been destroyed by the nucleases.
27. Highly repetitive DNA must be located in these regions. Since most highly repetitive DNA is not transcribed, the results suggest that centromeric and telomeric regions are not transcribed.
29. The C-value paradox involves the issues of the excessive amounts of DNA in eukaryotic cells and the difference between eukaryotic species that seem to have similar complexity. It is explained by the large amount of structural DNA in chromosomes as well as the large amounts of short and long interspersed elements (SINEs and LINEs).

Critical Thinking Question:

1. Comparative DNA studies can be helpful in understanding the roles of the various types of DNA in the eukaryotic chromosomes if there are cases in which there are remarkably large differences in the amount of DNA in similar species. It can then be inferred that the basic developmental plan of an organism is contained in the one with the lower amount of DNA, and the extra DNA in the species with more DNA may be superfluous. We do have cases in which amphibians differ by as much as one hundred times the amount of DNA found in similar species. The puffer fish has only one-sixth the amount of DNA as other higher eukaryotes.

Chapter 16 Gene Expression: Control in Eukaryotes

1. See figure 16.5.
3. Since 5-azacytidine prevents methylation, the observed increase in transcription suggests that the presence of methyl groups inhibits transcription.
5. Genomic equivalence means that all of the cells of a multicellular eukaryotic organism are genetically identical. Yet, cells in different tissues and different regions of the organism are phenotypically different, expressing different suites of genes. Explaining differences in gene expression among cells that are genetically identical is a major question of eukaryotic genetics.
7. The three classes of segmentation genes in *Drosophila* are gap, pair-rule, and segment polarity. Mutations in gap genes leave gaps of missing segments. Mutations in pair-rule genes leave gaps of even or odd sets of segments. Mutations of segment polarity genes cause changes in all segments, generally the change in anterior or posterior portions of each.
9. In the development of the early *Drosophila* embryo, a syncytial blastoderm stage is achieved after thirteen cell divisions. The nuclei are near the surface of the embryo but not surrounded by cell membranes. Thereafter, membranes form, creating a cellular blastoderm.
11. Maternal-effect genes determine four regions of the developing embryo (major gene in parentheses): anterior (*bicoid*), posterior (*nanos*), dorso-ventral (*Toll*), and terminal (*torso*).
13. The helix-turn-helix motif (see box 16.1, fig. 1) consists of two alpha helices separated by a short turn within the protein, providing the structure to interact with DNA. Two other motifs are the zinc finger and the leucine zipper. Another motif, a combination of helix-turn-helix and leucine zipper, is shown in box 16.1, figure 4.
15. Amphibians have very large eggs, development is external to the female, a ready supply of zygotes is available, and they are easily manipulated experimentally.
17. Assuming that each cancer might be controlled by a single locus and assuming that breast cancer appears only in women and prostate cancer appears only in men (sex limited), pancreatic and prostate cancer are probably controlled by autosomal recessive genes; colon cancer is probably controlled by a dominant gene (autosomal or sex linked); and breast cancer by a recessive gene, either autosomal or sex linked.
19. The protein product of the retinoblastoma gene, p105, binds with oncogene proteins. Thus, the protein may somehow suppress transformation; when bound by oncogene proteins, p105 may be rendered ineffective. Hence, p105 seems to act to suppress transformation and thus the gene is called an anti-oncogene.
21. Animal viruses can have DNA or RNA, either single- or double-stranded. They can be enveloped or nonenveloped. They can have simple or complex protein coats.
23. The following are translation mechanisms: normal translation; read-through translation; and splice and then translation.
25. The v and c refer to viral and cellular, respectively. A proto-oncogene is a cellular oncogene within a nontransformed cell.
27. The *v-src* gene has no introns and the virus can function without the gene.
29. We see one band that is common to both cell lines; this band must represent the normal oncogene. The fact that this band is present in both lines indicates that the insertion of a virus has occurred in only one of the two copies of the gene present in the clone 1 cell. If it had inserted within both genes, we should not have seen the normal band. We see a larger fragment in clone 1, indicating that the DNA of the virus does not contain a site for the restriction enzyme used and that the virus has inserted within the restriction sites that define the band, lengthening the region probed. Alternatively, the virus could contain a restriction site and has still inserted in such a way as to lengthen the band probed by inserting between the sequence probed and the original restriction site.
31. The components of an immunoglobulin light chain are V, J, and C regions; the components of an immunoglobulin heavy chain are V, D, J, and C regions.
33. The V-J joining recognition signal is a heptamer and nonamer separated by twenty-three and twelve base pairs; see figure 16.35.
35. A T-cell receptor is an immunoglobulinlike molecule located on the surfaces of T cells, enabling them to identify infected host cells.
37. One explanation is a defect in the maturation process of B cells. Another explanation is a defect in the process of V(D)J joining, which could involve five or more genes.
39. The simplest interpretation of these results is that something is different in the organization of antibody genes in embryonic cells and in B lymphocytes. If the genes were in the same place in both cases, we should have seen identical patterns for the two types of cells. The probe recognizes both variable and constant regions of the antibody gene, since it is made from the mature mRNA. In the B lymphocyte, the variable and constant regions are adjacent, but in the embryonic cells, there is some extra DNA between these two genes. This result led to the notion that variable genes are rearranged during the development of the immune system.

A-20

Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

Critical Thinking Question:

- Adenovirus attacks normal cells by binding to p53; it also attacks cancerous cells that lack p53. If we remove the gene for the protein that binds p53, the *E1B* gene, then the modified adenovirus will not be able to attack normal cells but will be able to attack cancerous ones lacking p53. This modification of adenovirus as a potential tool in treating cancer was published in 1996.

Chapter 17 Non-Mendelian Inheritance

- Persistence of an environmentally induced trait into later generations is known as dauermodification and does not imply genetic control. After a suitable number of generations, the phenotype returns to normal, indicating an environmental rather than genetic response.
- Maroon-like must affect the cytoplasm of the egg. The first cross is unusual and alerts us to maternal inheritance. A true Mendelian factor should produce one-half maroon-like males and females. The genotypes of the F_1 females are $ma-1^+/ma-1$ and $ma-1/ma-1$; half of the females should be of each genotype. If the wild-type allele produces wild-type cytoplasm for the progeny, regardless of the genotype of the progeny, any female that is $ma-1^+/ma-1$ will produce all wild-type progeny.
- Although the genetic scheme predicts shell coiling perfectly, one could do experiments involving the injection of cytoplasm into eggs to test the viral hypothesis.
- Female parent: *Dd*. Male parent: *d-*. (A): *dd*. Since selfing produces only sinistral snails, individual (A) must be homozygous for sinistral coiling, *dd*. The individual must get one *d* allele from each parent, so each parent must be at least heterozygous. Since (A) has dextral coiling, the mother must be heterozygous. The father could be either *D/d* or *d/d*.
- By looking at different species, it is clear that few genes for oxidative phosphorylation are found in all mitochondrial genomes.
- The rule of thumb is that suppressive petite mitochondria will dominate a cell, whereas neutral petite mitochondria will be lost in a competitive situation. Therefore
 segregational petite $\times$ segregational petite $\rightarrow$ segregational petites
 segregational petite $\times$ neutral petite $\rightarrow$ segregational petites
 segregational petite $\times$ suppressive petite $\rightarrow$ suppressive petites
 neutral petite $\times$ neutral petite $\rightarrow$ neutral petites
 neutral petite $\times$ suppressive petite $\rightarrow$ suppressive petites
 suppressive petite $\times$ suppressive petite $\rightarrow$ suppressive petites
- In 0.02% of the offspring cells, the mt^- allele of the streptomycin locus is inherited. In essence, these cells seem to have inherited chloroplast genes from the mt^- parent. These cells thus provide us with a window on the possibility of having chloroplast genotypes from both parents viable within the same cell. Thus, interaction among other chloroplast genes can be looked for in this class of offspring (0.02% of total). If recombination occurs, map distances can be calculated by the usual methods, keeping in mind that we are taking data only from within this 0.02% of offspring.
- Mitochondria and chloroplasts have prokaryotic affinities. They both have circular chromosomes, and their metabolism is affected by prokaryotic inhibitors (e.g., antibiotics). Certain prokaryotic messenger RNAs will hybridize with the organelle's DNA. There are numerous other aspects of biochemistry, morphology, and physiology that help to demonstrate affinities.
- One way to determine that two loci are involved is to look at the proportion of offspring that lose mu particles after autogamy. In some strains, one-half the offspring will lose mu particles, indicating one locus was initially segregating (autogamy of $M_1m_1M_2m_2$

yields $M_1M_1m_2m_2$ or $m_1m_1m_2m_2$ in a 1:1 ratio). In other strains, one-fourth of the offspring will lose mu after autogamy, indicating that two unlinked loci were segregating (autogamy in $M_1m_1M_2m_2$ yields $M_1M_1M_2M_2$, $M_1M_1m_2m_2$, $m_1m_1M_2M_2$, or $m_1m_1m_2m_2$ in a 1:1:1:1 ratio).

- Human mitochondrial DNA does not have introns. Finding an intron would suggest that the mitochondria had acquired a nuclear gene.
- a. All type 1, 1.5 and 3.7 kilobases. b. All type 2, 2.5 and 6.0 kilobases. Recall that the chloroplast DNA from mt^- cells does not appear in progeny.
- Conjugation produces exconjugants with the same genotype, but which haploid micronucleus survives in each cell is random. Therefore, one-fourth of the time, the genotypes of the exconjugants are expected to be *KK*, one-half of the time *Kk*, and one-fourth of the time *kk*. The sensitive exconjugant remains sensitive, regardless of which of the genotypes is present. Autogamy does not affect the genotype of homozygous exconjugants, but it does affect the heterozygote. Among the killer exconjugants we expect:

Killer Exconjugants	Autogamous Products	Phenotypic Ratio
1/4 <i>KK</i>	all <i>KK</i>	1/4 killer
1/2 <i>Kk</i>	1/2 <i>KK</i> 1/2 <i>kk</i>	1/4 killer 1/4 sensitive (Kappa are lost)
1/4 <i>kk</i>	all <i>kk</i>	1/4 sensitive (Kappa are lost)

- Use the striped plant as the egg parent, and get pollen from a plant with the following genotype: *IjIj jj*. If the striped plant is *iojap (ijij JJ)*, all progeny will be heterozygous for both genes and will also contain *iojap* cytoplasm. The F_1 plants will segregate green, striped, and white sections within the plant. If the original plant is *IjIj jj* (and thus *japonica*), all F_1 plants will have striped leaves.
- a. 2 normal:2 petite b. 0 normal:4 petite c. 4 normal:0 petite. A nuclear gene should segregate 2:2 for each allele. Cytoplasmic factors will produce four spores with identical cytoplasm.

Critical Thinking Question:

- We believe there are two mechanisms to ensure the distribution of cellular organelles during cytokinesis: stochastic and ordered inheritance. Stochastic inheritance simply means that no real mechanism exists; rather, the cell depends on the large number of the organelles to ensure an even distribution during the dividing of the cell. Ordered inheritance requires the even distribution of organelles in small numbers. This can be accomplished by special structures that divide a large organelle (e.g., a single, large chloroplast), or by other mechanisms that insert part of the mitochondrial system into new buds in budding yeast.

Chapter 18 Quantitative Inheritance

- Three in two hundred is approximately 1 in $64 = 1/(4)^3$; therefore, three loci (see table 18.1). Each effective allele contributes about 1/2 pound over the 2-pound base (3-pound difference divided by six effective alleles: $AA BB CC = 5$ pounds, $aa bb cc = 2$ pounds).
- Independent assortment: for example, $Aa Bb Cc Dd$ parents can have $AA BB CC DD$ offspring.
- Individuals of intermediate color can produce both lighter and darker offspring by independent assortment. That is, $Aa Bb Cc Dd$

parents can produce $AA\ BB\ CC\ DD$ and $aa\ bb\ cc\ dd$ offspring. However, if each effective allele adds color, then individuals with the base color (white) who mate with each other ($aabbccdd$) presumably cannot have children with darker skin.

7. The marker stock is exposed to DDT over many generations, selecting for DDT resistance. It is then crossed with the wild-type and F_1 offspring are backcrossed to generate flies with various combinations of selected chromosomes. These flies are then tested for their DDT resistance. For example:

$P_1\ Cy/Pm\ H/S\ Ce/M(4) \times +$ (wild-type)
 F_1 (male) $Cy/+ \ H/+ \ Ce/+ \times$ (backcross) + (wild-type female)
 $F_2\ Cy/+ \ +/+ \ +/+ \ +/+ \ H/+ \ +/+ \ +/+ \ +/+ \ Ce/+ \ +/+ \ M(4)/+$, etc.

or,

F_1 (male) $Cy/+ \ H/+ \ Ce/+ \times$ (backcross) $Cy/Pm\ H/S\ Ce/M(4)$
 (female)
 $F_2\ Cy/Pm\ H/+ \ Ce/+ \ +/Pm \ +/S \ +/M(4)$; etc.

9. 50 cm. The difference between the two heights is 20 cm. This difference must result from the presence of effective (uppercase) alleles. The tall plant has four effective alleles, so each effective allele contributes an average of 5 cm to the height of the plant. The heterozygote has two effective alleles; $2 \times 5 = 10$ cm above the base of 40 cm, or a total of 50 cm.
11. Eight. The frequency of individuals in the F_2 that resemble one parent is $1/4^n$, where n = the number of genes involved. In this case, 100 of 6,200,000 were like one parent. $100/6,200,000$ is approximately $1/64,000$, which approximates $1/4^8$. So we probably have eight genes involved.
13. $r = 0.43$; $H_N = 0.43/0.50 = 0.86$ (narrow-sense heritability); environmental variance, a component of the total phenotypic variance appears in the denominator of the heritability equations (18.11 and 18.12). Thus, environmental factors that lower environmental variance (more uniform environments or environmental effects) increase heritability. And, factors that raise the environmental variance (less uniform environments or environmental effects) decrease heritability.
15. $H = (4 - 0.9)/(5 + 5) = 0.31$; H (high line) $= (4 - 3)/5 = 0.2$; H (low line) $= (3 - 0.9)/5 = 0.42$. Part of the difference may be due to the number of alleles available for selection in each direction and nonadditive factors.
17. First, "outstanding athletic ability" must be defined. It can be defined subjectively by accomplishment or more objectively with a physiological measure. Then genetic effects must be assessed through heritability analyses such as twin studies and correlations among relatives.
19. Uniformly good nutrition should increase the heritability of height by eliminating some of the environmental variance; it affects only the denominator in a heritability equation.
21. Thorax length: $V_D + V_I = (100 - [43 + 51]) = 6$; $H_N = 43/100 = 0.43$; $H_B = 49/100 = 0.49$. Eggs laid: $V_D + V_I = 44$; $H_N = 0.18$; $H_B = 0.62$.
23. 4.5 g.

$$H_N = \frac{\text{gain}}{\text{selection differential}}$$

Thus (H_N) (selection differential) = gain. Since $H_N = 0.5$ and selective differential = 9, $(0.5)(9.0) = 4.5$ g.

25. 176 pounds. To solve this problem, use the formula for realized heritability:

$$H = \frac{\text{gain}}{\text{selection differential}} = \frac{(Y_0 - \bar{Y})}{(Y_p - \bar{Y})}$$

$$0.4 = \frac{(Y_0 - \bar{Y})}{(Y_p - \bar{Y})} = \frac{Y_0 - \bar{Y}}{185 - 170}$$

$$Y_0 - \bar{Y} = (0.4)(15) = 6.0$$

$$6.0 = Y_0 - 170$$

$$Y_0 = 176 \text{ lbs}$$

Critical Thinking Question:

1. The simplest cause for retraction of a study is the failure of that study to be replicated by others. That would come about when the conclusion isn't generally supportable, a phenomenon that could have two major causes. First, the study might have been done well but the phenomenon was specific to that study, due possibly to small sample sizes or a "private mutation," an effect found in one family or a small group of individuals but not a general phenomenon. Second, there could have been an inadvertent error in the study. For example, a size difference in a small region of the brain of homosexual and heterosexual males was reported but has not been verified. One investigator, who is following up the study, suggested that the difference could be an effect of differences in the way the brains were preserved and thus represent no real (biological) effect.

Chapter 19 Population Genetics: The Hardy-Weinberg Equilibrium and Mating Systems

1. The frequencies of the three genotypes are $f(MM) = 41/100 = 0.41$; $f(MN) = 38/100 = 0.38$; $f(NN) = 21/100 = 0.21$. The frequency of M , p , is the frequency of MM homozygotes plus half the frequency of heterozygotes:

$$p = f(M) + (1/2)f(MN) = 0.41 + (1/2)(0.38) = 0.41 + 0.19 = 0.60$$

$$q = 1 - p = 1 - 0.60 = 0.40$$

Alternatively

$$p = f(M) = \frac{2(\#MM) + \#MN}{2 \times \text{total}} = \frac{2(41) + 38}{200}$$

$$= \frac{120}{200} = 0.60$$

$$q = 1 - p = 0.40$$

We do the following chi-square test:

	<i>MM</i>	<i>MN</i>	<i>NN</i>	Total
Observed	41	38	21	100
Expected	$p^2 \times 100$ 36	$2pq \times 100$ 48	$q^2 \times 100$ 16	100
Chi-square	0.694	2.083	1.563	4.340

The critical chi-square (0.05, one degree of freedom) = 3.841. We thus reject the null hypothesis that this population is in Hardy-Weinberg proportions.

3. Here we must assume Hardy-Weinberg equilibrium because of dominance. The $f(tt) = 65/215 = 0.302$. Thus $q = f(t) = \sqrt{f(tt)} = \sqrt{0.302} = 0.55$; and $p = f(T) = 1 - 0.55 = 0.45$. Since there are zero degrees of freedom (number of phenotypes - number of alleles = $2 - 2 = 0$), we cannot do a chi-square test

A-22 Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

to determine if the population is mating at random (in Hardy-Weinberg proportions).

5. a. Most human populations should be in Hardy-Weinberg proportions at the taster locus regardless of what the allelic frequencies are. 3:1 is a family ratio when the parents are heterozygotes or a population ratio when $p = q = 0.5$, given dominance.
- b. With no other information, it is probably safest to assume $p = q = 0.5$. At equal frequencies of alleles, the dominant phenotype (tasting) occurs in 75% of people ($p^2 + 2pq$).
7. $p = 0.99$, $q = 0.01$. If the population is in equilibrium, there should be p^2 of AA + $2pq$ of Aa + q^2 of aa individuals. Since 1/10,000 shows the recessive trait, this is q^2 . Therefore,

$$q = \sqrt{1/10,000} = \sqrt{0.0001} = 0.01$$

Since $p + q = 1$, $p = 1 - 0.01 = 0.99$.

9. The expected frequency of brown-eyed individuals will depend on the allelic frequencies of the original population. If we assume that mating is random with respect to eye color, and there is no selection, allelic frequencies will not change with time. We can, for example, calculate the frequencies of brown-eyed individuals for two populations at equilibrium. Let p = frequency of the brown-eye allele and q = frequency of the blue-eye allele.

Population	p	q	Frequency of brown ($p^2 + 2pq$)
1.	0.7	0.3	0.91
2.	0.5	0.5	0.75

We see that the original premise will be met only if the alleles are equally frequent.

11. 0.187 MM , 0.491 MN , 0.321 NN . The next generation will achieve equilibrium and there will be $(0.43)^2 MM + 2(0.43)(0.57)MN + (0.57)^2 NN$.
13. $f(A) = 0.792$; $f(B) = 0.208$. The easiest way to calculate frequencies is to do it empirically. We have three hundred people, so we have six hundred alleles.

$$f(A) = \frac{2 \times 200(AA) + 75(AB)}{600} = \frac{475}{600} = 0.792$$

$$f(B) = \frac{2 \times 25(BB) + 75(AB)}{600} = \frac{125}{600} = 0.208$$

15. a. Hardy-Weinberg proportions are achieved in one generation of random mating (multiple-allelic extension) if the locus is autosomal. If the locus is autosomal but there are different frequencies in the two sexes, then Hardy-Weinberg proportions are achieved in two generations. If the locus is sex linked, with different initial frequencies in the two sexes, then approach to equilibrium is gradual.
- b. If the loci are not in equilibrium to begin with (linkage disequilibrium), then equilibrium is achieved asymptotically.
17. 0.63 type A, 0.08 type B, 0.28 type AB, 0.01 type O. Since the population is in equilibrium, the genotypic frequencies can be calculated as $(p + q + r)^2 = p^2 + 2pq + q^2 + 2pr + 2qr + r^2$. Let $p = 0.7$, $q = 0.2$, and $r = 0.1$. Blood types will be represented by the following:

$$\begin{array}{llll} A: p^2 + 2pr & B: q^2 + 2qr & AB: 2pq & O: r^2 \\ = 0.49 + 0.14 & = 0.04 + 0.04 & = 0.28 & = 0.01 \end{array}$$

19. Inbreeding is disadvantageous when it causes recessive deleterious alleles to become homozygous. This occurs in normally outbred populations of diploids that have built up these harmful alleles. In species that normally inbreed, these deleterious alleles are probably no longer present; they were either removed by selection long ago or cannot build up in the population because of the regular pattern of inbreeding.

21. There are four paths passing through A, the only common ancestor ($F_A = 0.01$). All have ABCI as one side of the path. The second legs of the four other paths are ADEHI, ADGHI, AFGHI, and AFEHI. (A path such as IHEDAFGHI is invalid, passing through H twice.) Since each path has six ancestors, the inbreeding coefficient is $F_1 = 4(1/2)^6(1.01) = 0.063$.

23. Using the formula $F = (2pq - H)/2pq$, we calculate that $2pq = 0.48$, and $H = 38/100 = 0.38$. Therefore, $F = (0.48 - 0.38)/0.48 = 0.208$.

$$25. 0.375. F = (2pq - H)/2pq = \frac{(0.32 - 0.20)}{0.32} = \frac{0.12}{0.32} = 0.375$$

Critical Thinking Question:

1. Let us use the superscripts m and p for male and female, respectively. Then, we can construct the following Punnett square creating the next generation:

	Males, $A; f(A) = p^m$	Males, $a; f(a) = q^m$
Females, $A; f(A) = p^f$	$f(AA) = p^m p^f$	$f(Aa) = p^f q^m$
Females, $a; f(a) = q^f$	$f(Aa) = p^m q^f$	$f(aa) = q^m q^f$

Since the distribution of offspring in the table is independent of sex, it is the same in both sexes. The frequency of the A allele, p (in both sexes), will be the sum of the frequencies of the homozygotes and half the heterozygotes, or:

$$p = p^m p^f + (1/2)(p^m q^f + p^f q^m)$$

We then substitute $(1 - p)$ for all q 's:

$$p = p^m p^f + (1/2)p^m (1 - p^f) + (1/2)p^f (1 - p^m)$$

which simplifies to:

$$p = (1/2)(p^m + p^f)$$

In other words, after one generation of random mating, the allelic frequencies in each sex are the averages for both sexes. Now the frequency is the same in both sexes, and a second generation of random mating will achieve Hardy-Weinberg proportions. The population is not in equilibrium after one generation because the proportion of genotypes is not p^2 , $2pq$, and q^2 if p^m does not equal p^f . In other words, $p^m p^f$ does not equal p^2 .

Chapter 20 Population Genetics: Processes That Change Allelic Frequencies

1. a. The equilibrium frequency of a is $\hat{q} = \frac{\mu}{(\mu + \nu)}$. Therefore

$$\hat{q} = (6 \times 10^{-5}) / (6 \times 10^{-5} + 7 \times 10^{-7}) = 0.00006 / 0.0000607 = 0.988$$

- b. If q = 0.90 at generation n , then

$$\begin{aligned} q_{n+1} &= q_n + \mu p_n - \nu q_n = \\ 0.90 &+ (6 \times 10^{-5})(0.10) - (7 \times 10^{-7})(0.90) = 0.900054 \end{aligned}$$

3. 0.714.

$$\hat{q} = \frac{\mu}{(\mu + \nu)} = \frac{5 \times 10^{-5}}{5 \times 10^{-5} + 2 \times 10^{-5}} = \frac{5}{7} = 0.714$$

5. We use equation 20.12 to calculate the migration rate, m :

$$m = (q_C - q_N)/(q_M - q_N)$$

In this case, $q_C = 0.45$, $q_N = 0.62$, and $q_M = 0.03$. Thus

$$m = (0.45 - 0.62)/(0.03 - 0.62) = (-0.17)/(-0.59) = 0.288$$

7. Fast: 0.56; slow: 0.44. With 900 butterflies we have 1,800 alleles. $0.6(1,800) = 1,080$ fast alleles, and $0.4(1,800) = 720$ slow alleles. In the migrant population, $(0.8)(180) = 144$ slow and $(0.2)(180) = 36$ fast alleles. Therefore, the frequency of the fast allele is $(1,080 + 36)/1,980 = 0.56$. The frequency of the slow allele is $1 - f(\text{fast allele}) = 1 - 0.56 = 0.44$.

9. 0.47. We again use equation 20.12:

$$m = \frac{q_C - q_N}{q_M - q_N}$$

where $m = 0.1$, $q_C = 0.45$, and $q_M = 0.25$.

$$0.1 = \frac{0.45 - q_N}{0.25 - q_N}$$

$$0.1(0.25 - q_N) = 0.45 - q_N$$

$$0.025 - 0.1q_N = 0.45 - q_N$$

$$0.9q_N = 0.425$$

$$q_N = \frac{0.425}{0.9} = 0.47$$

11. In stabilizing selection, extremes of a distribution are selected against. In directional selection, one extreme is favored over the other. In disruptive selection, both extremes are favored over the middle of the distribution.

13. Heterozygote disadvantage:

	AA	Aa	aa	Total
Before selection	p^2	$2pq$	q^2	1
Fitnesses (W)	1	$1 - s$	1	
Frequencies after selection	$p^2/\bar{W}$	$2pq(1 - s)/\bar{W}$	$q^2/\bar{W}$	$\bar{W} = 1 - 2pqs$

Then

$$q_{n+1} = (pq[1 - s] + q^2)/\bar{W}$$

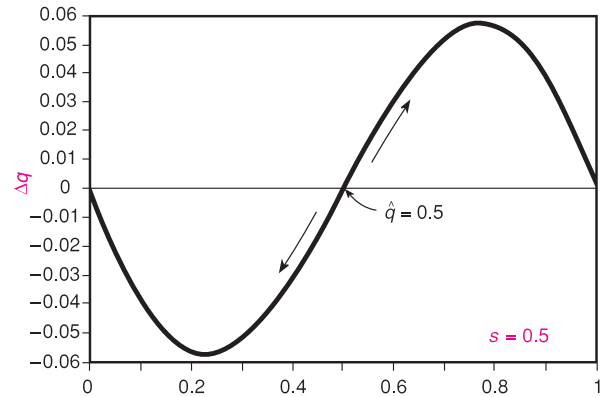
$$\Delta q = q_{n+1} - q = (pq[1 - s] + q^2)/\bar{W} - q\bar{W}/\bar{W}$$

$$= (pq[1 - s] + q^2 - q[1 - 2pqs])/\bar{W}$$

which simplifies to

$$\Delta q = spq(2q - 1)/\bar{W}$$

At $\Delta q = 0$, $\hat{q} = 0, 1$, or 0.5 ($2q - 1 = 0$, therefore $\hat{q} = 0.5$). The equilibrium points of zero and one are stable—if perturbed slightly (less than 0.5), the population will return to these values. The value $\hat{q} = 0.5$ is, however, unstable—if perturbed, it will continue away from the equilibrium point. This can be seen by either substituting into or graphing the Δq equation.

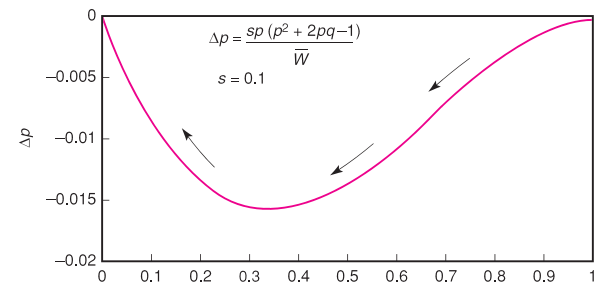


15.

	AA	Aa	aa	Total
Before selection	p^2	$2pq$	q^2	1
Fitnesses (W)	$1 - s$	$1 - s$	1	$\bar{W} = 1 - p^2s - 2pqs$
Frequencies after selection	$p^2(1 - s)/\bar{W}$	$2pq(1 - s)/\bar{W}$	$q^2/\bar{W}$	1

$$p_{n+1} = (p^2[1 - s] + pq[1 - s])/\bar{W}$$

$$\begin{aligned} \Delta p &= p_{n+1} - p = (p^2[1 - s] + pq[1 - s])/\bar{W} - p\bar{W}/\bar{W} \\ &= sp(p^2 + 2pq - 1)/\bar{W} = sp(-p^2 + 2p - 1)/\bar{W} \end{aligned}$$

And $\hat{p} = 0$ or 1 (from the root of the quadratic). Only zero is stable.

17. Since only heterozygotes survive, $\hat{q} = 0.5$. This can also be derived from equation 20.24:

$$\hat{q} = s_1/(s_1 + s_2)$$

$$\text{If } s_1 = s_2 = 1, \text{ then } \hat{q} = 1/2$$

19. The *ST* inversion seems to do best at lower elevations and the *AR* at higher elevations. *CH* (and others) do not appear to be affected by altitude. To test this hypothesis, we would grow caged populations of flies with different initial frequencies of the various inversions at different simulated elevations, simulated by temperature, pressure, oxygen content, or other. We predict that, regardless of initial conditions, they would eventually equilibrate at the values in the table for the given parameter of the altitude (temperature, pressure, oxygen content, or other) that is acting as a selective agent. We would thus identify the selective agent. Since the inversions isolate various allelic combinations, our next step (a potentially long-term step) would be to determine which loci the selective agent is acting on.

A-24 Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

21. 0.575. We can use the formula in equation 20.15, which is a simplified form of the sum of homozygotes + one-half the proportion of heterozygotes of the *b* allele.

$$q_{n+1} = \frac{q - sq^2}{1 - sq^2}$$

Since the relative fitness, $W = 0.4$, $s = 1 - W = 0.6$.

$$q_{n+1} = \frac{0.7 - (0.6)(0.49)}{1 - (0.6)(0.49)} = \frac{0.406}{0.706} = 0.575$$

23. 0.33, 0.25. Since the fitness is zero, $s = 1$, and we can use equation 20.17: $q_1 = q_0/(q_0 + 1) = 0.5/1.5 = 0.33$. For the second generation, we substitute the first generation numbers: $q_2 = 0.33/1.33 = 0.248$.

Critical Thinking Question:

- There could be several reasons why these systems are in existence. First, they could be in selection-mutation equilibrium. However, that would not account for the high frequencies of both alleles in human populations in the Rh system. Second, the polymorphism could be relatively new, somehow maintaining both alleles as the human population increased in recent times with natural selection not having enough time to eliminate one of the alleles. Third, although the Rh blood system could follow the heterozygous disadvantage model, selection could also be acting in other ways that might maintain the polymorphism. That is, aside from Rh incompatibility eliminating heterozygotes, other genotypic combinations could be favored under other circumstances. Finally, although one or the other allele is being eliminated in any one population due to heterozygous disadvantage, the constant mixing of human populations could be reintroducing the rarer allele.

Chapter 21 Evolution and Speciation

- Neo-Darwinism is the application of population genetics to Darwinian evolution. Darwinian evolution works as natural selection favors the most fit organisms in competition among the overproduced young of any species.
- Each process lets reproductive isolating mechanisms evolve while some barrier to breeding arises (see fig. 21.3).
- Constraint* refers to the limitations on changes that can take place. Some changes result in nonfunctional proteins and enzymes and thus cannot be in a successful lineage. For example, many base changes that lead to new amino acids in enzyme active sites disrupt enzymatic activity. If these mutations take place, they are eliminated by natural selection.
- Yes. Two distinct species should not yield fertile progeny. We see a great reduction in the numbers of offspring from hybrids, indicating that hybrid inviability is one isolating mechanism operating.
- Punctuated equilibrium proposes that species remain unchanged for long periods of time and that major changes occur only periodically. If species A existed for ten million years and suddenly (geologically speaking) changed dramatically to species X and Y, there would be few fossils because of the relatively short time in which intermediate forms were present. Another argument is, barring an incredibly detailed and complete sequence (of which there are almost none), there will always be gaps in the fossil record.
- Genetic variability can be maintained by heterozygote advantage (e.g., sickle-cell anemia in people); frequency-dependent selection

(rare-male mating advantage in *Drosophila*); transient polymorphism (industrial melanism in moths during an increase or decrease in industrialization); life-stage selection, which often happens when comparing egg, larval, pupal, and adult mortalities in *Drosophila*; differential selection in heterogeneous environments, common in some land snails; and neutrality.

- Presumably, in mammals, selection could involve types of substrates acted upon by electrophoretic variants; functioning at different pHs and ionic strengths in various cellular compartments; resistance to enzyme inhibitors; interaction with other proteins and membrane components; and others.

- Use the formula $K = -\ln(1 - d/n)$, in which d is the number of amino acid differences and n is the total number of amino acid sites. Thus:

$$\text{human being-dog: } K = -\ln(1 - 1/8) = 0.133$$

$$\text{human being-chicken: } K = -\ln(1 - 3/8) = 0.470$$

$$\text{and dog-chicken: } K = -\ln(1 - 3/8) = 0.470$$

These values place human beings and dogs very close and both equally far from chickens, which is consistent with the known evolutionary relationships.

- 0.33 *AA*, 0.49 *Aa*, 0.18 *aa*. The mean fitness of the population after selection is $0.5 + 2pq = 0.98$ (table 21.2). The new frequency of a genotype is its original frequency times its fitness, all divided by the mean fitness of the population. Or:

$$f(AA) = \frac{(0.36)(1.5 - 0.6)}{0.98} = 0.33$$

$$f(Aa) = 0.48/0.98 = 0.49$$

$$f(aa) = (0.16)(1.5 - 0.4)/0.98 = 0.18$$

- 0.023. $K = -\ln(1 - p)$

$$K = -\ln(1 - [23/1,000]) = -\ln(0.977) = 0.023$$

- The third. For many amino acids, the third position can be any of the four bases. In addition, wobble allows for some variation here as well. Changing the first or second base almost always produces a new amino acid.

- Eight first cousins, on average, carry the complete genome of an individual. Therefore, from an evolutionary point of view, an individual and his or her eight cousins include the same alleles.

Critical Thinking Question:

- Given that the gene for peppering is recessive and that one year equals one generation, the moths should be following a selection model in which natural selection acts against the recessive homozygote. In that case, we have already developed a selection model for this in chapter 20. Equation 20.15 relates the new allelic frequency to the old, given a particular selection coefficient:

$$q_{n+1} = q(1 - sq)/(1 - sq^2)$$

We can solve this equation for the selection coefficient:

$$s = (q - q_{n+1})/(q^2 - q^2q_{n+1})$$

Here, $q = 0.6$ and $q_{n+1} = 0.5$. When we solve the equation, we get: $s = 0.56$, or $W = 1 - s = 0.44$, which is the fitness of the peppered moth.

APPENDIX B

Suggestions for Further Reading

Chapter 1 Introduction

- Crow, J. 1993. N.I. Vavilov, martyr to genetic truth. *Genetics* 134:1–4.
- Hull, D. 1996. A revolutionary philosopher of science. *Nature* 382:203–4.
- Kuhn, T. 1962. *The Structure of Scientific Revolutions*. Chicago: University of Chicago Press.
- Maddox, J. 1999. The Unexpected Science to Come. *Scientific American*, December, 62–67.
- Moore, J. 1993. *Science as a Way of Knowing*. Cambridge, Mass.: Harvard University Press.
- Popper, K. 1962. *Conjectures and Refutations: The Growth of Scientific Knowledge*. New York: Basic Books.
- Ruse, M., ed. 1989. *Philosophy of Biology*. New York: Macmillan.
- Soyfer, V. 1994. *Lysenko and the Tragedy of Soviet Science*. New Brunswick, N.J.: Rutgers University Press.
- Sterelny, K., and P. Griffiths. 1999. *Sex and Death: An Introduction to Philosophy of Biology*. Chicago: University of Chicago Press.
- Stubbe, H. 1972. *History of Genetics*. Cambridge, Mass.: MIT Press.
- Sturtevant, A. 1965. *A History of Genetics*. New York: Harper & Row.

Chapter 2 Mendel's Principles

- Beadle, G., and E. Tatum. 1941. Genetic control of biochemical reactions in *Neurospora*. *Proceedings of the National Academy of Sciences, USA* 27:499–506.
- Bearn, A. 1994. Archibald Edward Garrod, the reluctant geneticist. *Genetics* 137:1–4.
- Brink, R., ed. 1966. *Heritage from Mendel*. Madison: University of Wisconsin Press.
- Corcos, A., and F. Monaghan. 1985. Role of de Vries in the recovery of Mendel's work. I: Was de Vries an independent discoverer of Mendel? *Journal of Heredity* 76:187–90.
- . 1987. Correns, an independent discoverer of Mendelism? I: An historical/critical note. *Journal of Heredity* 78:330.
- . 1993. *Gregor Mendel's Experiments on Plant Hybrids*. New Brunswick, N.J.: Rutgers University Press.

- Douglas, L., and E. Novitski. 1977. What chance did Mendel's experiments give him of noticing linkage? *Heredity* 38:253–57.
- Fisher, R. 1936. Has Mendel's work been rediscovered? *Annals of Science* 1:115–37.
- Garrod, A. 1909. *Inborn Errors of Metabolism*. London: Hodder and Stoughton.
- Greenspan, R. 1997. *Fly Pushing: The Theory and Practice of Drosophila Genetics*. Cold Spring Harbor, N.Y.: Cold Spring Harbor Laboratory Press.
- Horowitz, N. 1990. George Wells Beadle (1903–89). *Genetics* 124:1–6.
- . 1991. Fifty years ago: The *Neurospora* revolution. *Genetics* 127:631–35.
- . 1996. The sixtieth anniversary of biochemical genetics. *Genetics* 143:1–4.
- Jackson, I. 1994. Molecular and developmental genetics of mouse coat color. *Annual Review of Genetics* 28:189–217.
- Keightley, P. 1996. A metabolic basis for dominance and recessivity. *Genetics* 143:621–25.
- King, R., and W. Stansfield. 1996. *Dictionary of Genetics*, 5th ed. New York: Oxford University Press.
- Levy, H. 1999. Phenylketonuria: Old disease, new approach to treatment. *Proceedings of the National Academy of Sciences, USA* 96:1811–13.
- Martin, D., W. Proebsting, and P. Hedden. 1997. Mendel's dwarfing gene: cDNAs from the *Le* alleles and function of the expressed proteins. *Proceedings of the National Academy of Sciences, USA* 94:8907–11.
- Matton, D., N. Nass, A. Clarke, and E. Newbigin. 1994. Self-incompatibility: How plants avoid illegitimate offspring. *Proceedings of the National Academy of Sciences, USA* 91:1992–97.
- Nasrallah, J., et al. 1994. Signaling the arrest of pollen tube development in self-incompatible plants. *Science* 266:1505–8.
- Neel, J. 1994. *Physician to the Gene Pool: Genetic Lessons and Other Stories*. New York: Wiley.
- Orel, V. 1996. *Gregor Mendel. The First Geneticist*. New York: Oxford University Press.
- Perkins, D. 1992. *Neurospora: The organism behind the molecular revolution*. *Genetics* 130:687–701.

- Peters, J. 1959. *Classical Papers in Genetics*. Englewood Cliffs, N.J.: Prentice Hall.
- Pilgrim, I. 1986. A solution to the too-good-to-be-true paradox and Gregor Mendel. *Journal of Heredity* 77:218–20.
- Reaume, A., D. Knecht, and A. Chovnick. 1991. The rosy locus in *Drosophila melanogaster*: Xanthine dehydrogenase and eye pigments. *Genetics* 129:1099–109.
- Rick, C. 1991. Tomato paste: A concentrated review of genetic highlights from the beginnings to the advent of molecular genetics. *Genetics* 128:1–5.
- Stern, C., and E. Sherwood, eds. 1966. *The Origin of Genetics, A Mendel Source Book*. San Francisco: Freeman.
- Stubbe, H. 1972. *History of Genetics*. Cambridge, Mass.: MIT Press.
- Sturtevant, A. 1965. *A History of Genetics*. New York: Harper & Row.
- Thompson, R., and H.-H. Kirch. 1992. The S locus of flowering plants: When self-rejection is self-interest. *Trends in Genetics* 8:381–87.
- Weiling, F. 1971. Mendel's "too good" data in *Pisum* experiments. *Folia Mendeliana* 6:75–77.
- . 1986. What about Fisher's statement of the "too good" data of J. G. Mendel's *Pisum* paper? *Journal of Heredity* 77:281–83.
- Wellner, D., and A. Meister. 1981. A survey of inborn errors of amino acid metabolism and transport in man. *Annual Review of Biochemistry* 50:911–68.
- Woolf, C., and J. Swafford. 1988. Evidence for eumelanin- and pheomelanin-producing genotypes in the Arabian horse. *Journal of Heredity* 79:100–106.
- Yamamoto, E., et al. 1990. Molecular genetic basis of the histo-blood group ABO system. *Nature* 345:229–33.
- Yoshida, K., et al. 1995. Cause of blue petal colour. *Nature* 373:291.

Chapter 3 Mitosis and Meiosis

- Barton, N., and L. Goldstein. 1996. Going mobile: Microtubule motors and chromosome segregation. *Proceedings of the National Academy of Sciences, USA* 93:1735–42.

B-2 *Appendix B Suggestions for Further Reading*

- Carpenter, A. 1994. Chiasma function. *Cell* 77:959–62.
- Dawe, R. 1998. Meiotic chromosome organization and segregation in plants. *Annual Review of Plant Physiology and Plant Molecular Biology* 49:371–95.
- Dobie, K., et al. 1999. Centromere proteins and chromosome inheritance: A complex affair. *Current Opinion in Genetics & Development* 9:206–17.
- Glover, D., C. Gonzalez, and J. Raff. 1993. The centrosome. *Scientific American*, June, 62–68.
- Grancell, A., and P. Sorger. 1998. Chromosome movement: kinetochores motor along. *Current Biology* 8:R382–85.
- Heneen, W. 1982. The centromeric region in the scanning electron microscope. *Heredity* 97:311–14.
- Hirokawa, N., Y. Noda, and Y. Okada. 1998. Kinesin and dynein superfamily proteins in organelle transport and cell division. *Current Opinion in Cell Biology* 10:60–73.
- Holm, C. 1994. Coming undone: How to untangle a chromosome. *Cell* 77:955–57.
- Hoyt, M., and J. Geiser. 1996. Genetic analysis of the mitotic spindle. *Annual Review of Genetics* 30:7–33.
- Hyams, J., and C. Lloyd, eds. 1994. *Microtubules*. New York: Wiley-Liss.
- Kellogg, D., M. Moritz, and B. Alberts. 1994. The centrosome and cellular organization. *Annual Review of Biochemistry* 63:639–74.
- Marc, J. 1997. Microtubule-organizing centres in plants. *Trends in Plant Science* 2:223–30.
- Nasmyth, K. 1999. Separating sister chromatids. *Trends in Biochemical Sciences* 24:98–104.
- Nicklas, R. 1997. How cells get the right chromosomes. *Science* 275:632–37.
- Nurse, P. 2000. A long twentieth century of the cell cycle and beyond. *Cell* 100:71–78.
- Orr-Weaver, T. 1999. The ties that bind: Localization of the sister-chromatid cohesin complex on yeast chromosomes. *Cell* 99:1–4.
- Page, A., and P. Hieter. 1999. The anaphase-promoting complex: New subunits and regulators. *Annual Review of Biochemistry* 68:583–609.
- Pigozzi, M., and A. Solari. 1999. Recombination nodule mapping and chiasma distribution in spermatocytes of the pigeon, *Columba livia*. *Genome* 42:308–14.
- Rhoades, M. 1950. Meiosis in maize. *Journal of Heredity* 41:59–67.
- Schubert, I., and J. Oud. 1997. There is an upper limit of chromosome size for normal development of an organism. *Cell* 88:515–20.
- Skibbens, R., and P. Hieter. 1998. Kinetochores and the checkpoint mechanism that monitors for defects in the chromosome segregation machinery. *Annual Review of Genetics* 32:307–37.
- Sutton, W. 1903. The chromosomes in heredity. *Biological Bulletin* 4:231–51.
- Theriot, J., and L. Satterwhite. 1997. New wrinkles in cytokinesis. *Nature* 385:388–89.
- von Wettstein, D., S. Rasmussen, and P. Holm. 1984. The synaptonemal complex in genetic segregation. *Annual Review of Genetics* 18:331–413.
- Waters, J., and E. Salmon. 1997. Pathways of spindle assembly. *Current Opinion in Cell Biology* 9:37–43.
- Whittaker, R. 1969. New concepts of kingdoms of organisms. *Science* 163:150–60.
- Zickler, D., and N. Kleckner. 1998. The leptotene-zygotene transition of meiosis. *Annual Review of Genetics* 32:619–97.
- . 1999. Meiotic chromosomes: Integrating structure and function. *Annual Review of Genetics* 33:603–754.
- Zimmerman, W., C. Sparks, and S. Doxsey. 1999. Amorphous no longer: The centrosome comes into focus. *Current Opinion in Cell Biology* 11:122–28.
- Chapter 4 Probability and Statistics**
- Berger, J., and D. Berry. 1988. Statistical analysis and the illusion of objectivity. *American Scientist* 76:159–65.
- Dixon, W., and F. Massey, Jr. 1983. *Introduction to Statistical Analysis*, 4th ed. New York: McGraw-Hill.
- Howson, C., and P. Urbach. 1991. Bayesian reasoning in science. *Nature* 350:371–74.
- Jefferys, W., and J. Berger. 1992. Ockham's razor and Bayesian analysis. *American Scientist* 80:64–72.
- Maynard Smith, J. 1968. *Mathematical Ideas in Biology*. New York: Cambridge University Press.
- Moore, D. 1997. *Statistics: Concepts and Controversies*, 4th ed. San Francisco: Freeman.
- Ross, S. 1994. *A First Course in Probability*, 4th ed. New York: Macmillan.
- Siegel, S., and N. Castellan, Jr. 1988. *Nonparametric Statistics for the Behavioral Sciences*, 2d ed. New York: McGraw-Hill.
- Yoccoz, N. 1991. Use, overuse, and misuse of significance tests in evolutionary biology and ecology. *Bulletin of the Ecological Society of America* 72:106–11.
- Chapter 5 Sex Determination, Sex Linkage, and Pedigree Analysis**
- Baker, B., M. Gorman, and I. Marin. 1994. Dosage compensation in *Drosophila*. *Annual Review of Genetics* 28:491–521.
- Baker, B., et al. 1991. The doublesex locus of *Drosophila melanogaster* and its flanking regions: A cytogenetic analysis. *Genetics* 127:125–38.
- Barr, M., and E. Bertram. 1949. A morphological distinction between neurones of the male and female, and the behavior of the nucleolar satellite during accelerated nucleoprotein synthesis. *Nature* 163:676–77.
- Bashaw, G., and B. Baker. 1996. Dosage compensation and chromatin structure in *Drosophila*. *Current Opinion in Genetics and Development* 6:496–501.
- Bridges, C. 1932. The genetics of sex in *Drosophila*. In *Sex and Internal Secretions, A Survey of Recent Research*, edited by E. Allen, pp. 53–93. Baltimore: Williams & Wilkins.
- Brockdorff, N., et al. 1991. Conservation of position and exclusive expression of mouse *Xist* from the inactive X chromosome. *Nature* 351:329–31.
- Burgoyne, P. 1982. Genetic homology and crossing over in the X and Y chromosomes of mammals. *Human Genetics* 61:85–90.
- Carrel, L., et al. 1999. A first-generation X-inactivation profile of the human X chromosome. *Proceedings of the National Academy of Sciences, USA* 96:14440–44.
- Cline, T. 1988. Evidence that *sisterless-a* and *sisterless-b* are two of several discrete “numerator elements” of the X/A sex determination signal in *Drosophila* that switch *Sxl* between two alternative stable expression states. *Genetics* 119:829–62.
- Cline, T., and B. Meyer. 1996. Vive la difference: Males vs. females in flies vs. worms. *Annual Review of Genetics* 30:637–702.
- Corcos, A. 1983. Pattern baldness: Its genetics revisited. *American Biology Teacher* 45:371–75.
- Crow, J. 1994. Advantages of sexual reproduction. *Developmental Genetics* 15:205–13.
- Eicher, E., and L. Washburn. 1986. Genetic control of primary sex determination in mice. *Annual Review of Genetics* 20:327–60.
- Elbrecht, A., and R. Smith. 1992. Aromatase enzyme activity and sex determination in chickens. *Science* 255:467–70.
- Farrow, M., and R. Juberg. 1969. Genetics and laws prohibiting marriage in the United States. *Journal of the American Medical Association* 209:534–38.
- Gepner, J., and T. Hays. 1993. A fertility region on the Y chromosome of *Drosophila melanogaster* encodes a dynein microtubule motor. *Proceedings of the National Academy of Sciences, USA* 90:11132–36.
- Goodfellow, P., and R. Lovell-Badge. 1993. *SRY* and sex determination in mammals. *Annual Review of Genetics* 27:71–92.

- Gorman, M., and B. Baker. 1994. How flies make one equal two: Dosage compensation in *Drosophila*. *Trends in Genetics* 10:376–80.
- Graham, J., and R. Winters. 1961. Familial hypophosphatemia: An inherited demand for increased vitamin D. *Annals of the NY Academy of Sciences* 91:667–73.
- Haqq, C., et al. 1993. *SRY* recognizes conserved DNA sites in sex-specific promoters. *Proceedings of the National Academy of Sciences, USA* 90:1097–101.
- . 1994. Molecular basis of mammalian sexual determination: Activation of Müllerian inhibiting substance gene expression by *SRY*. *Science* 266:1494–1500.
- Harley, V., et al. 1992. DNA binding activity of recombinant *SRY* from normal males and XY females. *Science* 255:453–56.
- Heard, E., P. Clerc, and P. Avner. 1997. X-chromosome inactivation in mammals. *Annual Review of Genetics* 31:571–610.
- Herzing, L., et al. 1997. *Xist* has properties of the X-chromosome inactivation centre. *Nature* 386:272–75.
- Hines, P., and E. Culotta. 1998. The evolution of sex. *Science* 281:1979–2008. [Eight news items and articles on the evolution of sex.]
- Hunter, R. 1995. *Sex Determination, Differentiation and Intersexuality in Placental Mammals*. Cambridge, Mass.: Cambridge University Press.
- Kelley, R., et al. 1997. *Sex lethal* controls dosage compensation in *Drosophila* by a non-splicing mechanism. *Nature* 387:195–99.
- . 1999. Epigenetic spreading of the *Drosophila* dosage compensation complex from *roX* RNA genes into flanking chromatin. *Cell* 98:513–22.
- Kohler, R. 1994. *Lords of the Fly: Drosophila Genetics and the Experimental Life*. Chicago: University of Chicago Press.
- Koopman, P., et al. 1991. Male development of chromosomally female mice transgenic for *Sry*. *Nature* 351:117–21.
- Lahn, B., and D. Page. 1999. Four evolutionary strata on the human X chromosome. *Science* 286:964–67.
- Lindsley, D., and G. Zimm. 1992. *The genome of Drosophila melanogaster*. San Diego: Academic Press.
- Long, A., and R. Michod. 1995. Origin of sex for error repair. I: Sex, diploidy, and haploidy. *Theoretical Population Biology* 47:18–55.
- Lyon, M. 1962. Sex chromatin and gene action in the mammalian X chromosome. *American Journal of Human Genetics* 14:135–48.
- . 1992. Some milestones in the history of X-chromosome inactivation. *Annual Review of Genetics* 26:17–28.
- . 1999. X-chromosome inactivation. *Current Biology* 9:R235–37.
- McKusick, V.A. 1998. *Mendelian Inheritance in Man: A Catalog of Human Genes and Genetic Disorders*, 12th ed. Baltimore: Johns Hopkins University Press.
- Miller, O. 1995. The fifties and the renaissance in human and mammalian cytogenetics. *Genetics* 139:489–94.
- Morgan, T. 1910. Sex-limited inheritance in *Drosophila*. *Science* 32:120–22.
- Nicoll, M., C. Akerib, and B. Meyer. 1997. X-chromosome-counting mechanisms that determine nematode sex. *Nature* 388:200–204.
- Parkhurst, S., and P. Meneely. 1994. Sex determination and dosage compensation: Lessons from flies and worms. *Science* 264:924–32.
- Rice, W. 1994. Degeneration of a nonrecombining chromosome. *Science* 263:230–32.
- . 1996. Evolution of the Y sex chromosome in animals. *BioScience* 46:331–42.
- Scott, D., et al. 1995. Identification of a mouse male-specific transplantation antigen, H-Y. *Nature* 376:695–98.
- Shapiro, L., et al. 1979. Noninactivation of an X-chromosome locus in man. *Science* 204:1224–26.
- Smithies, O. 1995. Early days of gel electrophoresis. *Genetics* 139:1–4.
- Vainio, S., et al. 1999. Female development in mammals is regulated by Wnt-4 signaling. *Nature* 397:405–9.
- Walker, C., et al. 1991. The Barr body is a looped X chromosome formed by telomere association. *Proceedings of the National Academy of Sciences, USA* 88:6191–95.
- Willard, H. 1996. X chromosome inactivation, *XIST*, and pursuit of the X-inactivation center. *Cell* 86:5–7.
- Wright, W. 1988. Sex change in the Mollusca. *Trends in Ecology and Evolution* 3:137–40.
- Chapter 6 Linkage and Mapping in Eukaryotes**
- Beuler, E., et al. 1996. Mutation nomenclature: Nicknames, systematic names, and unique identifiers. *Human Mutation* 8:203–6.
- Bowring, E., and D. Catchside. 1999. Evidence for negative interference: Clustering of crossovers close to the *am* locus in *Neurospora crassa* among *am* recombinants. *Genetics* 152:965–69.
- Bridges, C. 1935. Salivary chromosome maps. *Journal of Heredity* 26:60–64.
- Carson, S., W. Henry, and T. Shows. 1985. Tissue factor gene localized to human chromosome 1 (1pter → 1p21). *Science* 229:991–93.
- Creighton, H., and B. McClintock. 1931. A correlation of cytological and genetical crossing over in *Zea mays*. *Proceedings of the National Academy of Sciences, USA* 17:492–97.
- Crow, J. 1988. A diamond anniversary: The first chromosome map. *Genetics* 118:1–3.
- . 1990. Mapping functions. *Genetics* 125:669–71.
- Emery, A., and D. Rimoin, eds. 1990. *Principles and Practice of Medical Genetics*, 2d ed. New York: Churchill Livingstone.
- Federoff, N. 1993. Barbara McClintock. *Evolución Biológica* 7:1–19.
- Fincham, J. 1998. Fungal genetics—Past and present. *Journal of Genetics* 77:55–63.
- Greenspan, R. 1997. *Fly Pushing: The Theory and Practice of Drosophila Genetics*. Cold Spring Harbor, N.Y.: Cold Spring Harbor Laboratory Press.
- Hall, M., and P. Linder. 1993. *The Early Days of Yeast Genetics*. Cold Spring Harbor, N.Y.: Cold Spring Harbor Laboratory Press.
- Hayles, J., and P. Nurse. 1992. Genetics of the fission yeast *Schizosaccharomyces pombe*. *Annual Review of Genetics* 26:373–402.
- Kaback, D., et al. 1999. Chromosome size-dependent control of meiotic reciprocal recombination in *Saccharomyces cerevisiae*: The role of crossover interference. *Genetics* 152:1475–86.
- Kohler, R. 1994. *Lords of the Fly: Drosophila Genetics and the Experimental Life*. Chicago: University of Chicago Press.
- Lefevre, G., and W. Watkins. 1986. The question of the total gene number in *Drosophila melanogaster*. *Genetics* 113:869–95.
- Lewis, E. 1995. Remembering Sturtevant. *Genetics* 141:1227–30.
- Lindsley, D., and G. Zimm. 1992. *The genome of Drosophila melanogaster*. San Diego: Academic Press.
- Littlefield, J. 1964. Selection of hybrids from matings of fibroblasts in vitro and their presumed recombinants. *Science* 145:709–10.
- Lyon, M. 1990. L. C. Dunn and mouse genetic mapping. *Genetics* 125:231–36.
- Manney, T., and M. Manney. 1993. Using yeast genetics to generate a research environment. *Genetics* 134:387–91.
- McKusick, V. 1998. *Mendelian Inheritance in Man: A Catalog of Human Genes and Genetic Disorders*, 12th ed. Baltimore: Johns Hopkins University Press.
- McPeck, M., and T. Speed. 1995. Modeling interference in genetic recombination. *Genetics* 139:1031–44.
- Morgan, T. 1919. *The Physical Basis of Heredity*. Philadelphia: Lippincott.
- Morton, N. 1955. Sequential test for the detection of linkage. *American Journal of Human Genetics* 7:277–318.
- . 1995. LODs past and present. *Genetics* 140:7–12.
- Munz, P. 1994. An analysis of interference in the fission yeast *Schizosaccharomyces pombe*. *Genetics* 137:701–7.

B-4 *Appendix B Suggestions for Further Reading*

- O'Brien, S. 1993. *Genetic Maps, 6th ed. Book 3: Lower Eukaryotes. Book 4: Nonhuman Vertebrates. Book 5: The Human Maps. Book 6: Plants.* Cold Spring Harbor, N.Y.: Cold Spring Harbor Laboratory Press.
- Panthier, J., et al. 1990. Evidence for mitotic recombination in *Wei/+* heterozygous mice. *Genetics* 125:175–82.
- Preuss, D., S. Rhee, and R. Davis. 1994. Tetrad analysis possible in *Arabidopsis* with mutation of the QARTET (QRT) genes. *Science* 264:1458–60.
- Risch, N. 1992. Genetic linkage: Interpreting Lod scores. *Science* 255:803–4.
- Roman, H. 1986. The early days of yeast genetics: A personal narrative. *Annual Review of Genetics* 20:1–12.
- Ruddle, F., and R. Kucherlapati. 1974. Hybrid cells and human genes. *Scientific American*, July, 36–44.
- Sorsa, V. 1988. *Chromosome Maps of Drosophila*. Two vols. Boca Raton, FL: CRC Press.
- Sowers, A., ed. 1987. *Cell Fusion*. New York: Plenum Press.
- Stahl, F., and R. Lande. 1995. Estimating interference and linkage map distance from two-factor tetrad data. *Genetics* 139:1449–54.
- Stern, C. 1936. Somatic crossing over and segregation in *Drosophila melanogaster*. *Genetics* 21:625–730.
- Sturtevant, A. 1913. The linear arrangement of six sex-linked factors in *Drosophila*, as shown by their mode of association. *Journal of Experimental Zoology* 14:43–59.
- Sugawara, O., et al. 1990. Induction of cellular senescence in immortalized cells by human chromosome 1. *Science* 247:707–10.
- Svinarich, D., and S. Krawetz. 1993. Gene assignment by quantitative hybridization analysis of somatic cell hybrids. *BioTechniques* 14:82–86.
- Verma, R., and A. Babu. 1994. *Human Chromosomes. Manual of Basic Techniques*. Elmsford, N.Y.: Pergamon Press.
- Zallen, D., and R. Burian. 1992. On the beginnings of somatic cell hybridization: Boris Ephrussi and chromosome transplantation. *Genetics* 132:1–8.
- Zhao, H., and T. Speed. 1996. On genetic map functions. *Genetics* 142:1369–77.
- Chapter 7 Linkage and Mapping in Prokaryotes and Bacterial Viruses**
- American Association for the Advancement of Science. 1997. Frontiers in microbiology. *Science* 276:2. May, special issue.
- Ankenbauer, R. 1997. Reassessing forty years of genetic doctrine: Retrotransfer and conjugation. *Genetics* 145:543–49.
- Atlas, R. 1995. *Principles of Microbiology*. Dubuque, Iowa: Wm. C. Brown Publishers.
- Bachmann, B. 1990. Linkage map of *Escherichia coli* K-12, 8th ed. *Microbiological Review* 54:130–97.
- Berlyn, M., and S. Letovsky. 1992. COTRANS: A program for cotransduction analysis. *Genetics* 131:235–41.
- Birge, E. 1994. *Bacterial and Bacteriophage Genetics*, 3d ed. New York: Springer-Verlag.
- Blum, P., et al. 1999. Forward to the special issue on archaical genetics. *Genetics* 152:1243. (Followed by six articles on the genetics of the archaea.)
- Cairns, J., G. Stent, and J. Watson, eds. 1966. Phage and the origins of molecular biology. *Cold Spring Harbor Symposium on Quantitative Biology* 31.
- Cavalli-Sforza, L. 1992. Forty years ago in *Genetics*: The unorthodox mating behavior of bacteria. *Genetics* 132:635–37.
- Cerritelli, M., et al. 1997. Encapsidated conformation of bacteriophage T7 DNA. *Cell* 91:271–80.
- Clewell, D. 1993. *Bacterial Conjugation*. New York: Plenum Press.
- Eigen, M. 1993. Viral quaspecies. *Scientific American*, July, 42–49.
- Garfield, E. 1990. A tribute to Joshua Lederberg: Highlights of a remarkable scientific career. *Current Contents, Life Sciences* 33:5–11.
- Hoch, J. 1991. Genetic analysis in *Bacillus subtilis*. *Methods in Enzymology* 204:305–20.
- Hoppert, M., and F. Mayer. 1999. Prokaryotes. *Scientific American*, November-December, 518–25.
- Ippen-Ihler, K., and E. Minkley, Jr. 1986. The conjugation system of F, the fertility factor of *Escherichia coli*. *Annual Review of Genetics* 20:593–624.
- Jacob, F., and E. Wollman. 1961. *Sexuality and the Genetics of Bacteria*. New York: Academic Press.
- Lanka, E., and B. Wilkins. 1995. DNA processing reactions in bacterial conjugation. *Annual Review of Biochemistry* 64:141–69.
- Lederberg, J. 1987. Genetic recombination in bacteria: A discovery account. *Annual Review of Genetics* 21:23–46.
- . 1989. Replica plating and indirect selection of bacterial mutants: Isolation of preadaptive mutants in bacteria by sib selection. *Genetics* 121:395–99.
- Lederberg, J., et al. 1951. Recombination analysis of bacterial heredity. *Cold Spring Harbor Symposium on Quantitative Biology* 16:413–43.
- Lloyd, R., and C. Buckman. 1995. Conjugational recombination in *Escherichia coli*: Genetic analysis of recombinant formation in Hfr X F⁻ crosses. *Genetics* 139:1123–48.
- Low, K. 1991. Conjugational methods for mapping with Hfr and F-prime strains. *Methods in Enzymology* 204:43–62.
- Mackenzie, C., A. Simmons, and S. Kaplan. 1999. Multiple chromosomes in bacteria: The Yin and Yang of *trp* gene localization in *Rhodobacter sphaeroides* 2.4.1. *Genetics* 153:525–38.
- Marciano, D., M. Russel, and S. Simon. 1999. An aqueous channel for filamentous phage export. *Science* 284:1516–19.
- Miller, J. 1992. *A Short Course in Bacterial Genetics*. Cold Spring Harbor, N.Y.: Cold Spring Harbor Laboratory Press.
- Miller, R. 1998. Bacterial gene swapping in nature. *Scientific American*, January, 66–71.
- Oldstone, M., and A. Levine. 2000. Virology in the next millennium. *Cell* 100:139–42.
- Redfield, R., M. Schrag, and A. Dean. 1997. The evolution of bacterial transformation: Sex with poor relations. *Genetics* 146:27–38.
- Stahl, F. 1989. The linkage map of phage T4. *Genetics* 123:245–48.
- Stent, G., and R. Calendar. 1978. *Molecular Genetics, An Introductory Narrative*, 2d ed. San Francisco: Freeman.
- Sternberg, N., and R. Maurer. 1991. Bacteriophage-mediated generalized transduction in *Escherichia coli* and *Salmonella typhimurium*. *Methods in Enzymology* 204:18–43.
- Susman, M. 1995. The Cold Spring Harbor phage course (1945–70): A 50th anniversary remembrance. *Genetics* 139:1101–6.
- Teleman, A., et al. 1998. Chromosome arrangement within a bacterium. *Current Biology* 8:1102–9.
- Wikoff, W., and J. Johnson. 1999. Virus assembly: Imaging a molecular machine. *Current Biology* 9:R296–300.
- Zinder, N. 1992. Forty years ago: The discovery of bacterial transduction. *Genetics* 132:291–94.
- Chapter 8 Cytogenetics**
- Antonarakis, S., et al. 1993. Mitotic errors in somatic cells cause trisomy 21 in about 4.5% of cases and are not associated with advanced maternal age. *Nature Genetics* 3:146–50.
- Arnold, M. 1994. Natural hybridization and Louisiana irises. *BioScience* 44:141–47.
- Ashley, C., Jr., and S. Warren. 1995. Trinucleotide repeat expansion and human disease. *Annual Review of Genetics* 29:703–28.
- Ashworth, A., et al. 1991. X-chromosome inactivation may explain the difference in viability of XO humans and mice. *Nature* 351:406–8.
- Barch, M., ed. 1997. *ACT Cytogenetics Laboratory Manual*. Philadelphia: Lippincott.

- Bedell, M., N. Jenkins, and N. Copeland. 1996. Good genes in bad neighbourhoods. *Nature Genetics* 12:229–32.
- Benavente, E., and J. Orellana. 1991. Chromosome differentiation and pairing behavior of polyploids: An assessment on preferential metaphase I associations in colchicine-induced autotetraploid hybrids within the genus *Secale*. *Genetics* 128:433–42.
- Borgaonkar, D. 1997. *Chromosomal Variation in Man: A Catalogue of Chromosomal Variants and Anomalies*, 8th ed. New York: Alan R. Liss, Inc.
- Brewster, T., and P. Gerald. 1978. Chromosome disorders associated with mental retardation. *Pediatric Annals* 7:82–89.
- Chenuil, A., N. Galtier, and P. Berrebi. 1999. A test of the hypothesis of an autopolyploid vs. allopolyploid origin for a tetraploid lineage: Application to the genus *Barbus* (Cyprinidae). *Heredity* 82:373–80.
- Cochran, D. 1983. Alternate-2 disjunction in the German cockroach. *Genetics* 104:215–17.
- Coyne, J., et al. 1993. The fertility effects of pericentric inversions in *Drosophila melanogaster*. *Genetics* 134:487–96.
- Cribiu, E., et al. 1989. Identification of chromosomes involved in a Robertsonian translocation in cattle. *Genetique, Selection, Evolution* 21:555–60.
- De Vicente, M., and P. Arus. 1996. Tetrasomic inheritance of isozymes in sainfoin (*Onobrychis viciaefolia* Scop.). *Journal of Heredity* 87:54–62.
- Dewald, G., et al. 1980. Origin of chi46,XX/46,XY chimerism in a human true hermaphrodite. *Science* 207:321–23.
- Epstein, C. 1988. Mechanisms of the effects of aneuploidy in mammals. *Annual Review of Genetics* 22:51–75.
- Futch, D. 1966. A study of speciation in South Pacific populations of *Drosophila ananassae*. *University of Texas Studies in Genetics* 6615:79–120.
- Gallardo, M., et al. 1999. Discovery of tetraploidy in a mammal. *Nature* 401:341.
- Jacobs, P., and T. Hassold. 1995. The origin of numerical chromosome abnormalities. *Advances in Genetics* 33:101–33.
- Jones, G. 1994. Meiosis in autopolyploid *Crepis capillaris*. III: Comparison of triploids and tetraploids; evidence for non-independence of autonomous pairing sites. *Heredity* 73:215–19.
- Karpechenko, G. 1928. Polyploid hybrids of *Raphanus sativus* L. X *Brassica oleracea* L. *Zeitschrift fuer Induktive Abstamungslehre und Vererbungslehre* 48:1–85.
- Khush, G., et al. 1984. Primary trisomics of rice: Origin, morphology, cytology and use in linkage mapping. *Genetics* 107:141–63.
- Koller, A., J. Heitman, and M. Hall. 1996. Regional bivalent-univalent pairing versus trivalent pairing of a trisomic chromosome in *Saccharomyces cerevisiae*. *Genetics* 144:957–66.
- Leitch, I., and M. Bennett. 1997. Polyploidy in angiosperms. *Trends in Plant Science* 2:470–76.
- Lejeune, J., M. Gautier, and R. Turpin. 1959. Etude des chromosomes somatiques de neuf enfants mongoliens. *Comptes Rendus de L'Academie des Sciences (Paris)* 248:1721–22.
- Loidl, J. 1995. Meiotic chromosome pairing in triploid and tetraploid *Saccharomyces cerevisiae*. *Genetics* 139:1511–20.
- Lukaszewski, A. 1995. Chromatid and chromosome-type breakage-fusion-bridge cycles in wheat (*Triticum aestivum* L.). *Genetics* 140:1069–85.
- Martin, A., et al. 1999. A fertile amphiploid between diploid wheat (*Triticum tauschii*) and crested wheatgrass (*Agropyron cristatum*). *Genome* 42:519–24.
- Orr, H. 1990. “Why polyploidy is rarer in animals than in plants” revisited. *American Naturalist* 136:759–70.
- Page, S., and L. Shaffer. 1997. Nonhomologous Robertsonian translocations form predominantly during female meiosis. *Nature Genetics* 15:231–32.
- Penrose, L. 1933. The relative effects of paternal and maternal age in mongolism. *Journal of Genetics* 27:219–24.
- Pokholkova, G., et al. 1993. Observations on the induction of position effect variegation of euchromatic genes in *Drosophila melanogaster*. *Genetics* 133:231–42.
- Richards, R., et al. 1992. Evidence of founder chromosomes in fragile X syndrome. *Nature Genetics* 1:257–60.
- Sable, J., and S. Henikoff. 1996. Copy number and orientation determine the susceptibility of a gene to silencing by nearby heterochromatin in *Drosophila*. *Genetics* 142:447–58.
- Sturtevant, A. 1925. The effects of unequal crossing over at the Bar locus in *Drosophila*. *Genetics* 10:117–47.
- Talbert, P., and S. Henikoff. 2000. A reexamination of spreading of position-effect variegation in the *white-rough* region of *Drosophila melanogaster*. *Genetics* 154:259–72.
- Therman, E., and B. Susman. 1995. *Human Chromosomes*, 3d ed. New York: Springer-Verlag.
- Trottier, Y., D. Devys, and J. Mandel. 1993. An expanding story (Fragile-X syndrome). *Current Biology* 3:783–86.
- Warburton, D., J. Byrne, and N. Canki. 1991. *An Atlas of Prenatal Development in Conceptions with Chromosomal Anomalies*. New York: Oxford University Press.
- Witkin, H., et al. 1976. Criminality in XYY and XXY men. *Science* 193:547–55.
- Zheng, Y., R. Roseman, and W. Carlson. 1999. Time course study of the chromosome-type breakage-fusion-bridge cycle in maize. *Genetics* 153:1435–44.

Chapter 9 Chemistry of the Gene

- Abdurashidova, G., et al. 2000. Start sites of bi-directional DNA synthesis at the human lamin B2 origin. *Science* 287:2023–26.
- Adams, D., et al. 1992. The role of topoisomerase IV in partitioning bacterial replicons and the structure of catenated intermediates in DNA replication. *Cell* 71:277–88.
- Anderson, R., et al. 1996. Transmission dynamics and epidemiology of BSE in British cattle. *Nature* 382:779–88.
- Arscott, P., et al. 1989. Scanning tunnelling microscopy of Z-DNA. *Nature* 339:484–86.
- Avery, O., C. MacLeod, and M. McCarty. 1944. Studies on the chemical nature of the substance inducing transformation of Pneumococcal types. *Journal of Experimental Medicine* 79:137–58.
- Baker, T., and S. Bell. 1998. Polymerases and the replisomes: machines within machines. *Cell* 92:295–305.
- Baker, T., and S. Wickner. 1992. Genetics and enzymology of DNA replication in *Escherichia coli*. *Annual Review of Genetics* 26:447–77.
- Berger, J., et al. 1996. Structure and mechanism of DNA topoisomerase II. *Nature* 379:225–32.
- Cairns, J. 1963. The chromosomes of *E. coli*. *Cold Spring Harbor Symposium on Quantitative Biology* 28:43–46.
- Cann, I., and Y. Ishino. 1999. Archaeal DNA replication: Identifying the pieces to solve a puzzle. *Genetics* 152:1249–67.
- Chargaff, E., and J. Davidson, eds. 1955. *The Nucleic Acids*. New York: Academic Press.
- Clayton, D. 1991. Replication and transcription of vertebrate mitochondrial DNA. *Annual Review of Cell Biology* 7:453–78.
- Dickerson, R., et al. 1982. The anatomy of A-, B-, and Z-DNA. *Science* 216:475–85.
- Donovan, S., and J. Diffley. 1996. Replication origins in eukaryotes. *Current Opinion in Genetics and Development* 6:203–7.
- Driscoll, R., M. Youngquist, and D. Baldeschwieler. 1990. Atomic-scale imaging of DNA using scanning tunneling microscopy. *Nature* 346:294–96.
- Falkenberg, M., I. Lehman, and P. Elias. 2000. Leading- and lagging-strand DNA synthesis *in vitro* by a reconstituted herpes simplex virus type 1 replisome. *Proceedings of the National Academy of Sciences, USA* 97:3896–3900.

B-6

Appendix B Suggestions for Further Reading

- Fraenkel-Conrat, H., and B. Singer. 1957. Virus reconstitution. II: Combination of protein and nucleic acid from different strains. *Biobchimica et Biophysica Acta* 24:540–48.
- Frank-Kamenetskii, M., and S. Mirkin. 1995. Triplex DNA structures. *Annual Review of Biochemistry* 64:65–95.
- Griffith, F. 1928. Significance of pneumococcal types. *Journal of Hygiene* 27:113–59.
- Hershey, A., and M. Chase. 1952. Independent functions of viral protein and nucleic acid in growth of bacteriophage. *Journal of General Physiology* 36:39–56.
- Holt, I., H. Lorimer, and H. Jacobs. 2000. Coupled leading- and lagging-strand synthesis of mammalian mitochondrial DNA. *Cell* 100:515–24.
- Hübscher, U., H. Nasheuer, and J. Syväoja. 2000. Eukaryotic DNA polymerases, a growing family. *Trends in Biochemical Sciences* 25:143–47.
- Jacobs, C., and L. Shapiro. 1999. Bacterial cell division: A moveable feast. *Proceedings of the National Academy of Sciences, USA* 96:5891–93.
- Joyce, C., and T. Steitz. 1994. Function and structure relationships in DNA polymerases. *Annual Review of Biochemistry* 63:777–822.
- Kamada, K., et al. 1996. Structure of a replication-terminator protein complexed with DNA. *Nature* 383:598–603.
- Keck, J., and J. Berger. 1999. Enzymes that push DNA around. *Nature Structural Biology* 6:900–902.
- Kelman, Z., and M. O'Donnell. 1995. DNA polymerase III holoenzyme: Structure and function of a chromosomal replicating machine. *Annual Review of Biochemistry* 64:171–200.
- Kornberg, A. 1989. *For the Love of Enzymes: The Odyssey of a Biochemist*. Cambridge, Mass.: Harvard University Press.
- Kramer, M., et al. 1998. Lagging-strand replication of rolling-circle plasmids: Specific recognition of the *ssaA*-type origins in different gram-positive bacteria. *Proceedings of the National Academy of Sciences, USA* 95:10505–10.
- Lederberg, J. 1991. The gene (H. J. Muller, 1947). *Genetics* 129:313–16.
- Lindahl, T., and D. Barnes. 1992. Mammalian DNA ligases. *Annual Review of Biochemistry* 61:251–81.
- Lohman, T., and M. Ferrari. 1994. *Escherichia coli* single-stranded DNA-binding protein: Multiple DNA-binding modes and cooperativities. *Annual Review of Biochemistry* 63:527–70.
- Manna, A., et al. 1996. The dimer-dimer interaction surface of the replication terminator protein of *Bacillus subtilis* and termination of DNA replication. *Proceedings of the National Academy of Sciences, USA* 93:3253–58.
- . 1996. Helicase-contrahelicase interaction and the mechanism of termination of DNA replication. *Cell* 87:881–91.
- Marahens, Y., and B. Stillman. 1992. A yeast chromosomal origin of DNA replication defined by multiple functional elements. *Science* 255:817–23.
- Marians, K. 1992. Prokaryotic DNA replication. *Annual Review of Biochemistry* 61:673–719.
- Marmur, J., and P. Doty. 1962. Determination of the base composition of deoxyribonucleic acid from its thermal denaturation temperature. *Journal of Molecular Biology* 5:109–18.
- Marx, J. 1995. How DNA replication originates. *Science* 270:1585–87.
- Meselson, M., and F. Stahl. 1958. The replication of DNA in *Escherichia coli*. *Proceedings of the National Academy of Sciences, USA* 44:671–82.
- Moon, K., et al. 1999. Identification and reconstitution of the origin of recognition complex from *Schizosaccharomyces pombe*. *Proceedings of the National Academy of Sciences, USA* 96:12367–72.
- Morais-Cabral, J., et al. 1997. Crystal structure of the breakage-reunion domain of DNA gyrase. *Nature* 388:903–6.
- Nelson, J., C. Lawrence, and D. Hinkle. 1996. Thymine-thymine dimer bypass by yeast DNA polymerase ζ . *Science* 272:1646–49.
- Niki, H., and S. Hiraga. 1998. Polar localization of the replication origin and terminus in *Escherichia coli* nucleoids during chromosome partitioning. *Genes & Development* 12:1036–45.
- Novick, R. 1998. Contrasting lifestyles of rolling-circle phages and plasmids. *Trends in Biochemical Sciences* 23:434–38.
- Prusiner, S., ed. 1999. *Prion Biology and Diseases*. Cold Spring Harbor, N. Y.: Cold Spring Harbor Laboratory Press.
- Rangarajan, S., et al. 1997. *Escherichia coli* DNA polymerase II catalyzes chromosomal and episomal DNA synthesis *in vivo*. *Proceedings of the National Academy of Sciences, USA* 94:946–51.
- Rich, A., A. Nordheim, and A. Wang. 1984. The chemistry of left-handed Z-DNA. *Annual Review of Biochemistry* 53:791–846.
- Roca, J. 1995. The mechanisms of DNA topoisomerases. *Trends in Biochemical Sciences* 20:156–60.
- Rothfield, L., S. Justice, and J. Garcia-Lara. 1999. Bacterial cell division. *Annual Review of Genetics* 33:423–48.
- Shamoo, Y., and T. Steitz. 1999. Building a replisomes from interacting pieces: Sliding clamp complexed to a peptide from DNA polymerase and a polymerase editing complex. *Cell* 99:155–66.
- Shapiro, L., and R. Losick. 2000. Dynamic spatial regulation in the bacterial cell. *Cell* 100:89–98.
- Sharpe, M., and J. Errington. 1999. Upheaval in the bacterial nucleoid. *Trends in Genetics* 15:70–74.
- Stillman, B. 1994. Smart machines at the DNA replication fork. *Cell* 78:725–28.
- Strobel, S., and P. Dervan. 1990. Site-specific cleavage of a yeast chromosome by oligonucleotide-directed triple-helix formation. *Science* 249:73–75.
- Subramanya, H., et al. 1996. Crystal structure of an ATP-dependent DNA ligase from bacteriophage T7. *Cell* 85:607–15.
- Tessman, I., and M. Kennedy. 1994. DNA polymerase II of *Escherichia coli* in the bypass of abasic sites *in vivo*. *Genetics* 136:439–48.
- Toyn, J., et al. 1995. The activation of DNA replication in yeast. *Trends in Biochemical Sciences* 20:70–73.
- Turchi, J., et al. 1994. Enzymatic completion of mammalian lagging-strand DNA replication. *Proceedings of the National Academy of Sciences, USA* 91:9803–7.
- Vlieghe, D., et al. 1996. Parallel and antiparallel (G•GC)₂ triple helix fragments in a crystal structure. *Science* 273:1702–5.
- Waga, S., and B. Stillman. 1994. Anatomy of a DNA replication fork revealed by a reconstitution of SV40 DNA replication *in vitro*. *Nature* 369:207–12.
- . 1998. The DNA replication fork in eukaryotic cells. *Annual Review of Biochemistry* 67:721–51.
- Wang, A., et al. 1981. Left-handed double helical DNA: Variations in the backbone conformation. *Science* 211:171–76.
- Wang, J. 1996. DNA topoisomerases. *Annual Review of Biochemistry* 65:635–92.
- Watson, J. 1968. *The Double Helix*. New York: Signet.
- Watson, J., and F. Crick. 1953. Molecular structure of nucleic acids: A structure for deoxyribose nucleic acid. *Nature* 171:737–38.
- West, S. 1996. DNA helicase: New breeds of translocating motors and molecular pumps. *Cell* 86:177–80.
- Westheimer, F. 1987. Why nature chose phosphates. *Science* 235:1173–78.
- Wheeler, R., and L. Shapiro. 1997. Bacterial chromosome segregation: Is there a mitotic apparatus? *Cell* 88:577–79.
- Wickner, R. 1996. Prions and RNA viruses of *Saccharomyces cerevisiae*. *Annual Review of Genetics* 30:109–139.
- Yuzhakov, A., Z. Kelman, and M. O'Donnell. 1999. Trading places on DNA—A three-point switch underlies primer handoff from primase to the replicative DNA polymerase. *Cell* 96:153–63.
- Zubay, G. 1997. *Biochemistry*, 4th ed. Dubuque, Iowa: Wm. C. Brown Publishers.

**Chapter 10 Gene Expression:
Transcription**

- Arkhipova, I. 1995. Promoter elements in *Drosophila melanogaster* revealed by sequence analysis. *Genetics* 139: 1359–69.
- Belfort, M. 1990. Phage T4 introns: Self-splicing and mobility. *Annual Review of Genetics* 24:363–85.
- Bell, S., et al. 1999. Orientation of the transcription preinitiation complex in Archaea. *Proceedings of the National Academy of Sciences, USA* 96:13662–67.
- Bentley, D. 1999. Coupling RNA polymerase II transcription with pre-mRNA processing. *Current Opinion in Cell Biology* 11:347–51.
- Brewer, B. 1988. When polymerases collide: Replication and the transcriptional organization of the *E. coli* chromosome. *Cell* 53:679–86.
- Brody, E., and J. Abelson. 1985. The “spliceosome”: Yeast pre-messenger RNA associates with a 40S complex in a splicing-dependent reaction. *Science* 228:963–67.
- Burley, S., and R. Roeder. 1996. Biochemistry and structural biology of transcription factor IID (TFIID). *Annual Review of Biochemistry* 65:769–99.
- Busby, S., and R. Ebright. 1994. Promoter structure, promoter recognition, and transcription activation in prokaryotes. *Cell* 79:743–46.
- Cattaneo, R. 1991. Different types of messenger RNA editing. *Annual Review of Genetics* 28:71–88.
- Cech, T. 1986. RNA as an enzyme. *Scientific American*, November, 64–75.
- . 1990. Self-splicing of group I introns. *Annual Review of Biochemistry* 59:543–68.
- Cech, T., and O. Uhlenbeck. 1994. Hammerhead nailed down. *Nature* 372:39–40.
- Cheng, S.-W., et al. 1991. Functional importance of sequence in the stem-loop of a transcription terminator. *Science* 254:1205–7.
- Conaway, J., and R. Conaway. 1999. Transcription elongation and human disease. *Annual Review of Biochemistry* 68:301–19.
- Cramer, P., et al. 2000. Architecture of RNA polymerase II and implications for the transcription mechanism. *Science* 288:640–49.
- Crick, F. 1970. Central dogma of molecular biology. *Nature* 227:561–63.
- Das, A. 1993. Control of transcription termination by RNA-binding proteins. *Annual Review of Biochemistry* 62:893–930.
- Di Nicola Negri, E., et al. 1997. The eucaryal tRNA splicing endonuclease recognizes a tripartite set of RNA elements. *Cell* 89:859–66.
- Dinter-Gottlieb, G. 1986. Viroids and virusoids are related to group I introns. *Proceedings of the National Academy of Sciences, USA* 83:6250–54.
- Dombroski, A., et al. 1996. The sigma subunit of *Escherichia coli* RNA polymerase senses promoter spacing. *Proceedings of the National Academy of Sciences, USA* 93:8858–62.
- Dorit, R., L. Schoenbach, and W. Gilbert. 1990. How big is the universe of exons? *Science* 250:1377–82.
- Dove, S., J. Joung, and A. Hochschild. 1997. Activation of prokaryotic transcription through arbitrary protein-protein contacts. *Nature* 386:627–30.
- Eckstein, F., and D. Lilley, eds. 1997. *Catalytic RNA*. New York: Springer.
- Estrem, S., et al. 1999. Bacterial promoter architecture: Subsite structure of UP elements and interactions with the carboxy-terminal domain of the RNA polymerase subunit. *Genes & Development* 13:2134–47.
- Ferat, J.-L., and F. Michel. 1993. Group II self-splicing introns in bacteria. *Nature* 364:358–61.
- Frank, D., and N. Pace. 1998. Ribonuclease P: Unity and diversity in a tRNA-processing ribozyme. *Annual Review of Biochemistry* 67:153–80.
- Fu, J., et al. 1999. Yeast RNA polymerase II at 5 Å resolution. *Cell* 98:799–810.
- Garrett, R. 1999. Mechanics of the ribosome. *Nature* 400:811–12. (Introducing two crystallographic studies.)
- Geiselman, J., et al. 1993. A physical model for the translocation and helicase activities of *Escherichia coli* transcription termination protein rho. *Proceedings of the National Academy of Sciences, USA* 90:7754–58.
- Gesteland, R., T. Cech, and J. Atkins, eds. 1999. *The RNA World*, 2d ed. Cold Spring Harbor, N. Y.: Cold Spring Harbor Laboratory Press.
- Goodrich, J., G. Cutler, and R. Tjian. 1996. Contacts in context: Promoter specificity and macromolecular interactions in transcription. *Cell* 84:825–30.
- Guthold, M., et al. 1994. Following the assembly of RNA polymerase-DNA complexes in aqueous solutions with the scanning force microscope. *Proceedings of the National Academy of Sciences, USA* 91:12927–31.
- Hahn, S. 1998. The role of TAFs in RNA polymerase II transcription. *Cell* 95:579–82.
- Hall, B., and S. Spiegelman. 1961. Sequence complementarity of T2-DNA and T2-specific RNA. *Proceedings of the National Academy of Sciences, USA* 47:137–46.
- Helmann, J., and M. Chamberlain. 1988. Structure and function of bacterial sigma factors. *Annual Review of Biochemistry* 57:839–72.
- Holley, R., et al. 1965. Structure of a ribonucleic acid. *Science* 147: 1462–65.
- Kable, M., et al. 1996. RNA editing: A mechanism for gRNA-specified uridylyl insertion into precursor mRNA. *Science* 273:1189–95.
- Kim, Y., et al. 1993. Crystal structure of a yeast TBP/TATA-box complex. *Nature* 365:512–20.
- Koleske, A., and R. Young. 1995. The RNA polymerase II holoenzyme and its implications for gene regulation. *Trends in Biochemical Sciences* 20:113–17.
- Krämer, A. 1996. The structure and function of proteins involved in mammalian pre-mRNA splicing. *Annual Review of Biochemistry* 65:367–409.
- Lake, J. 1976. Ribosome structure determined by electron microscopy of *Escherichia coli* small subunits, large subunits and monomeric ribosomes. *Journal of Molecular Biology* 105:131–59.
- Liu, B., and B. Alberts. 1995. Head-on collision between a DNA replication apparatus and RNA polymerase transcription complex. *Science* 267:1131–37.
- Liu, B., et al. 1993. The DNA replication fork can pass RNA polymerase without displacing the nascent transcript. *Nature* 366:33–39.
- Lopez, A. 1998. Alternative splicing of pre-mRNA: Developmental consequences and mechanisms of regulation. *Annual Review of Genetics* 32:279–305.
- Lykke-Anderson, J., et al. 1997. Archaeal introns: Splicing, intercellular mobility and evolution. *Trends in Biochemical Sciences* 22:326–31.
- Maréchal-Drouard, L., J. Weil, and A. Dietrich. 1993. Transfer RNAs and transfer RNA genes in plants. *Annual Review of Plant Physiology and Plant Molecular Biology* 44:13–32.
- Marmorstein, R., et al. 1992. DNA recognition by GAL4: Structure of a protein-DNA complex. *Nature* 356:408–14.
- McCarthy, B., and J. Holland. 1965. Denatured DNA as a direct template for in vitro protein synthesis. *Proceedings of the National Academy of Sciences, USA* 54:880–86.
- McCracken, S., et al. 1997. The C-terminal domain of RNA polymerase II couples mRNA processing to transcription. *Nature* 385:357–61.
- McKeown, M. 1992. Alternative mRNA splicing. *Annual Review of Cell Biology* 8:133–55.
- Michel, F., and J.-L. Ferat. 1995. Structure and activities of group II introns. *Annual Review of Biochemistry* 64:435–61.

B-8 *Appendix B Suggestions for Further Reading*

- Miller, O., Jr., et al. 1970. Electron microscopic visualization of transcription. *Cold Spring Harbor Symposium on Quantitative Biology* 35:505–12.
- Mistell, T., J. Caceres, and D. Spector. 1997. The dynamics of pre-mRNA splicing factor in living cells. *Nature* 387:523–27.
- Murray, H., and K. Jarrell. 1999. Flipping the switch to an active spliceosome. *Cell* 96:599–602.
- Naryshkin, N., et al. 2000. Structural organization of the RNA polymerase-promoter open complex. *Cell* 101:601–11.
- Nikolov, D., and S. Burley. 1997. RNA polymerase II transcription initiation: a structural view. *Proceedings of the National Academy of Sciences, USA* 94:15–22.
- Olson, M., M. Dundr, and A. Szebeni. 2000. The nucleolus: An old factory with unexpected capabilities. *Trends in Cell Biology* 10:189–96.
- Pabo, C., and R. Sauer. 1992. Transcription factors: Structural families and principles of DNA recognition. *Annual Review of Biochemistry* 61:1053–95.
- Peterson, M., et al. 1991. Structure and functional properties of human general transcription factor IIE. *Nature* 354:369–73.
- Pley, H., K. Flaherty, and D. McKay. 1994. Three-dimensional structure of a hammerhead ribozyme. *Nature* 372:68–74.
- Proudfoot, N. 1996. Ending the message is not so simple. *Cell* 87:779–81.
- Ptashne, M., and A. Gann. 1997. Transcriptional activation by recruitment. *Nature* 386:569–77.
- Pyle, A. 1993. Ribozymes: A distinct class of metalloenzymes. *Science* 261:709–14.
- Rich, A. 1993. The structure and biology of transfer RNA. *Structure* 1:vi–vii.
- Riesner, D., and H. Gross. 1985. Viroids. *Annual Review of Biochemistry* 54:531–64.
- Roeder, R. 1996. The role of general initiation factors in transcription by RNA polymerase II. *Trends in Biochemical Sciences* 21:327–35.
- Ross, W., et al. 1993. A third recognition element in bacterial promoters: DNA binding by the α subunit of RNA polymerase. *Science* 262:1407–13.
- Schafer, D., et al. 1991. Transcription by single molecules of RNA polymerase observed by light microscopy. *Nature* 352:444–48.
- Scheer, U., and R. Hock. 1999. Structure and function of the nucleolus. *Current Opinion in Cell Biology* 11:385–90.
- Scott, W., and A. Klug. 1996. Ribozymes: Structure and mechanism in RNA catalysis. *Trends in Biochemical Sciences* 21:220–24.
- Sharp, P. 1994. Split genes and RNA splicing. *Cell* 77:805–15.
- Shih, M.-C., P. Heinrich, and H. Goodman. 1988. Intron existence predated the divergence of eukaryotes and prokaryotes. *Science* 242:1164–66.
- Shimamoto, A., et al. 1996. A unique human gene that spans over 230 kb in the human chromosome 8p11-12 and codes multiple family proteins sharing RNA-binding motifs. *Proceedings of the National Academy of Sciences, USA* 93:10913–17.
- Sleeman, J., and A. Lamond. 1999. Nuclear organization of pre-mRNA splicing factors. *Current Opinion in Cell Biology* 11:372–77.
- Smith, C., and J. Steitz. 1997. Sno storm in the nucleolus: new roles for myriad small RNPs. *Cell* 89:669–72.
- Smith, C., J. Patton, and B. Nadal-Ginard. 1989. Alternative splicing in the control of gene expression. *Annual Review of Genetics* 23:527–77.
- Steinmetz, E., and T. Platt. 1994. Evidence supporting a tethered tracking model for helicase activity of *Escherichia coli* rho factor. *Proceedings of the National Academy of Sciences, USA* 91:1401–5.
- Steitz, J. 1988. “Snurps.” *Scientific American*, June, 56–63.
- Stillman, B., ed. 1998. *Mechanisms of Transcription*. Cold Spring Harbor, N.Y.: Cold Spring Harbor Laboratory Press.
- Stoltzfus, A., et al. 1994. Testing the exon theory of genes: The evidence from protein structure. *Science* 265:202–7.
- Tarn, W.-Y., and J. Steitz. 1997. Pre-mRNA splicing: The discovery of a new spliceosome doubles the challenge. *Trends in Biochemical Sciences* 22:132–37.
- Tennyson, C., H. Klamut, and R. Worton. 1995. The human dystrophin gene requires 16 hours to be transcribed and is cotranscriptionally spliced. *Nature Genetics* 9:184–90.
- Tjian, R. 1995. Molecular machines that control genes. *Scientific American*, February, 54–61.
- Tjian, R., and T. Maniatis. 1994. Transcriptional activation: A complex puzzle with few easy pieces. *Cell* 77:5–8.
- Uptain, S., C. Kane, and M. Chamberlin. 1997. Basic mechanisms of transcript elongation and its regulation. *Annual Review of Biochemistry* 66:117–72.
- Venema, J., and D. Tollervey. 1999. Ribosome synthesis in *Saccharomyces cerevisiae*. *Annual Review of Genetics* 33:261–311.
- Von Hippel, P. 1994. Passing lanes for polymerases? *Current Biology* 4:333–36.
- Waldrop, M. 1992. Finding RNA makes proteins gives “RNA world” a big boost. *Science* 256:1396–97.
- Wang, J.-F., and T. Cech. 1992. Tertiary structure around the guanosine-binding site of the *Tetrahymena* ribozyme. *Science* 256:526–29.
- Wedekind, J., and D. McKay. 1998. Crystallographic structures of the hammerhead ribozyme: Relationship to ribozyme folding and catalysis. *Annual Review of Biophysics and Biomolecular Structure* 27:475–502.
- White, R. 1998. *RNA Polymerase III Transcription*. New York: Springer.
- Wieczorek, E., et al. 1998. Function of TAF_{II}-containing complex without TBP in transcription by RNA polymerase II. *Nature* 393:187–91.
- Will, C., et al. 1999. Identification of both shared and distinct proteins in the major and minor spliceosomes. *Science* 284:2003–5.
- Wolin, S., and A. Matera. 1999. The trials and travels of tRNA. *Genes & Development* 13:1–10.
- Woodson, S., and N. Leontis. 1998. Structure and dynamics of ribosomal RNA. *Current Opinion in Structural Biology* 8:294–300.
- Zaug, A., and T. Cech. 1986. The intervening sequence RNA of *Tetrahymena* is an enzyme. *Science* 231:470–75.
- Zawel, L., and D. Reinberg. 1995. Common themes in assembly and function of eukaryotic transcription complexes. *Annual Review of Biochemistry* 64:533–61.
- Zeigler, D., and D. Dean. 1990. Orientation of genes in the *Bacillus subtilis* chromosome. *Genetics* 125:703–8.
- Zorio, D., et al. 1994. Operons as a common form of chromosomal organization in *C. elegans*. *Nature* 372:270–72.

Chapter 11 Gene Expression: Translation

- Arnez, J., and D. Moras. 1997. Structural and functional considerations of the aminoacylation reaction. *Trends in Biochemical Sciences* 22:211–16.
- Batey, R., et al. 2000. Crystal structure of the ribonucleoprotein core of the signal recognition particle. *Science* 287:1232–39.
- Berchtold, H., et al. 1993. Crystal structure of active elongation factor Tu reveals major domain rearrangements. *Nature* 365:126–32.
- Bernabeu, C., and J. Lake. 1982. Nascent polypeptide chains emerge from the exit domain of the large ribosomal subunit: Immune mapping of the nascent chains. *Proceedings of the National Academy of Sciences, USA* 79:3111–15.
- Boyd, L., and C. Thummel. 1993. Selection of CUG and AUG initiator codons for *Drosophila* E74A translation depends on downstream sequences. *Proceedings of the National Academy of Sciences, USA* 90:9164–67.

- Bukau, B., et al. 2000. Getting newly synthesized proteins into shape. *Cell* 101:119–22.
- Capel, M., et al. 1987. A complete mapping of the proteins in the small ribosomal subunit of *Escherichia coli*. *Science* 238:1403–6.
- Caras, L., and G. Weddell. 1989. Signal peptide for protein secretion directing glycopospholipid membrane anchor attachment. *Science* 243:1196–98.
- Cate, J., et al. 1999. X-ray crystal structures of 70S ribosome functional complexes. *Science* 285:2095–104.
- Chapeville, F., et al. 1962. On the role of soluble ribonucleic acid in coding for amino acids. *Proceedings of the National Academy of Sciences, USA* 48:1086–92.
- Chen, C.-Y., and P. Sarnow. 1995. Initiation of protein synthesis by the eukaryotic translational apparatus on circular RNAs. *Science* 268:415–17.
- Chin, G., and J. Marx, eds. 1994. Resistance to antibiotics: Frontiers in biotechnology. *Science* 264:359–93.
- Chothia, C., and A. Finkelstein. 1990. The classification and origins of protein folding patterns. *Annual Review of Biochemistry* 59:1007–39.
- Cigan, A., L. Feng, and T. Donahue. 1988. tRNA^{Met} functions in directing the scanning ribosome to the start site of translation. *Science* 242:93–97.
- Crick, F. 1966. Codon-anticodon pairing: The wobble hypothesis. *Journal of Molecular Biology* 19:548–55.
- Curnow, A., et al. 1997. Glu-tRNA^{Gln} amidotransferase: A novel heterotrimeric enzyme required for correct decoding of glutamine codons during translation. *Proceedings of the National Academy of Sciences, USA* 94:11819–26.
- Ellis, R. 1996. *The Chaperonins*. San Diego: Academic Press.
- Feltham, J., and L. Gierasch. 2000. GroEL-substrate interactions: Molding the fold, or folding the mold? *Cell* 100:193–96.
- Frank, J. 1998. How the ribosome works. *American Scientist* 86:428–39.
- Freyman, D., et al. 1997. Structure of the conserved GTPase domain of the signal recognition particle. *Nature* 385:361–64.
- Gabashvili, I., et al. 2000. Solution structure of the *E. coli* 70S ribosome at 11.5 Å resolution. *Cell* 100:537–49.
- Gerber, A., and W. Keller. 1999. An adenosine deaminase that generates inosine at the wobble position of tRNAs. *Science* 286:1146–49.
- Gingras, A., B. Raught, and N. Sonenberg. 1999. eIF4 initiation factors: Effectors of mRNA recruitment to ribosomes and regulators of translation. *Annual Review of Biochemistry* 68:913–63.
- Green, R., and H. Noller. 1997. Ribosomes and translation. *Annual Review of Biochemistry* 66:679–716.
- Grunberg-Manago, M. 1963. Polynucleotide phosphorylase. *Progress in Nucleic Acid Research* 1:93–133.
- Hendrick, J., and F. Hartl. 1993. Molecular chaperone functions of heat-shock proteins. *Annual Review of Biochemistry* 62:349–84.
- Houry, W., et al. 1999. Identification of *in vivo* substrates of the chaperonin GroEL. *Nature* 402:147–54.
- Ibba, M., and D. Söll. 1999. Quality control mechanisms during translation. *Science* 286:1893–97.
- Ito, K., M. Uno, and Y. Nakamura. 2000. A tripeptide ‘anticodon’ deciphers stop codons in messenger RNA. *Nature* 403:680–84.
- Knight, R., S. Freeland, and L. Landweber. 1999. Selection, history and chemistry: The three faces of the genetic code. *Trends in Biochemical Sciences* 24:241–47.
- Kurland, C. 1992. Translational accuracy and the fitness of bacteria. *Annual Review of Genetics* 26:29–50.
- Lee, V., G. Miller, and M. Yagisawa. 1999. What’s new in the antibiotic pipeline. *Current Opinion in Microbiology* 2:475–82.
- Levy, S. 1998. The challenge of antibiotic resistance. *Scientific American*, March, 46–53.
- Lodmell, J., and A. Dahlberg. 1997. A conformational switch in *Escherichia coli* 16S ribosomal RNA during decoding of messenger RNA. *Science* 277:1262–67.
- Low, S., and M. Berry. 1996. Knowing when not to stop: Selenocysteine incorporation in eukaryotes. *Trends in Biochemical Sciences* 21:203–8.
- Macejak, D., and P. Sarnow. 1991. Internal initiation of translation mediated by the 5′ leader of a cellular mRNA. *Nature* 353:90–94.
- Mande, S., et al. 1996. Structure of the heat shock protein chaperonin-10 of *Mycobacterium leprae*. *Science* 271:203–7.
- McCarthy, J., and R. Brimacombe. 1994. Prokaryotic translation: The interactive pathway leading to initiation. *Trends in Genetics* 10:402–7.
- Moazed, D., and H. Noller. 1989. Interaction of tRNA with 23S rRNA in the ribosomal A, P, and E sites. *Cell* 57:585–97.
- Nakamura, Y., K. Ito, and L. Isaksson. 1996. Emerging understanding of translation termination. *Cell* 87:147–50.
- Neupert, W. 1997. Protein import into mitochondria. *Annual Review of Biochemistry* 66:863–917.
- Nirenberg, M., and P. Leder. 1964. RNA code-words and protein synthesis: The effect of trinucleotides upon the binding of tRNA to ribosomes. *Science* 145:1399–1407.
- Nirenberg, M., and J. Matthaei. 1961. The dependence of cell-free protein synthesis in *E. coli* upon naturally occurring or synthetic polyribonucleotides. *Proceedings of the National Academy of Sciences, USA* 47:1588–1602.
- Nissen, P., et al. 1998. The crystal structure of Cys-tRNA^{Cys}-EF-Tu-GDPNP reveals general and specific features in the ternary complex and in tRNA. *Structure* 7:143–56.
- Normanly, J., and J. Abelson. 1989. tRNA identity. *Annual Review of Biochemistry* 58:1029–49.
- Ochoa, S. 1980. The pursuit of a hobby. *Annual Review of Biochemistry* 49:1–30.
- Preer, J., et al. 1985. Deviation from the universal code shown by the gene for surface protein 51A in *Paramecium*. *Nature* 314:188–90.
- Richards, F. 1991. The protein folding problem. *Scientific American*, January, 54–63.
- Rodnina, M., et al. 1997. Hydrolysis of GTP by elongation factor G drives tRNA movement on the ribosome. *Nature* 385:37–41.
- Ruff, M., et al. 1991. Class II aminoacyl transfer RNA synthetases: Crystal structure of yeast aspartyl-tRNA synthetase complexed with tRNA^{Asp}. *Science* 252:1682–89.
- Ryabova, L., and T. Hohn. 2000. Ribosome shunting in the cauliflower mosaic virus 35S RNA leader is a special case of reinitiation of translation functioning in plant and animal systems. *Genes & Development* 14:817–29.
- Sachs, A. 2000. Cell cycle-dependent translation initiation: IRES elements prevail. *Cell* 101:243–45.
- Sachs, A., P. Sarnow, and M. Hentze. 1997. Starting at the beginning, middle, and end: Translation initiation in eukaryotes. *Cell* 89:831–38.
- Saks, M., J. Sampson, and J. Abelson. 1994. The transfer RNA identity problem: A search for rules. *Science* 263:191–97.
- Samaha, R., R. Green, and H. Noller. 1995. A base pair between tRNA and 23S rRNA in the peptidyl transferase centre of the ribosome. *Nature* 377:309–14.
- Sanger, F. 1988. Sequences, sequences, and sequences. *Annual Review of Biochemistry* 57:1–28.
- Schimmel, P. 1987. Aminoacyl tRNA synthetases: General scheme of structure-function relationships in the polypeptides and recognition of transfer RNAs. *Annual Review of Biochemistry* 56:125–58.

B-10 *Appendix B Suggestions for Further Reading*

- Scott, J., et al. 1983. Structure of a mouse submaxillary messenger RNA encoding epidermal growth factor and seven related proteins. *Science* 221:236–40.
- Selmer, M., et al. 1999. Crystal structure of *Thermotoga maritima* ribosome recycling factor: a tRNA mimic. *Science* 286:2349–52.
- Shtilerman, M., G. Lorimer, and S. Englander. 1999. Chaperonin function: Folding by forced unfolding. *Science* 284:822–25.
- Sigler, P., et al. 1998. Structure and function in Gro-EL-mediated protein folding. *Annual Review of Biochemistry* 67:581–608.
- Song, H., et al. 2000. The crystal structure of human eukaryotic release factor eRF1—Mechanism of stop codon recognition and peptidyl-tRNA hydrolysis. *Cell* 100:311–21.
- Sprinzl, M. 1994. Elongation factor Tu: A regulatory GTPase with an integrated effector. *Trends in Biochemical Sciences* 19:245–50.
- Stadtman, T. 1996. Selenocysteine. *Annual Review of Biochemistry* 65:83–100.
- Stansfield, I., and M. Tuite. 1994. Polypeptide chain termination in *Saccharomyces cerevisiae*. *Current Genetics* 25:385–95.
- Stark, H., et al. 1997. Arrangement of tRNAs in pre- and posttranslational ribosomes revealed by electron cryomicroscopy. *Cell* 88:19–28.
- . 2000. Large-scale movement of elongation factor G and extensive conformational change of the ribosome during translocation. *Cell* 100:301–9.
- Stathopoulos, C., et al. 2000. One polypeptide with two aminoacyl-tRNA synthetase activities. *Science* 287:479–82.
- Stern, S., et al. 1989. RNA-protein interactions in 30S ribosomal subunits: Folding and function of 16S rRNA. *Science* 244:783–90.
- Vogel, G. 1998. Tracking the history of the genetic code. *Science* 281:329–31.
- Wilson, K., and H. Noller. 1998. Molecular movement inside the translational engine. *Cell* 92:337–49.
- Wolin, S. 1994. From the elephant to *E. coli*: SRP-dependent protein targeting. *Cell* 77:787–90.
- Xu, Z., A. Horwich, and P. Sigler. 1997. The crystal structure of the asymmetric GroEL-GroES-(ADP)₇ chaperonin complex. *Nature* 388:741–50.
- Amrein, M., et al. 1988. Scanning tunneling microscopy of RecA-DNA complexes coated with a conducting film. *Science* 240:514–16.
- Anderson, D., and S. Kowalczykowski. 1997. The recombination hot spot χ is a regulatory element that switches the polarity of DNA degradation by the RecBCD enzyme. *Genes & Development* 11:571–81.
- Ariyoshi, M., et al. 1994. Atomic structure of the RuvC resolvase: A Holliday junction-specific endonuclease from *E. coli*. *Cell* 78:1063–72.
- Benzer, S. 1961. On the topography of the genetic fine structure. *Proceedings of the National Academy of Sciences, USA* 47:403–15.
- Bruner, S., D. Norman, and G. Verdine. 2000. Structural basis for recognition and repair of the endogenous mutagen 8-oxoguanine in DNA. *Nature* 403:859–66.
- Bruner, S., et al. 1998. Repair of oxidatively damaged guanine in *Saccharomyces cerevisiae* by an alternative pathway. *Current Biology* 8:393–403.
- Buermeyer, A., et al. 1999. Mammalian DNA mismatch repair. *Annual Review of Genetics* 33:533–64.
- Cairns, J., J. Overbaugh, and S. Miller. 1988. The origin of mutants. *Nature* 335:142–45.
- Camerini-Otero, R., and P. Hsieh. 1995. Homologous recombination proteins in prokaryotes and eukaryotes. *Annual Review of Genetics* 29:509–52.
- Cox, M., et al. 2000. The importance of repairing stalled replication forks. *Nature* 404:37–41.
- Craig, N. 1988. The mechanism of conservative site-specific recombination. *Annual Review of Genetics* 22:77–105.
- Culotta, E., and D. Koshland, Jr. 1994. Molecule of the year: DNA repair works its way to the top. *Science* 266:1926–29.
- de Laat, W., N. Jaspers, and J. Hoeijmakers. 1999. Molecular mechanisms of nucleotide excision repair. *Genes & Development* 13:768–85.
- Drake, J., et al. 1998. Rates of spontaneous mutation. *Genetics* 148:1667–86.
- Eggleston, A., and S. West. 1996. Exchanging partners: recombination in *E. coli*. *Trends in Genetics* 12:20–26.
- Epstein, R., et al. 1963. Physiological studies of conditional lethal mutations of bacteriophage T4D. *Cold Spring Harbor Symposium on Quantitative Biology* 28:375–92.
- Featherstone, C., and S. Jackson. 1999. Ku, a DNA repair protein with multiple cellular functions? *Mutation Research* 434:3–15.
- Foster, P. 1999. Mechanisms of stationary phase mutation: A decade of adaptive mutation. *Annual Review of Genetics* 33:57–88.
- Friedberg, E. 1996. Relationships between DNA repair and transcription. *Annual Review of Biochemistry* 65:15–42.
- Friedberg, E., G. Walker, and W. Siede. 1995. *DNA Repair and Mutagenesis*. Herndon, Va.: ASM Press.
- Goodman, M., and B. Tiffin. 2000. Sloopier copier DNA polymerases involved in genome repair. *Current Opinion in Genetics & Development* 10:162–68.
- Green, M. 1990. The foundations of genetic fine structure: A retrospective from memory. *Genetics* 124:793–96.
- Hall, B. 1997. On the specificity of adaptive mutations. *Genetics* 145:39–44.
- Hanawalt, P., B. Donahue, and K. Sweder. 1994. Repair and transcription: Collision or collusion? *Current Biology* 4:518–21.
- Hoeijmakers, J. 1993. Nucleotide excision repair I: From *E. coli* to yeast. *Trends in Genetics* 9:173–77.
- . 1993. Nucleotide excision repair II: From yeast to mammals. *Trends in Genetics* 9:211–17.
- Holliday, R. 1974. Molecular aspects of genetic exchange and gene conversion. *Genetics* 78:273–87.
- Hopfner, K.-P., et al. 2000. Structural biology of the Rad50 ATPase: ATP-driven conformational control in DNA double-strand break repair and the ABC-ATPase superfamily. *Cell* 101:789–800.
- Johnson, R., S. Prakash, and L. Prakash. 1999. Efficient bypass of a thymine-thymine dimer by yeast DNA polymerase, Pol η . *Science* 283:1001–4.
- Kanaar, R., J. Hoeijmakers, and D. van Gent. 1998. Molecular mechanisms of DNA double-strand break repair. *Trends in Cell Biology* 8:483–89.
- Karran, P. 2000. DNA double-strand break repair in mammalian cells. *Current Opinion in Genetics & Development* 10:144–50.
- Keeney, S., C. Giroux, and N. Kleckner. 1997. Meiosis-specific DNA double-strand breaks are catalyzed by Spo11, a member of a widely conserved protein family. *Cell* 88:375–84.
- Kobayashi, T., et al. 1997. Mutations in the *XPD* gene leading to xeroderma pigmentosum symptoms. *Human Mutation* 9:322–31.
- Kowalczykowski, S., and A. Eggleston. 1994. Homologous pairing and DNA strand-exchange proteins. *Annual Review of Biochemistry* 63:991–1043.
- Lee, C., and M. Levitt. 1991. Accurate prediction of the stability and activity effects of site-directed mutagenesis on a protein core. *Nature* 352:448–51.
- Ley, R. 1993. Photoreactivation in humans. *Proceedings of the National Academy of Sciences, USA* 90:4337.

Chapter 12 DNA: Its Mutation, Repair, and Recombination

- Aboussekhr, A., et al. 1995. Mammalian DNA nucleotide excision repair reconstituted with purified protein components. *Cell* 80:859–68.
- Ames, B., R. Magaw, and L. Gold. 1987. Ranking possible carcinogenic hazards. *Science* 236:271–80.

- Lindahl, T., and R. Wood. 1999. Quality control by DNA repair. *Science* 286:1897–1905.
- Luria, S., and M. Delbrück. 1943. Mutations of bacteria from virus sensitivity to virus resistance. *Genetics* 28:491–511.
- McCullough, A. M., Dodson, and R. Lloyd. 1999. Initiation of base excision repair: Glycosylase mechanisms and structures. *Annual Review of Biochemistry* 68:255–85.
- Modrich, P., and R. Lahue. 1996. Mismatch repair in replication fidelity, genetic recombination, and cancer biology. *Annual Review of Biochemistry* 65:101–33.
- Mol, C., et al. 1999. DNA repair mechanisms for the recognition and removal of damaged DNA bases. *Annual Review of Biophysics and Biomolecular Structure* 28:101–28.
- Morgan, A. 1993. Base mismatches and mutagenesis: How important is tautomerism? *Trends in Biochemical Sciences* 18:160–63.
- Muller, H. 1927. Artificial transmutation of the gene. *Science* 66:84–87.
- Neel, J. 1994. *Physician to the Gene Pool: Genetic Lessons and Other Stories*. New York: Wiley.
- Ortiz-Lombardia, M., et al. 1999. Crystal structure of a DNA Holliday junction. *Nature Structural Biology* 6:913–17.
- Parikh, S., C. Mol, and J. Tainer. 1997. Base excision repair enzyme family portrait: Integrating the structure and chemistry of an entire DNA repair pathway. *Structure* 5:1543–50.
- Potter, H., and D. Dressler. 1976. On the mechanism of genetic recombination: Electron microscopic observation of recombination intermediates. *Proceedings of the National Academy of Sciences, USA* 73:3000–3004.
- Puchta, H., and B. Hohn. 1996. From centimorgans to base pairs: homologous recombination in plants. *Trends in Plant Science* 1:340–48.
- Rafferty, J., et al. 1996. Crystal structure of DNA recombination protein RuvA and a model for its binding to the Holliday junction. *Science* 274:415–21.
- Richardson, C., and M. Jasin. 2000. Frequent chromosomal translocations induced by DNA double-strand breaks. *Nature* 405:697–700.
- Ripley, L. 1990. Frameshift mutation: Determinants of specificity. *Annual Review of Genetics* 24:189–213.
- Roberts, R., and X. Cheng. 1998. Base flipping. *Annual Review of Biochemistry* 67:181–98.
- Roman, H., and M. Ruzinski. 1990. Mechanisms of gene conversion in *Saccharomyces cerevisiae*. *Genetics* 124:7–25.
- Rosche, W., P. Foster, and J. Cairns. 1999. The role of transient hypermutators in adaptive mutation in *Escherichia coli*. *Proceedings of the National Academy of Sciences, USA* 96:6862–67.
- Sancar, A. 1996. DNA excision repair. *Annual Review of Biochemistry* 65:43–81.
- Singer, B., and J. Kusmierak. 1982. Chemical mutagenesis. *Annual Review of Biochemistry* 52:655–93.
- Smith, M. 1985. In vitro mutagenesis. *Annual Review of Genetics* 19:423–62.
- Stadler, L. 1928. Mutations in barley induced by X rays and radium. *Science* 68:186–87.
- Stahl, F. 1994. The Holliday junction on its thirtieth anniversary. *Genetics* 138:241–46.
- . 1996. Meiotic recombination in yeast: coronation of the double-strand-break repair model. *Cell* 87:965–68.
- Story, R., I. Weber, and T. Steitz. 1992. The structure of *E. coli* RecA protein monomer and polymer. *Nature* 355:318–25 (Erratum, p. 567).
- Streisinger, G., and J. Owen. 1985. Mechanisms of spontaneous and induced frameshift mutation in bacteriophage T4. *Genetics* 109:633–59.
- Tanaka, K., and R. Wood. 1994. Xeroderma pigmentosum and nucleotide excision repair of DNA. *Trends in Biochemical Sciences* 19:83–86.
- Tang, M., et al. 2000. Roles of *E. coli* DNA polymerases IV and V in lesion-targeted and untargeted SOS mutagenesis. *Nature* 404:1014–18.
- Tennant, R., et al. 1987. Prediction of chemical carcinogenicity in rodents from in vitro genetic toxicity assays. *Science* 236:933–41.
- Van Dyck, E., et al. 1999. Binding of double-strand breaks in DNA by human Rad52 protein. *Nature* 398:728–31.
- Washington, M., et al. 2000. Accuracy of thymine-thymine dimer bypass by *Saccharomyces cerevisiae* DNA polymerase η . *Proceedings of the National Academy of Sciences, USA* 97:3094–99.
- West, S. 1994. The processing of recombination intermediates: Mechanistic insights from studies of bacterial proteins. *Cell* 76:9–15.
- . 1997. Processing of recombination intermediates by the RuvABC proteins. *Annual Review of Genetics* 31:213–44.
- Wood, R. 1996. DNA repair in eukaryotes. *Annual Review of Biochemistry* 65:135–67.
- Yanofsky, C., et al. 1967. The complete amino-acid sequence of the tryptophan synthetase A protein (a subunit) and its colinear relationship with the genetic map of the A gene. *Proceedings of the National Academy of Sciences, USA* 57:296–98.

Chapter 13 Genomics, Biotechnology, and Recombinant DNA

- Abelson, P., and P. Hines. 1999. The plant revolution. *Science* 285:367–68. (Followed by seven articles on plant genomics.)
- Adams, M., et al. 2000. The genome sequence of *Drosophila melanogaster*. *Science* 287:2185–95. (Followed by seven related articles.)
- Adelman, J., et al. 1987. Two mammalian genes transcribed from opposite strands of the same DNA locus. *Science* 235:1514–17.
- Andreason, G., and G. Evans. 1988. Introduction and expression of DNA molecules in eukaryotic cells by electroporation. *BioTechniques* 6:650–60.
- Armour, J., and A. Jeffreys. 1992. Biology and applications of human minisatellite loci. *Current Opinion in Genetics and Development* 2:850–56.
- Arnheim, N., and H. Erlich. 1992. Polymerase chain reaction strategy. *Annual Review of Biochemistry* 61:131–56.
- Ayala, F., and B. Black. 1993. Science and the courts. *American Scientist* 81:230–39.
- Berg, P., and M. Singer. 1995. The recombinant DNA controversy: Twenty years later. *Proceedings of the National Academy of Sciences, USA* 92:9011–13.
- Blattner, F., et al. 1997. The complete genome sequence of *Escherichia coli* K12. *Science* 277:1453–62.
- Bolivar, F., et al. 1977. Construction and characterization of new cloning vehicles. II: A multipurpose cloning system. *Gene* 2:95–113.
- Brandon, E., R. Idzerda, and G. McKnight. 1995. Targeting the mouse genome: A compendium of knockouts. *Current Biology* 5:625–34 (part I); 5:758–65 (part II); 5:873–81 (part III).
- Cai, W., et al. High-resolution restriction maps of bacterial artificial chromosomes constructed by optical mapping. *Proceedings of the National Academy of Sciences, USA* 95:3390–95.
- Capecchi, M. 1994. Targeted gene replacement. *Scientific American*, March, 52–59.
- Caruthers, M. 1985. Gene synthesis machines: DNA chemistry and its uses. *Science* 230:281–85.
- C. elegans* Sequencing Consortium. 1998. Genome sequence of the nematode *C. elegans*: A platform for investigating biology. *Science* 282:2012–18. (Followed by five articles.)
- Chu, T.-H., et al. 1995. Highly efficient eukaryotic gene expression vectors for peptide secretion. *BioTechniques* 18:890–99.
- Cohen, J., and M. Hogan. 1994. The new genetic medicines. *Scientific American*, December, 76–82.

B-12 *Appendix B Suggestions for Further Reading*

- Cohen, S., et al. 1973. Construction of biologically functional bacterial plasmids in vitro. *Proceedings of the National Academy of Sciences, USA* 70:3240–44.
- Danna, K., and D. Nathans. 1971. Specific cleavage of Simian Virus 40 DNA by restriction endonuclease of *Haemophilus influenzae*. *Proceedings of the National Academy of Sciences, USA* 68:2913–17.
- DePamphilis, M., et al. 1988. Microinjecting DNA into mouse ova to study DNA replication and gene expression and to produce transgenic animals. *BioTechniques* 6:662–80.
- Dhand, R., ed. 2000. Functional genomics. *Nature* 405:619–65. (Several articles on genomics and proteomics.)
- Dib, C., et al. 1996. A comprehensive genetic map of the human genome based on 5,264 microsatellites. *Nature* 380:152–54.
- Dickson, R., et al. 1997. On/off blinking and switching behaviour of single molecules of green fluorescent protein. *Nature* 388:355–58.
- Dunham, I., et al. 1999. The DNA sequence of human chromosome 22. *Nature* 402:489–95.
- Elder, J., R. Spritz, and S. Weissman. 1981. Simian virus 40 as a eukaryotic cloning vehicle. *Annual Review of Genetics* 15:295–340.
- Ferber, D. 1999. GM crops in the cross hairs. *Science* 286:1662–66.
- Fleischmann, R., et al. 1995. Whole-genome random sequencing and assembly of *Haemophilus influenzae* Rd. *Science* 269:496–512.
- Fodor, S. 1997. DNA sequencing: Massively parallel genomics. *Science* 277:393–95.
- Friedmann, T., ed. 1999. *The Development of Human Gene Therapy*. Cold Spring Harbor, N.Y.: Cold Spring Harbor Laboratory Press.
- Gasser, C., and R. Fraley. 1992. Transgenic crops. *Scientific American*, June, 62–69.
- Gerhold, D., T. Rushmore, and C. Caskey. 1999. DNA chips: Promising toys have become powerful tools. *Trends in Biochemical Sciences* 24:168–73.
- Gheysen, G., G. Angenon, and M. Van Montagu. 1992. Transgenic plants: *Agrobacterium tumefaciens*-mediated transformation and its use for crop improvement. In *Transgenesis*, edited by J. Murray, pp. 187–232. West Sussex, England: John Wiley.
- Gilbert, W. 1981. DNA sequencing and gene structure. *Science* 214:1305–12.
- Hagelberg, E., I. Gray, and A. Jeffreys. 1991. Identification of the skeletal remains of a murder victim by DNA analysis. *Nature* 352:427–29.
- Hall, J., et al. 1990. Linkage of early-onset familial breast cancer to chromosome 17q21. *Science* 250:1684–89.
- Hattori, M., et al. 2000. The DNA sequence of human chromosome 21. *Nature* 405:311–19.
- Horsch, R., et al. 1985. A simple and general method for transferring genes into plants. *Science* 227:1229–31.
- Jasny, B., and P. Hines. 1999. Genome prospecting. *Science* 286:443. (Followed by eight articles.)
- Johnson, I. 1983. Human insulin from recombinant DNA technology. *Science* 219:632–37.
- Johnston, S., et al. 1988. Mitochondrial transformation in yeast by bombardment with microprojectiles. *Science* 240:1538–41.
- Johnston-Dow, L., et al. 1987. Optimized method for fluorescent and radio labeled DNA sequencing. *BioTechniques* 5:754–65.
- Kato, Y., et al. 1998. Eight calves cloned from somatic cells of a single adult. *Science* 282:2095.
- Kilbane, J., II, and B. Bielaga. 1991. Instantaneous gene transfer from donor to recipient microorganisms via electroporation. *BioTechniques* 10:354–65.
- Kmieć, E. 1999. Gene therapy. *American Scientist* 87:240–47.
- Kreiner, T. 1996. Rapid genetic sequence analysis using a DNA probe array system. *American Laboratory*, March: 39–43.
- Kubota, C., et al. 2000. Six cloned calves produced from adult fibroblast cells after long-term culture. *Proceedings of the National Academy of Sciences, USA* 97:990–95.
- Lander, E., and R. Weinberg. 2000. Genomics: Journey to the center of biology. *Science* 287:1777–82.
- Lasic, D. 1992. Liposomes. *American Scientist* 80:20–31.
- Lim, K., and C.-B. Chae. 1989. A simple assay for DNA transfection by incubation of the cells in culture dishes with substrates for beta-galactosidase. *BioTechniques* 7:576–79.
- Malakoff, D. 2000. The rise of the mouse, biomedicine's model mammal. *Science* 288:248–53. (Followed by two related articles.)
- Mannino, R., and S. Gould-Fogerite. 1988. Liposome mediated gene transfer. *BioTechniques* 6:682–90.
- Marshall, E. 1999. Gene therapy death prompts review of adenovirus vector. *Science* 286:2244–45.
- Martin, C., et al. 1991. Improved chemiluminescent DNA sequencing. *BioTechniques* 11:110–13.
- Marx, J. 1982. Building bigger mice through gene transfer. *Science* 218:1298.
- . 1988. DNA fingerprinting takes the witness stand. *Science* 240:1616–18.
- . 1989. The cystic fibrosis gene is found. *Science* 245:923–25.
- McElfresh, K., D. Vining-Forde, and I. Balazs. 1993. DNA-based identity testing in forensic science. *BioScience* 43:149–57.
- McLaren, A. 2000. Cloning: Pathways to a pluripotent future. *Science* 288:1775–80.
- Meinke, D., et al. 1998. *Arabidopsis thaliana*: A model plant for genome analysis. *Science* 282:662–82. (Followed by related articles on genomes.)
- Miki, Y., et al. 1994. A strong candidate for the breast and ovarian cancer susceptibility gene BRCA1. *Science* 266:66–71.
- Mullis, K. 1990. The unusual origin of the polymerase chain reaction. *Scientific American*, April, 56–65.
- Nakamura, Y., et al. 1987. Variable number of tandem repeat (VNTR) markers for human gene mapping. *Science* 235:1616–22.
- Neufeld, P., and N. Colman. 1990. When science takes the witness stand. *Scientific American*, May, 46–53.
- Normark, S., et al. 1983. Overlapping genes. *Annual Review of Genetics* 17:499–525.
- Nowak, R. 1994. Breast cancer gene offers surprises. *Science* 265:1796–99.
- Oliver, S., et al. 1992. The complete DNA sequence of yeast chromosome III. *Nature* 357:38–46.
- Ow, D., et al. 1986. Transient and stable expression of the firefly luciferase gene in plant cells and transgenic plants. *Science* 234:856–59.
- Palmiter, R., and R. Brinster. 1986. Germ-line transformation of mice. *Annual Review of Genetics* 20:465–99.
- Paoletti, M., and D. Pimentel. 1996. Genetic engineering in agriculture and the environment. *BioScience* 46:665–73.
- Pestka, S. 1983. The purification and manufacture of human interferons. *Scientific American*, August, 37–43.
- Rennie, J. 1994. Grading the gene tests. *Scientific American*, June, 88–97.
- Richa, J., and C. Lo. 1989. Introduction of human DNA into mouse eggs by injection of dissected chromosome fragments. *Science* 245:175–77.
- Roberts, L. 1993. Zeroing in on a breast cancer susceptibility gene. *Science* 259:622–25.
- Sanger, F. 1988. Sequences, sequences, and sequences. *Annual Review of Biochemistry* 57:1–28.
- Sanger, F., et al. 1978. The nucleotide sequence of bacteriophage ϕ X174. *Journal of Molecular Biology* 125:225–46.
- Schell, J. 1987. Transgenic plants as tools to study the molecular organization of plant genes. *Science* 237:1176–83.
- Schuler, G., et al. 1996. A gene map of the human genome. *Science* 274:540–46.
- Scientific American*. 1997. Special report: Making gene therapy work. June, 95–123 (5 articles).
- Shapiro, H. 1997. Ethical and policy issues of human cloning. *Science* 277:195–96.

- Shigekawa, K., and W. Dower. 1988. Electroporation of eukaryotes and prokaryotes: A general approach to the introduction of macromolecules into cells. *BioTechniques* 6:742–51.
- Snow, A., and P. Palma. 1997. Commercialization of transgenic plants: Potential ecological risks. *BioScience* 47:86–96.
- Snyder, R., et al. 1999. Correction of hemophilia B in canine and murine models using recombinant adeno-associated viral vectors. *Nature Medicine* 5:64–70.
- Southern, E. 1975. Detection of specific sequences among DNA fragments separated by gel electrophoresis. *Journal of Molecular Biology* 98:503–17.
- Stix, G. 1995. A recombinant feast. *Scientific American*, March, 38–40.
- Takada, T., et al. 1997. Selective production of transgenic mice using green fluorescent protein as a marker. *Nature Biotechnology* 15:458–61.
- Tanksley, S., et al. 1992. High-density molecular linkage maps of the tomato and potato genomes. *Genetics* 132:1141–60.
- Tsien, R. 1998. The green fluorescent protein. *Annual Review of Biochemistry* 67:509–44.
- Velander, W., H. Lubon, and W. Drohan. 1997. Transgenic livestock as drug factories. *Scientific American*, January, 70–74.
- Venter, J., H. Smith, and L. Hood. 1996. A new strategy for genome sequencing. *Nature* 381:364–66.
- Venter, J., et al. 1998. Shotgun sequencing of the human genome. *Science* 280:1540–42.
- Wakayama, T., et al. 1998. Full-term development of mice from enucleated oocytes injected with cumulus cell nuclei. *Nature* 394:369–74.
- Wang, D., et al. 1998. Large-scale identification, mapping, and genotyping of single-nucleotide polymorphisms in the human genome. *Science* 280:1077–82.
- White, R., and J. Lalouel. 1988. Chromosome mapping with DNA markers. *Scientific American*, February, 40–48.
- Williams, R., et al. 1991. Introduction of foreign genes into tissues of living mice by DNA-coated microprojectiles. *Proceedings of the National Academy of Sciences, USA* 88:2726–30.
- Wilmot, I. 1998. Cloning for medicine. *Scientific American*, December, 58–63.
- Wilmot, I., et al. 1997. Viable offspring derived from fetal and adult mammalian cells. *Nature* 385:810–13.
- Wilson, M., and S. Lindow. 1993. Release of recombinant microorganisms. *Annual Review of Microbiology* 47:913–44.
- Wilson, M., et al. 1995. Extraction, PCR amplification and sequencing of mitochondrial DNA from human hair shafts. *BioTechniques* 18:662–69.
- Wong, W., et al. 1988. Wood hydrolysis by *Cellulomonas fimi* endoglucanase and exoglucanase coexpressed as secreted enzymes in *Saccharomyces cerevisiae*. *Bio/Technology* 6:713–19.
- Wood, K., et al. 1989. Complementary DNA coding click beetle luciferases can elicit bioluminescence of different colors. *Science* 244:700–702.
- Ye, X., et al. 2000. Engineering the provitamin A (β -carotene) biosynthetic pathway into (carotenoid-free) rice endosperm. *Science* 287:303–5.
- Yuan, R. 1981. Structure and mechanism of multifunctional restriction endonucleases. *Annual Review of Biochemistry* 50:285–315.

Chapter 14 Gene Expression: Control in Prokaryotes and Phages

- Albright, R., and B. Matthews. 1998. How Cro and λ -repressor distinguish between operators: The structural basis underlying a genetic switch. *Proceedings of the National Academy of Sciences, USA* 95:3431–36.
- Antson, A., et al. 1999. Structure of the *trp* RNA-binding attenuation protein, TRAP, bound to RNA. *Nature* 401:235–42.
- Bachmair, A., D. Finley, and A. Varshavsky. 1986. In vivo half-life of a protein is a function of its amino-terminal residue. *Science* 234:179–86.
- Beckwith, J., and T. Silvahy. 1992. *The Power of Bacterial Genetics: A Literature-Based Course*. Cold Spring Harbor, N.Y.: Cold Spring Harbor Laboratory Press.
- Bell, C., et al. 2000. Crystal structure of the λ repressor C-terminal domain provides a model for cooperative operator binding. *Cell* 101:801–11.
- Casjens, S. 1998. The diverse and dynamic structure of bacterial genomes. *Annual Review of Genetics* 32:339–77.
- Charlebois, R., ed. 1999. *Organization of the Prokaryotic Genome*. Washington, D.C.: ASM Press.
- Chen, Y., Y. Ebricht, and R. Ebricht. 1994. Identification of the target of a transcription activator protein by protein-protein photocrosslinking. *Science* 265:90–92.
- Dickson, R., et al. 1975. Genetic regulation: The *lac* control region. *Science* 187:27–35.
- Eguchi, Y., T. Itoh, and J.-I. Tomazawa. 1991. Antisense RNA. *Annual Review of Biochemistry* 60:631–52.
- Erie, D.A., et al. 1994. DNA bending by Cro protein in specific and nonspecific complexes: Implications for protein site recognition and specificity. *Science* 266:1562–66.
- Gallant, J., and D. Lindsley. 1998. Ribosomes can slide over and beyond “hungry”

- codons, resuming protein chain elongation many nucleotides downstream. *Proceedings of the National Academy of Sciences, USA* 95:13771–76.
- Geiduschek, E. 1991. Regulation of expression of the late genes of bacteriophage T4. *Annual Review of Genetics* 25:437–60.
- Ghosh, S., and M. Deutscher. 1999. Oligoribonuclease is an essential component of the mRNA decay pathway. *Proceedings of the National Academy of Sciences, USA* 96:4372–77.
- Gilbert, W., and B. Müller-Hill. 1966. Isolation of the *Lac* repressor. *Proceedings of the National Academy of Sciences, USA* 56:1891–98.
- Gold, L. 1988. Posttranscriptional regulatory mechanisms in *Escherichia coli*. *Annual Review of Biochemistry* 57:199–233.
- Gottesman, S. 1999. Regulation by proteolysis: Developmental switches. *Current Opinion in Microbiology* 2:142–47.
- Greenblatt, J., J. Nodwell, and W. Mason. 1993. Transcriptional antitermination. *Nature* 364:401–5.
- Grindley, N. 1997. Site-specific recombination: Synapsis and strand exchange revealed. *Current Biology* 7:R608–12.
- Haren, L., B. Ton-Hoang, and M. Chandler. Integrating DNA: Transposases and retroviral integrases. *Annual Review of Microbiology* 53:245–81.
- Helmann, J. 1999. Anti-sigma factors. *Current Opinion in Microbiology* 2:135–41.
- Holzman, D. 1991. A “jumping gene” caught in the act. *Science* 254:1728–29.
- Hughes, K., and K. Mathee. 1998. The anti-sigma factors. *Annual Review of Microbiology* 52:231–86.
- Jacob, F. 1988. *The Statue Within: An Autobiography*. Translated from the French by F. Philip, Alfred P. Sloan Foundation Series. New York: Basic Books.
- Jacob, F., and J. Monod. 1961. Genetic regulatory mechanisms in the synthesis of proteins. *Journal of Molecular Biology* 3:318–56.
- Johnson, W., C. Moran, Jr., and R. Losick. 1983. Two RNA polymerase sigma factors from *Bacillus subtilis* discriminate between overlapping promoters for a developmentally regulated gene. *Nature* 302:800–804.
- Judson, H. 1979. *The Eighth Day of Creation: Makers of the Revolution in Biology*. New York: Simon and Schuster.
- Kantrowitz, E., and W. Lipscomb. 1988. *Escherichia coli* aspartate transcarbamylase: The relation between structure and function. *Science* 241:669–74.

B-14 *Appendix B Suggestions for Further Reading*

- Kihara, A., Y. Akiyama, and K. Ito. 1997. Host regulation of lysogenic decision in bacteriophage λ : transmembrane modulation of FtsH (HflB), the cII degrading protease, by HflKC (HflA). *Proceedings of the National Academy of Sciences, USA* 94:5544–49.
- Kleckner, N. 1990. Regulating Tn10 and IS10 transposition. *Genetics* 124:449–54.
- Kolter, R., and C. Yanofsky. 1982. Attenuation in amino acid biosynthetic operons. *Annual Review of Genetics* 16:113–34.
- Konigsberg, W., and G. Godson. 1983. Evidence for use of rare codons in the *dnaG* gene and other regulatory genes of *Escherichia coli*. *Proceedings of the National Academy of Sciences, USA* 80:687–91.
- Kröger, M., and G. Hobom. 1982. Structural analysis of insertion sequence IS5. *Nature* 297:159–62.
- Lewis, M., et al. 1996. Crystal structure of the lactose operon repressor and its complexes with DNA and inducer. *Science* 271:1247–54.
- Li, M., H. Moyle, and M. Susskind. 1994. Target of the transcriptional activator function of phage λ cI protein. *Science* 263:75–77.
- McGlynn, P., and R. Lloyd. 2000. Modulation of RNA polymerase by (p)ppGpp reveals a RecG-dependent mechanism for replication fork progression. *Cell* 101:35–45.
- Müller-Hill, B. 1996. *The lac Operon*. Berlin: Walter de Gruyter.
- Murialdo, H. 1991. Bacteriophage lambda DNA maturation and packaging. *Annual Review of Biochemistry* 60:125–53.
- Oxender, D., G. Zurawski, and C. Yanofsky. 1979. Attenuation in the *Escherichia coli* tryptophan operon: Role of RNA secondary structure involving the tryptophan codon region. *Proceedings of the National Academy of Sciences, USA* 76:5524–28.
- Ptashne, M. 1992. *A Genetic Switch: Gene Control and Phage λ* , 2d ed. Cambridge, Mass.: Cell Press and Blackwell Scientific Publications.
- Rechsteiner, M., and S. Rogers. 1996. PEST sequences and regulation by proteolysis. *Trends in Biochemical Sciences* 21:267–71.
- Rogers, S., R. Wells, and M. Rechsteiner. 1986. Amino acid sequences common to rapidly degraded proteins: The PEST hypothesis. *Science* 234:364–68.
- Schultz, S., G. Shields, and T. Steitz. 1991. Crystal structure of a CAP-DNA complex: The DNA is bent by 90°. *Science* 253:1001–7.
- Sherratt, D. 1995. *Mobile Genetic Elements*. Oxford: Oxford University Press.
- Varshavsky, A. 1996. The N-end rule: functions, mysteries, uses. *Proceedings of the National Academy of Sciences, USA* 93:12142–49.
- Weintraub, H. 1990. Antisense RNA and DNA. *Scientific American*, January, 40–46.
- Yanofsky, C. 1984. Comparison of regulatory and structural regions of genes of tryptophan metabolism. *Molecular Biology and Evolution* 1:143–61.
- Yura, T., and K. Nakahigashi. 1999. Regulation of the heat-shock response. *Current Opinion in Microbiology* 2:153–58.

Chapter 15 The Eukaryotic Chromosome

- Bank, A., J. Mears, and F. Ramirez. 1980. Disorders of human hemoglobin. *Science* 207:486–93.
- Berendes, H., F. van Breugel, and T. Holt. 1965. Experimental puffs in salivary gland chromosomes of *Drosophila hydei*. *Chromosoma (Berlin)* 16:35–46.
- Berger, S. 1999. Gene activation by histone and factor acetyltransferases. *Current Opinion in Cell Biology* 11:336–41.
- Blackburn, E. 1992. Telomerases. *Annual Review of Biochemistry* 61:113–29.
- Bonne-Andrea, C., M. Wong, and B. Alberts. 1990. In vitro replication through nucleosomes without histone displacement. *Nature* 343:719–26.
- Britten, R., and D. Kohne. 1968. Repeated sequences in DNA. *Science* 161:529–40.
- Broda, P., S. Oliver, and P. Sims, eds. 1993. The eukaryotic genome. *Society for General Microbiology*; Symposium 50:1–407.
- Callan, H. 1986. *Lampbrush Chromosomes, vol. 36: Molecular Biology, Biochemistry, and Biophysics*. New York: Springer-Verlag.
- Clarke, L., and J. Carbon. 1985. The structure and function of yeast centromeres. *Annual Review of Genetics* 19:29–56.
- Collins, K. 1999. Ciliate telomerase biochemistry. *Annual Review of Biochemistry* 68:187–218.
- Comings, D. 1978. Mechanisms of chromosome banding and implications for chromosome structure. *Annual Review of Genetics* 12:25–46.
- Copenhaver, G., et al. 1999. Genetic definition and sequence analysis of *Arabidopsis* centromeres. *Science* 286:2468–74.
- Costanzi, C., and J. Pehrson. 1998. Histone macroH2A1 is concentrated in the inactive X chromosome of female mammals. *Nature* 393:599–601.
- Courties, C., et al. 1994. Smallest eukaryotic organism. *Nature* 370:255.
- Fitzgerald, M., T. McKnight, and D. Shippen. 1996. Characterization and developmental patterns of telomerase expression in plants. *Proceedings of the National Academy of Sciences, USA* 93:14422–27.
- Gall, J., E. Cohen, and D. Atherton. 1974. The satellite DNAs of *Drosophila virilis*. *Cold Spring Harbor Symposium on Quantitative Biology* 38:417–21.

- Greider, C. 1996. Telomere length regulation. *Annual Review of Biochemistry* 65:337–65.
- . 1999. Telomeres do D-loop-T-loop. *Cell* 97:419–22.
- Greider, C., and E. Blackburn. 1996. Telomeres, telomerase and cancer. *Scientific American*, February, 92–97.
- Gross, D., and W. Garrard. 1988. Nuclease hypersensitive sites in chromatin. *Annual Review of Biochemistry* 57:159–97.
- Grunstein, M. 1992. Histones as regulators of genes. *Scientific American*, October, 68–74B.
- Hardison, R. 1999. The evolution of hemoglobin. *American Scientist* 87:126–37.
- Horvath, M., et al. 1998. Crystal structure of the *Oxytricha nova* telomere end binding protein complexed with single strand DNA. *Cell* 95:963–74.
- Kavenoff, R., L. Klotz, and B. Zimm. 1974. On the nature of chromosome-sized DNA molecules. *Cold Spring Harbor Symposium on Quantitative Biology* 38:1–8.
- Kimura, K., et al. 1999. 13S condensin actively reconfigures DNA by introducing global positive writhe: Implications for chromosome condensation. *Cell* 98:239–48.
- Krude, T. 1995. Nucleosome assembly during DNA replication. *Current Biology* 5:1232–34.
- Kumar, A., and J. Bennetzen. 1999. Plant retrotransposons. *Annual Review of Genetics* 33:479–532.
- Lowary, P., and J. Widom. 1997. Nucleosome packaging and nucleosome positioning of genomic DNA. *Proceedings of the National Academy of Sciences, USA* 94:1183–88.
- Luger, K., et al. 1997. Crystal structure of the nucleosome core particle at 2.8 Å resolution. *Nature* 389:251–60.
- Maniatis, T., et al. 1980. The molecular genetics of human hemoglobins. *Annual Review of Genetics* 14:145–78.
- Marcand, S., E. Gilson, and D. Shore. 1997. A protein-counting mechanism for telomere length regulation in yeast. *Science* 275:986–90.
- Maxson, R., R. Cohn, and L. Kedes. 1983. Expression and organization of histone genes. *Annual Review of Genetics* 17:239–77.
- Melamed, M., T. Lindmo, and M. Mendelsohn, eds. 1990. *Flow Cytometry and Sorting*, 2d ed. New York: Wiley-Liss.
- Mitton, J., and M. Grant. 1996. Genetic variation and the natural history of quaking aspen. *BioScience* 46:25–31.
- Morse, R. 1993. Nucleosome disruption by transcription factor binding in yeast. *Science* 262:1563–66.

- Osheim, Y., and O. Miller, Jr. 1983. Novel amplification and transcriptional activity of chorion genes in *Drosophila melanogaster* follicle cells. *Cell* 33:543–53.
- Owen-Hughes, T., et al. 1996. Persistent site-specific remodeling of a nucleosome array by transient action of the SWI/SNF complex. *Science* 273:513–16.
- Paranjape, S., R. Kamakaka, and J. Kadonaga. 1994. Role of chromatin structure in the regulation of transcription by RNA polymerase II. *Annual Review of Biochemistry* 63:265–97.
- Paulson, J., and U. Laemmli. 1977. The structure of histone depleted metaphase chromosomes. *Cell* 12:817–28.
- Pereira, S., et al. 1997. Archaeal nucleosomes. *Proceedings of the National Academy of Sciences, USA* 94:12633–37.
- Peterson, C., and J. Workman. 2000. Promoter targeting and chromatin remodeling by the SWI/SNF complex. *Current Opinion in Genetics & Development* 10:187–92.
- Petrov, D., et al. 2000. Evidence for DNA loss as a determinant of genome size. *Science* 287:1060–62.
- Pruss, D., et al. 1996. An asymmetric model for the nucleosome: A binding site for linker histones inside the DNA gyres. *Science* 274:614–17.
- Schulz, H., et al. 1999. Dense populations of a giant sulfur bacterium in Namibian shelf sediments. *Science* 284:493–95.
- Shippen-Lentz, D., and E. Blackburn. 1990. Functional evidence for an RNA template in telomerase. *Science* 247:546–52.
- Smit, A., and A. Riggs. 1996. *Tiggers* and other DNA transposon fossils in the human genome. *Proceedings of the National Academy of Sciences, USA* 93:1443–48.
- Smith, M., J. Bruhn, and J. Anderson. 1992. The fungus *Armillaria bulbosa* is among the largest and oldest living organisms. *Nature* 356:428–31.
- Stamatoyannopoulos, G. 1991. Human hemoglobin switching. *Science* 252:383.
- Stark, G., and G. Wahl. 1984. Gene amplification. *Annual Review of Biochemistry* 53:447–91.
- Strahl, B., and C. Allis. 2000. The language of covalent histone modifications. *Nature* 403:41–45.
- Strunnikov, A. 1998. SMC proteins and chromosome structure. *Trends in Cell Biology* 8:454–59.
- Studitsky, V., D. Clark, and G. Felsenfeld. 1994. A histone octamer can step around a transcribing polymerase without leaving the template. *Cell* 76:371–82.
- Studitsky, V., et al. 1997. Mechanism of transcription through the nucleosome by eukaryotic RNA polymerase. *Science* 278:1960–63.
- Sumner, A. 1990. *Chromosome Banding*. Cambridge, Mass.: Unwin Hyman.
- Taylor, J., P. Woods, and W. Hughes. 1957. The organization and duplication of chromosomes as revealed by autoradiographic studies using tritium-labeled thymidine. *Proceedings of the National Academy of Sciences, USA* 43:122–28.
- Therman, E., and B. Susman. 1995. *Human Chromosomes*, 3d ed. New York: Springer-Verlag.
- Thomas, J. 1999. Histone H1: location and role. *Current Opinion in Cell Biology* 11:312–17.
- Tsukiyama, T., and C. Wu. 1997. Chromatin remodeling and transcription. *Current Opinion in Genetics & Development* 7:182–91.
- van Steensel, B., and T. de Lange. 1997. Control of telomere length by the human telomeric protein TRF1. *Nature* 385:740–43.
- Wade, P., D. Pruss, and A. Wolffe. 1997. Histone acetylation: chromatin in action. *Trends in Biochemical Sciences* 22:128–32.
- Wang, Y., and D. Patel. 1993. Solution structure of the human telomeric repeat d[AG₃(T₂AG₃)₃] G-tetraplex. *Structure* 1:263–82.
- Whitehouse, I., et al. 1999. Nucleosome mobilization catalysed by the yeast SWI/SNF complex. *Nature* 400:784–87.
- Williamson, J. 1994. G-quartet structures in telomeric DNA. *Annual Review of Biophysics and Biomolecular Structure* 23:703–30.
- Wolffe, A. 1994. Nucleosome positioning and modification: Chromatin structures that potentiate transcription. *Trends in Biochemical Sciences* 19:240–44.
- . 1995. Histone deviants. *Current Biology* 5:452–54.
- Workman, J., and R. Kingston. 1998. Alteration of nucleosome structure as a mechanism of transcriptional regulation. *Annual Review of Biochemistry* 67:545–79.
- Yu, G.-L., et al. 1990. In vivo alteration of telomere sequences and senescence caused by mutated *Tetrahymena* telomerase RNAs. *Nature* 344:126–32.
- Zhang, X., et al. 1999. Telomere shortening and apoptosis in telomerase-inhibited human tumor cells. *Genes & Development* 13:2388–99.
- Chapter 16 Gene Expression: Control in Eukaryotes**
- Akira, S., K. Okazaki, and H. Sakano. 1987. Two pairs of recombination signals are sufficient to cause immunoglobulin V-(D)-J joining. *Science* 238:1134–38.
- Amedeo, P., et al. 2000. Disruption of the plant gene *MOM* releases transcriptional silencing of methylated genes. *Nature* 405:203–6.
- Amig, S., et al. 1994. Transient accumulation of new class II MHC molecules in a novel endocytic compartment in B lymphocytes. *Nature* 369:113–20.
- Amit, A., et al. 1986. Three-dimensional structure of an antigen-antibody complex at 2.8 Å resolution. *Science* 233:747–53.
- Arstila, T., et al. 1999. A direct estimate of the human αβ T cell receptor diversity. *Science* 286:958–61.
- Autexier, C., and C. Greider. 1996. Telomerase and cancer: revisiting the telomere hypothesis. *Trends in Biochemical Sciences* 21:387–91.
- Bashirullah, A., R. Cooperstock, and H. Lipshitz. 1998. RNA localization in development. *Annual Review of Biochemistry* 67:335–94.
- Bate, M., and A. Martinez Arias, eds. 1993. *The Development of Drosophila melanogaster*. Cold Spring Harbor, N.Y.: Cold Spring Harbor Laboratory Press.
- Bhattacharyya, M., et al. 1990. The wrinkled-seed character of peas described by Mendel is caused by a transposon-like insertion in a gene encoding starch-branching enzyme. *Cell* 60:115–22.
- Bird, A., and A. Wolffe. 1999. Methylation-induced repression—Belts, braces, and chromatin. *Cell* 99:451–54.
- Bischoff, J., et al. 1996. An adenovirus mutant that replicates selectively in p53-deficient human tumor cells. *Science* 274:373–76.
- Brown, J., et al. 1993. Three-dimensional structure of the human class II histocompatibility antigen HLA-DR1. *Nature* 364:33–39.
- Brown, M., and E. Solomon. 1997. Studies on inherited cancers: Outcomes and challenges of 25 years. *Trends in Genetics* 13:202–6.
- Burton, D., and J. Moore. 1998. Why do we not have an HIV vaccine and how can we make one? *Nature Medicine Vaccine Supplement* 4:495–98.
- Campos-Ortega, J., and V. Hartenstein. 1985. *The Embryonic Development of Drosophila melanogaster*. Berlin: Springer-Verlag.
- Carroll, S. 1995. Homeotic genes and the evolution of arthropods and chordates. *Nature* 376:479–85.
- Casares, F., and R. Mann. 1998. Control of antennal versus leg development in *Drosophila*. *Nature* 392:723–26.
- Caspari, T. 2000. Checkpoints: How to activate p53. *Current Biology* 10:R315–17.
- Cavance, W., and R. White. 1995. The genetic basis of cancer. *Scientific American*, March, 72–79.
- Cohen, L. 1987. Diet and cancer. *Scientific American*, November, 42–48.
- Cold Spring Harbor Laboratory Press. 1998. *Mechanisms of Transcription*. Cold Spring Harbor Symposia on Quantitative Biology, Volume 63. New York: Cold Spring Harbor.

B-16

Appendix B Suggestions for Further Reading

- Collins, E., D. Garboczi, and D. Wiley. 1994. Three-dimensional structure of a peptide extending from one end of a class I MHC site. *Nature* 371:626–29.
- Cullen, B., ed. 1993. *Human Retroviruses*. Oxford: IRL Press at Oxford University Press.
- Davies, D., E. Padlan, and S. Sheriff. 1990. Antibody-antigen complexes. *Annual Review of Biochemistry* 59:439–73.
- Davis, M. 1990. T cell receptor gene diversity and selection. *Annual Review of Biochemistry* 59:475–96.
- De Robertis, E., G. Oliver, and C. Wright. 1990. Homeobox genes and the vertebrate body plan. *Scientific American*, July, 46–52.
- Doolittle, R. 1995. The multiplicity of domains in proteins. *Annual Review of Biochemistry* 64:287–314.
- Driever, W., et al. 1994. Zebrafish: Genetic tools for studying vertebrate development. *Trends in Genetics* 10:152–59.
- Duboule, D., ed. 1994. *Guidebook to the Homeobox Genes*. New York: Oxford University Press.
- Duke, R., D. Ojcius, and J. Young. 1996. Cell suicide in health and disease. *Scientific American*, December, 80–87.
- Dyson, N., et al. 1989. The human papilloma virus-16 E7 oncoprotein is able to bind to the retinoblastoma gene product. *Science* 243:934–37.
- Ellenberger, T. 1994. Getting a grip on DNA recognition: Structures of the basic region leucine zipper, and the basic region helix-loop-helix DNA-binding domains. *Current Opinion in Structural Biology* 4:12–21.
- Engelhard, V. 1994. How cells process antigens. *Scientific American*, August, 54–61.
- Evans, R., and S. Hollenberg. 1988. Zinc fingers: Gilt by association. *Cell* 52:1–3.
- Fearon, E. 1997. Human cancer syndromes: Clues to the origin and nature of cancer. *Science* 278:1043–50. (Plus other articles in this section on Frontiers in Cancer Research.)
- Federoff, N. 1984. Transposable genetic elements in maize. *Scientific American*, June, 84–98.
- Ferré-D'Amaré, A., et al. 1993. Recognition by Max of its cognate DNA through a dimeric b/HLH/Z domain. *Nature* 363:38–45.
- Fraser, S., and R. Harland. 2000. The molecular metamorphosis of experimental embryology. *Cell* 100, 41–55.
- French, D., R. Laskov, and M. Scharff. 1989. The role of somatic hypermutation in the generation of antibody diversity. *Science* 244:1152–57.
- Fugmann, S., et al. 2000. The RAG proteins and V(D)J recombination: Complexes, ends, and transposition. *Annual Review of Immunology* 18:495–527.
- Gao, F., et al. 1999. Origin of HIV-1 in the chimpanzee *Pan troglodytes troglodytes*. *Nature* 397:436–41.
- Garcia, K., L. Teyton, and I. Wilson. 1999. Structural basis of T cell recognition. *Annual Review of Immunology* 17:369–97.
- Gehring, W. 1999. *Master Control Genes in Development and Evolution: The Homeobox Story*. New Haven, Connecticut: Yale University Press.
- Gehring, W., M. Affolter, and T. Bürglin. 1994. Homeodomain proteins. *Annual Review of Biochemistry* 64:487–526.
- Gehring, W., et al. 1994. Homeodomain-DNA recognition. *Cell* 78:211–23.
- Gierl, A., H. Saedler, and P. Peterson. 1989. Maize transposable elements. *Annual Review of Genetics* 23:71–85.
- Goff, S. 1992. Genetics of retroviral integration. *Annual Review of Genetics* 26:527–44.
- Goodrich, J., et al. 1997. A Polycomb-group gene regulates homeotic gene expression in *Arabidopsis*. *Nature* 386:44–51.
- Groll, M., et al. 1997. Structure of 20S proteasome from yeast at 2.4 Å resolution. *Nature* 386:463–71.
- Haber, J. 1998. Mating-type gene switching in *Saccharomyces cerevisiae*. *Annual Review of Genetics* 32:561–99.
- Halder, G., P. Callaerts, and W. Gehring. 1995. Induction of ectopic eyes by targeted expression of the eyeless gene in *Drosophila*. *Science* 267:1788–92.
- Haluska, E., Y. Tsujimoto, and C. Croce. 1987. Oncogene activation by chromosome translocation in human malignancy. *Annual Review of Genetics* 21:321–45.
- Hanahan, D., and R. Weinberg. 2000. The hallmarks of cancer. *Cell* 100:57–70.
- Harrison, S. 1991. A structural taxonomy of DNA-binding domains. *Nature* 353:715–19.
- Hastie, N. 1994. The genetics of Wilm's tumor—A case of disrupted development. *Annual Review of Genetics* 28:523–58.
- Heemels, M.-T., and H. Ploegh. 1995. Generation, translocation, and presentation of MHC class I-restricted peptides. *Annual Review of Biochemistry* 64:463–91.
- Heinlein, M. 1996. Excision patterns of *Activator (Ac)* and *Dissociation (Ds)* elements in *Zea mays* L.: implications for the regulation of transposition. *Genetics* 144:1851–69.
- Hershko, A., and A. Ciechanover. 1998. The ubiquitin system. *Annual Review of Biochemistry* 67:425–79.
- Hirao, A., et al. 2000. DNA damage-induced activation of p53 by the checkpoint kinase Chk2. *Science* 287:1824–27.
- Horvitz, H., and J. Sulston. 1990. "Joy of the worm." *Genetics* 126:287–92.
- Hunter, T. 2000. Signaling—2000 and beyond. *Cell* 100:113–27.
- Johnson, P., and S. McKnight. 1989. Eukaryotic transcriptional regulatory proteins. *Annual Review of Biochemistry* 58:799–839.
- Jürgens, G., R. Torres Ruiz, and T. Berleth. 1994. Embryonic pattern formation in flowering plants. *Annual Review of Genetics* 28:351–71.
- Karin, M., and T. Hunter. 1995. Transcriptional control by protein phosphorylation: Signal transmission from the cell surface to the nucleus. *Current Biology* 5:747–57.
- Katz, R., and A. Skalka. 1994. The retroviral enzymes. *Annual Review of Biochemistry* 63:133–73.
- Kennel, S., et al. 1984. Monoclonal antibodies in cancer detection and therapy. *BioScience* 34:150–56.
- Kinzler, K., and B. Vogelstein. 1996. Lessons from hereditary colorectal cancer. *Cell* 87:159–70.
- Knudson, A., Jr. 1993. Antioncogenes and human cancer. *Proceedings of the National Academy of Sciences, USA* 90:10914–21.
- Kussie, P. 1996. Structure of the MDM2 oncoprotein bound to the p53 tumor suppressor transactivation domain. *Science* 274:948–53.
- Laird, P., and R. Jaenisch. 1996. The role of DNA methylation in cancer genetics and epigenetics. *Annual Review of Genetics* 30:441–64.
- Landschulz, W., P. Johnson, and S. McKnight. 1988. The leucine zipper: A hypothetical structure common to a new class of DNA binding proteins. *Science* 240:1759–64.
- Lawrence, P. 1992. *The Making of a Fly*. Oxford: Blackwell.
- Lawrence, P., and G. Morata. 1994. Homeobox genes: Their function in *Drosophila* segmentation and pattern formation. *Cell* 78:181–89.
- Lawrence, P., and G. Struhl. 1996. Morphogens, compartments, and pattern: Lessons from *Drosophila*? *Cell* 85:951–61.
- Lebel-Hardenack, S., and S. Grant. 1997. Genetics of sex determination in flowering plants. *Trends in Plant Science* 2:130–36.
- Lengauer, C., K. Kinzler, and B. Vogelstein. 1998. Genetic instabilities in human cancers. *Nature* 396:643–49.
- Levine, A. 1993. The tumor suppressor genes. *Annual Review of Biochemistry* 62:623–51.
- . 1997. p53, the cellular gatekeeper for growth and division. *Cell* 88:323–31.
- Ma, H., and C. dePamphilis. 2000. The ABCs of floral evolution. *Cell* 101:5–8.

- Malkin, D. 1994. Germline p53 mutations and heritable cancer. *Annual Review of Genetics* 28:443–65.
- Mannervik, M., et al. 1999. Transcriptional coregulators in development. *Science* 284:606–9.
- Marx, J. 1984. What do oncogenes do? *Science* 223:673–76.
- . 1994. Oncogenes reach a milestone. *Science* 266:1942–44.
- Max, E., J. Seidman, and P. Leder. 1979. Sequences of five potential recombination sites encoded close to an immunoglobulin k constant region gene. *Proceedings of the National Academy of Sciences, USA* 76:3450–54.
- McGinnis, W. 1994. A century of homeosis, a decade of homeoboxes. *Genetics* 137:607–11.
- McGinnis, W., and R. Krumlauf. 1992. Homeobox genes and axial patterning. *Cell* 68:283–302.
- McGinnis, W., and M. Kuziora. 1994. The molecular architects of body design. *Scientific American*, February, 58–66.
- McKnight, S. 1991. Molecular zippers in gene regulation. *Scientific American*, April, 54–64.
- Melek, M., and M. Gellert. 2000. RAG1/2-mediated resolution of transposition intermediates: two pathways and possible consequences. *Cell* 101:625–33.
- Meyerowitz, E. 1994. The genetics of flower development. *Scientific American*, November, 56–65.
- MHC Sequencing Consortium. 1999. Complete sequence and gene map of a human major histocompatibility complex. *Nature* 401:921–23 (plus insert).
- Milstein, C. 1986. From antibody structure to immunological diversification of immune response. *Science* 231:1261–68.
- Mitelman, F., F. Mertens, and B. Johansson. 1997. A breakpoint map of recurrent chromosomal rearrangements in human neoplasia. *Nature Genetics Special Issue* 15:417–74.
- Ng, H.-H., and A. Bird. 1999. DNA methylation and chromatin modification. *Current Opinion in Genetics & Development* 9:158–63.
- Niehrs, C., and N. Pollet. 1999. Synexpression groups in eukaryotes. *Nature* 402:483–87.
- Nüsslein-Volhard, C. 1996. Gradients that organize embryo development. *Scientific American*, August, 54–61.
- Nüsslein-Volhard, C., and E. Wieschaus. 1980. Mutations affecting segment number and polarity in *Drosophila*. *Nature* 287:795–801.
- O'Brien, S., and M. Dean. 1997. In search of AIDS-resistance genes. *Scientific American*, September, 44–51.
- Oettinger, M. 1999. V(D)J recombination: on the cutting edge. *Current Opinion in Cell Biology* 11:325–29.
- Oshima, Y., and I. Takano. 1971. Mating types in *Saccharomyces*: Their convertibility and homothallism. *Genetics* 67:327–35.
- Pidkowich, M., J. Klenz, and G. Haughn. 1999. The making of a flower: Control of floral meristem identity in *Arabidopsis*. *Trends in Plant Science* 4:64–70.
- Pieters, J. 1997. MHC class II restricted antigen presentation. *Current Opinion in Immunology* 9:89–96.
- Rahman, N., and M. Stratton. 1998. The genetics of breast cancer susceptibility. *Annual Review of Genetics* 32:95–121.
- Ray, L. 1999. The science of signal transduction. *Science* 284:755–56. (Introduces several articles in a signal transduction review section.)
- Reinherz, E., et al. 1999. The crystal structure of a T cell receptor in complex with peptide and MHC class II. *Science* 286:1913–21.
- Rhodes, D., and A. Klug. 1993. Zinc fingers. *Scientific American*, February, 56–65.
- Riddle, D., et al. 1997. *C. elegans* II: Cold Spring Harbor, New York: Cold Spring Harbor Laboratory Press.
- Rowen, L., B. Koop, and L. Hood. 1996. The complete 685-kilobase DNA sequence of the human β T cell receptor locus. *Science* 272:1755–62.
- Ruddle, E., et al. 1994. Evolution of HOX genes. *Annual Review of Genetics* 28:423–42.
- Rudolf, K., et al. 1999. Longevity, stress response, and cancer in aging telomerase-deficient mice. *Cell* 96:701–12.
- Sánchez-García, I. 1997. Consequences of chromosomal abnormalities in tumor development. *Annual Review of Genetics* 31:429–53.
- Scientific American*. 1993. Life, death, and the immune system. September (entire issue, ten articles).
- . 1996. Making headway against cancer. September (entire issue, ten articles).
- . 1998. Defeating AIDS: What will it take? Special Report. July, 81–107 (several articles).
- Scott, J., and T. Pawson. 2000. Cell communication: The inside story. *Scientific American*, June, 72–79.
- Scott, M. 2000. Development: The natural history of genes. *Cell* 100:27–40.
- Smith, G., et al. 1999. Purification and DNA binding properties of the ataxia-telangiectasia gene product ATM. *Proceedings of the National Academy of Sciences, USA* 111:34–39.
- Stanbridge, E. 1990. Human tumor suppressor genes. *Annual Review of Genetics* 24:615–57.
- Stern, L., et al. 1994. Crystal structure of the human class II MHC protein HLA-DR1 complexed with an influenza virus peptide. *Nature* 368:215–21.
- St. Johnston, D., and C. Nüsslein-Volhard. 1992. The origin of pattern and polarity in the *Drosophila* embryo. *Cell* 68:201–19.
- Strominger, J. 1989. Developmental biology of T-cell receptors. *Science* 244:943–50.
- Struhl, K. 1999. Fundamentally different logic of gene regulation in eukaryotes and prokaryotes. *Cell* 98:1–4.
- Sugden, B. 1993. How some retroviruses got their oncogenes. *Trends in Biochemical Sciences* 18:233–35.
- Sulston, J., et al. 1983. The embryonic cell lineage of the nematode, *Caenorhabditis elegans*. *Developmental Biology* 100:64–119.
- Tanaka, H., et al. 2000. A ribonucleotide reductase gene involved in a p53-dependent cell-cycle checkpoint for DNA damage. *Nature* 404:42–49.
- Tang, H., K. Kuhen, and F. Wong-Staal. 1999. Lentivirus replication and regulation. *Annual Review of Genetics* 33:133–70.
- Toyota, M., and J.-P. Issa. 2000. The role of DNA hypermethylation in human neoplasia. *Electrophoresis* 21:329–33.
- Travers, A. 1989. DNA conformation and protein binding. *Annual Review of Biochemistry* 58:427–52.
- Van Noorden, C., L. Meade-Tollin, and F. Bosman. 1998. Metastasis. *American Scientist* 86:130–41.
- Varmus, H. 1988. Retroviruses. *Science* 240:1427–35.
- Varmus, H., and N. Nathanson. 1998. Science and the control of AIDS. *Science* 280:1815. (Introducing several articles on AIDS.)
- Von Boehmer, H., and P. Kisielow. 1991. How the immune system learns about self. *Scientific American*, October, 74–81.
- Waldman, T. 1991. Monoclonal antibodies in diagnosis and therapy. *Science* 252:1657–62.
- Weigel, D. 1995. The genetics of flower development: from floral induction to ovule morphogenesis. *Annual Review of Genetics* 29:19–39.
- Weigel, D., and O. Nilsson. 1995. A developmental switch sufficient for flower initiation in diverse plants. *Nature* 377:495–500.
- Weissman, B., et al. 1987. Introduction of a normal human chromosome 11 into a Wilm's tumor cell line controls its tumorigenic expression. *Science* 236:175–80.
- Zhu, T., et al. 1998. An African HIV-1 sequence from 1959 and implications for the origin of the epidemic. *Nature* 391:594–97.
- Ziegler, A., et al. 1994. Sunburn and p53 in the onset of skin cancer. *Nature* 372:773–76.

B-18

Appendix B Suggestions for Further Reading

Chapter 17 Non-Mendelian Inheritance

- Armstrong, M., V. Blok, and M. Phillips. 2000. A multipartite mitochondrial genome in the potato cyst nematode *Globodera pallida*. *Genetics* 154:181–92.
- Avise, J. 1991. Ten unorthodox perspectives on evolution prompted by comparative population genetic findings on mitochondrial DNA. *Annual Review of Genetics* 25:45–69.
- Birger, Y., et al. 1999. The imprinting box of the mouse *Igf2r* gene. *Nature* 397:84–88.
- Blackburn, E., and K. Karrer. 1986. Genomic reorganization in ciliated protozoans. *Annual Review of Genetics* 20:501–21.
- Brannan, C., and M. Bartolomei. 1999. Mechanisms of genomic imprinting. *Current Opinion in Genetics & Development* 9:164–70.
- Buiting, K., et al. 1995. Inherited microdeletions in the Angelman and Prader-Willi syndromes define an imprinting centre on human chromosome 15. *Nature Genetics* 9:395–400.
- Chang, D., and D. Clayton. 1987. A mammalian mitochondrial RNA processing activity contains nucleus-encoded RNA. *Science* 235:1178–84.
- Christian, J., and C. Lemunyan. 1958. Adverse effects of crowding on lactation and reproduction of mice and two generations of their progeny. *Endocrinology* 63:517–29.
- Ciferri, O., and L. Dure, III, eds. 1983. *Structure and Function of Plant Genomes*. New York: Plenum Press.
- Clewell, D., ed. 1993. *Bacterial Conjugation*. New York: Plenum Press.
- Dilts, J., and R. Quackenbush. 1986. A mutation in the R body-coding sequence destroys expression of the killer trait in *P. tetraurelia*. *Science* 232:641–43.
- Evans, R., K. Oakley, and G. Clark-Walker. 1985. Elevated levels of petite formation in strains of *Saccharomyces cerevisiae* restored to respiratory competence. I: Association of both high and moderate frequencies of petite mutant formation with the presence of aberrant mitochondrial DNA. *Genetics* 111:389–402.
- Gyllenstein, U., et al. 1991. Paternal inheritance of mitochondrial DNA in mice. *Nature* 352:255–57.
- Hardy, K. 1994. *Plasmids*. Oxford: Oxford University Press.
- Hoeh, W., K. Blakley, and W. Brown. 1991. Heteroplasmy suggests limited biparental inheritance of *Mytilus* mitochondrial DNA. *Science* 251:1488–90.
- Hurt, E., and G. Schatz. 1987. A cytosolic protein contains a cryptic mitochondrial targeting signal. *Nature* 325:499–503.
- Ippen-Ihler, K., and E. Minkley, Jr. 1986. The conjugation system of F, the fertility factor of *Escherichia coli*. *Annual Review of Genetics* 20:593–624.
- Johns, D. 1996. The other human genome: mitochondrial DNA and disease. *Nature Medicine* 2:1065–68.
- Kiberstis, P., ed. 1999. Mitochondria make a comeback. *Science* 283:1475–97. (Four articles on evolution, diseases, function, and inheritance.)
- Kuroiwa, T., and H. Uchida. 1996. Organelle division and cytoplasmic inheritance. *BioScience* 46:827–35.
- Lalande, M. 1996. Parental imprinting and human disease. *Annual Review of Genetics* 30:173–95.
- Landis, W. 1987. Factors determining the frequency of the killer trait within populations of the *Paramecium aurelia* complex. *Genetics* 115:197–205.
- Lang, B., M. Gray, and G. Burger. 1999. Mitochondrial genome evolution and the origin of eukaryotes. *Annual Review of Genetics* 33:351–97.
- Lang, B., et al. 1997. An ancestral mitochondrial DNA resembling a eubacterial genome in miniature. *Nature* 387:493–97.
- Larsson, N., and D. Clayton. 1995. Molecular genetic aspects of human mitochondrial disorders. *Annual Review of Genetics* 29:151–78.
- Lewin, B. 1998. The mystique of epigenetics. *Cell* 93:301–3. (Followed by seven articles on epigenetic inheritance.)
- Lightowers, R., et al. 1997. Mammalian mitochondrial genetics: Heredity, heteroplasmy and disease. *Trends in Genetics* 13:450–55.
- Luft, R. 1994. The development of mitochondrial medicine. *Proceedings of the National Academy of Sciences, USA* 91:8731–38.
- Lunt, D., and B. Hyman. 1997. Animal mitochondrial DNA recombination. *Nature* 387:247.
- Mannella, C., M. Marko, and K. Buttle. 1997. Reconsidering mitochondrial structure: New views of an old organelle. *Trends in Biochemical Sciences* 22:37–38.
- Margulis, L. 1995. *Symbiosis in Cell Evolution*, 2d ed. San Francisco: Freeman.
- McKusick, V. 1998. *Mendelian Inheritance in Man: A Catalog of Human Genes and Genetic Disorders*, 12th ed. Baltimore: Johns Hopkins University Press.
- Neale, D., K. Marshall, and R. Sederoff. 1989. Chloroplast and mitochondrial DNA are paternally inherited in *Sequoia sempervirens* D. Don Endl. *Proceedings of the National Academy of Sciences, USA* 86:9347–49.
- Perrimon, N., L. Engstrom, and A. Mahowald. 1989. Zygotic lethals with specific maternal effect phenotypes in *Drosophila melanogaster*. I. Loci on the X chromosome. *Genetics* 121:333–52.
- Pfanner, N., and W. Neupert. 1990. The mitochondrial protein import apparatus. *Annual Review of Biochemistry* 59:331–53.
- Preer, J., Jr. 1997. Whatever happened to *Paramecium* genetics? *Genetics* 145:217–25.
- Preer, J., Jr., L. Preer, and A. Jurand. 1974. Kappa and other endosymbionts in *Paramecium aurelia*. *Bacteriological Reviews* 38:113–63.
- Razin, A., and H. Cedar. 1994. DNA methylation and genomic imprinting. *Cell* 77:473–76.
- Rhoades, M. 1946. Plastid mutations. *Cold Spring Harbor Symposium on Quantitative Biology* 11:202–7.
- Rodgers, J. 1991. Mechanisms Mendel never knew. *Mosaic* 22:2–11. (Entire issue on nontraditional inheritance.)
- Sager, R. 1972. *Cytoplasmic Genes and Organelles*. New York: Academic Press.
- Saville, B., Y. Kohli, and J. Anderson. 1998. mtDNA recombination in a natural population. *Proceedings of the National Academy of Sciences, USA* 95:1331–35.
- Shadel, G., and D. Clayton. 1997. Mitochondrial DNA maintenance in vertebrates. *Annual Review of Biochemistry* 66:409–35.
- Sonneborn, T. 1950. Methods in the general biology and genetics of *Paramecium aurelia*. *Journal of Experimental Zoology* 113:87–148.
- Stephanou, G., and S. Alahiotis. 1983. Non-Mendelian inheritance of “heat-sensitivity” in *Drosophila melanogaster*. *Genetics* 103:93–107.
- Stoneking, M., and H. Soodyall. 1996. Human evolution and the mitochondrial genome. *Current Opinion in Genetics and Development* 6:731–36.
- Summers, D. 1996. *The Biology of Plasmids*. Oxford: Blackwell Scientific.
- Thomas, C., ed. 1989. *Promiscuous Plasmids of Gram-Negative Bacteria*. London: Academic Press.
- Tilghman, S. 1999. The sins of the fathers and mothers: Genomic imprinting in mammalian development. *Cell* 96:185–93.
- Unsel, M., et al. 1997. The mitochondrial genome of *Arabidopsis thaliana* contains 57 genes in 366,924 nucleotides. *Nature Genetics* 15:57–61.
- Walbot, V., and E. Coe, Jr. 1979. Nuclear gene *iojap* conditions a programmed change to ribosome-less plastids in *Zea mays*. *Proceedings of the National Academy of Sciences, USA* 76:2760–64.
- Wallace, D. 1992. Diseases of the mitochondrial DNA. *Annual Review of Biochemistry* 61:1175–212.
- . 1997. Mitochondrial DNA in aging and disease. *Scientific American*, August, 40–47.

- Wallace, D., et al. 1988. Mitochondrial DNA mutation associated with Leber's hereditary optic neuropathy. *Science* 242:1427–30.
- Warren, G., and W. Wickner. 1996. Organelle inheritance. *Cell* 84:395–400.
- Wickner, R., 1986. Double-stranded RNA replication in yeast: The killer system. *Annual Review of Biochemistry* 55:373–95.
- Wiener, M., et al. 1997. Crystal structure of colicin Ia. *Nature* 385:461–64.
- Williamson, D., and D. Poulson. 1979. Sex ratio organisms (spiroplasmas) of *Drosophila*. *The Mycoplasmas* 3:175–208.
- Yaffe, M., et al. 1996. Microtubules mediate mitochondrial distribution in fission yeast. *Proceedings of the National Academy of Sciences, USA* 93:11664–68.

Chapter 18 Quantitative Inheritance

- American Association for the Advancement of Science. 1994. Genes and behavior. *Science* 264:1685–739.
- Bennett, S., and J. Todd. 1996. Human type 1 diabetes and the insulin gene: Principles of mapping polygenes. *Annual Review of Genetics* 30:343–70.
- Bouchard, T., Jr., and M. McGue. 1981. Familial studies of intelligence: A review. *Science* 212:1055–59.
- Carson, R., and M. Rothstein, eds. 1999. *Behavioral Genetics: The Clash of Culture and Biology*. Baltimore: Johns Hopkins University Press.
- Cheverud, J., and E. Routman. 1995. Epistasis and its contribution to genetic variance components. *Genetics* 139:1455–61.
- Cheverud, J., et al. 1996. Quantitative trait loci for murine growth. *Genetics* 142:1305–19.
- Crow, J. 1957. Genetics of insect resistance to chemicals. *Annual Review of Entomology* 2:227–46.
- . 1997. Birth defects, Jimson weed and bell curves. *Genetics* 147:1–6.
- Daniels, S., et al. 1996. A genome-wide search for quantitative trait loci underlying asthma. *Nature* 383:247–50.
- Devlin, B., M. Daniels, and K. Roeder. 1997. The heritability of IQ. *Nature* 388:468–71.
- de Waal, F. 1999. The end of nature versus nurture. *Scientific American*, December, 94–99.
- Eshed, Y., and D. Zamir. 1996. Less-than-additive epistatic interactions of quantitative trait loci in tomato. *Genetics* 143:1807–17.
- Falconer, D. 1992. Early selection experiments. *Annual Review of Genetics* 26:1–14.
- . 1995. *Introduction to Quantitative Genetics*, 4th ed. London: Longman.
- Greenspan, R. 1995. Understanding the genetic construction of behavior. *Scientific American*, April, 72–78.

- Hall, J. 1996. Twinning: Mechanisms and genetic implications. *Current Opinion in Genetics and Development* 6:343–47.
- Harrison, G., and J. Owen. 1964. Studies on the inheritance of human skin color. *Annals of Human Genetics* 28:27–37.
- Holt, S. 1961. Inheritance of dermal ridge patterns. In *Recent Advances in Human Genetics*, edited by L. Penrose, pp. 101–19. London: J. and A. Churchill.
- Horgan, J. 1993. Eugenics revisited. *Scientific American*, June, 122–31.
- Keightley, P., et al. 1996. A genetic map of quantitative trait loci for body weight in the mouse. *Genetics* 142:227–35.
- Laxova, R. 1998. Lionel Sharples Penrose, 1898–1972: A personal memoir in celebration of the centenary of his birth. *Genetics* 150:1333–40.
- Li, A., et al. 1997. Epistasis for three grain yield components in rice (*Oryza sativa* L.). *Genetics* 145:453–65.
- Lynch, M., and B. Walsh. 1997. *Genetics and Analysis of Quantitative Traits*. Sunderland, Mass.: Sinauer Associates.
- Mackintosh, N., ed. 1995. *Cyril Burt: Fraud or Framed?* Oxford: Oxford University Press.
- Paterson, A., et al. 1991. Mendelian factors underlying quantitative traits in tomato: Comparison across species, generations, and environments. *Genetics* 127:181–97.
- Plomin, R. 1990. The role of inheritance in behavior. *Science* 248:183–88.
- . 1999. Genetics and general cognitive ability. *Nature* 402:c25–c29.
- Plomin, R., and J. DeFries. 1998. The genetics of cognitive abilities and disabilities. *Scientific American*, May, 62–69.
- Powledge, T. 1993. The genetic fabric of human behavior. *BioScience* 43:362–67.
- . 1993. The inheritance of behavior in twins. *BioScience* 43:420–24.
- Rose, R., et al. 1990. Social contact and sibling similarity: Facts, issues, and red herrings. *Behavioral Genetics* 20:763–78.
- Tanksley, S. 1993. Mapping polygenes. *Annual Review of Genetics* 27:205–33.
- Velmalal, R., et al. 1999. A search for quantitative trait loci for milk production traits on chromosome 6 in Finnish Ayrshire cattle. *Animal Genetics* 30:136–43.
- Vieira, C., et al. 2000. Genotype-environment interaction for quantitative trait loci affecting life span in *Drosophila melanogaster*. *Genetics* 154:213–27.
- Woolf, C. 1990. Multifactorial inheritance of common white markings in the Arabian horse. *Journal of Heredity* 81:250–56.
- Yashin, A., I. Iachine, and J. Harris. 1999. Half of the variation in susceptibility to mortality is genetic: Findings from Swedish twin survival data. *Behavior Genetics* 29:11–19.

Chapter 19 Population Genetics: The Hardy-Weinberg Equilibrium and Mating Systems

- Bittles, A., and J. Neel. 1994. The costs of human inbreeding and their implications for variations at the DNA level. *Nature Genetics* 8:117–21.
- Bittles, A., et al. 1991. Reproductive behavior and health in consanguineous marriages. *Science* 252:789–94.
- Buss, D. 1985. Human mate selection. *American Scientist* 73:47–51.
- Cavalli-Sforza, L., P. Menozzi, and A. Piazza. 1996. *The History and Geography of Human Genes*. Princeton, N.J.: Princeton University Press.
- Crow, J. 1987. Population genetics history: A personal view. *Annual Review of Genetics* 21:1–22.
- . 1988. Eighty years ago: The beginnings of population genetics. *Genetics* 119:473–76.
- . 1988. Sewall Wright (1889–1988). *Genetics* 119:1–4.
- De Finetti, B. 1926. Considerazioni matematiche sul l'ereditarieta mendeliana. *Metron* 6:1–41.
- Dudash, M., and D. Carr. 1998. Genetics underlying inbreeding depression in *Mimulus* with contrasting mating systems. *Nature* 393:682–84.
- Fisher, R. 1930. *The Genetical Theory of Natural Selection*. Oxford: Clarendon Press.
- Frankham, R. 1995. Conservation genetics. *Annual Review of Genetics* 29:305–27.
- Haldane, J. 1932. *The Causes of Evolution*. New York: Harper & Row.
- Hardy, G. 1908. Mendelian proportions in a mixed population. *Science* 28:49–50.
- Hill, W. 1995. Sewall Wright's "systems of mating." *Genetics* 143:1499–1506.
- Jimenez, J., et al. 1994. An experimental study of inbreeding depression in a natural habitat. *Science* 266:271–73.
- Kimura, M. 1988. Thirty years of population genetics with Dr. Crow. *Japanese Journal of Genetics* 63:1–10.
- May, R. 1995. The cheetah controversy. *Nature* 374:309–10.
- Maynard Smith, J. 1989. *Evolutionary Genetics*. Oxford: Oxford University Press.
- Ralls, K., J. Ballou, and A. Templeton. 1988. Estimates of lethal equivalents and the cost of inbreeding in mammals. *Conservation Biology* 2:185–93.
- Russell, E. 1989. Sewall Wright's contributions to physiological genetics and to inbreeding theory and practice. *Annual Review of Genetics* 23:1–18.
- Saccheri, I., et al. 1998. Inbreeding and extinction in a butterfly metapopulation. *Nature* 392:491–94.

B-20

Appendix B Suggestions for Further Reading

- Sarkar, S. 1992. A centenary reassessment of J.B.S. Haldane, 1892–1964. *BioScience* 42:777–85.
- Spieess, E. 1989. *Genes in Populations*, 2d ed. New York: Wiley-Liss.
- Weinberg, W. 1908. Über den Nachweis der Vererbung beim Menschen. *Jahreshefte Verein fuer Vaterlaendische Naturkunde in Wuerttemberg* 64:368–82.
- Wright, S. 1931. Evolution in Mendelian populations. *Genetics* 16:97–159.
- . 1968–78. *Evolution and the Genetics of Populations: A Treatise*. 4 vols. Chicago: University of Chicago Press.
- Chapter 20 Population Genetics: Processes That Change Allelic Frequencies**
- Cameron, R. 1992. Change and stability in *Cepaea* populations over 25 years: A case of climatic selection. *Proceedings of the Royal Society of London, series B* 248:181–87.
- Caro, T., and M. Laurenson. 1994. Ecological and genetic factors in conservation: A cautionary tale. *Science* 263:485–86.
- Carson, H. 1997. Sexual selection: A driver of genetic change in Hawaiian *Drosophila*. *The Journal of Heredity* 88:343–52.
- Castle, W. 1903. The laws of heredity of Galton and Mendel, and some laws governing race improvement by selection. *Proceedings of the American Academy of Arts and Sciences* 39:223–42.
- Crow, J. 1987. Population genetics history: A personal view. *Annual Review of Genetics* 21:1–22.
- . 1990. Fisher's contributions to genetics and evolution. *Theoretical Population Biology* 38:263–75.
- . 1992. Centennial: J.B.S. Haldane, 1892–1964. *Genetics* 130:1–6.
- . 1992. Erwin Schrödinger and the hornless cattle problem. *Genetics* 130:237–39.
- Diamond, J. 1988. Founding fathers and mothers. *Natural History*, June, 10–15.
- . 1994. Pitcairn before the Bounty. *Nature* 369:608–9.
- Dobzhansky, T. 1958. Genetics of natural populations. XXVII. The genetic changes in populations of *Drosophila pseudoobscura* in the American Southwest. *Evolution* 12:385–401.
- Ganetzky, B. 1999. Yuichiro Hiraizumi and forty years of segregation distortion. *Genetics* 152:1–4.
- Gibbons, A. 1994. Genes point to a new identity for Pacific pioneers. *Science* 263:32–33.
- Hill, G. 1991. Plumage coloration is a sexually selected indicator of male quality. *Nature* 350:337–39.
- Houle, D., et al. 1992. The genomic mutation rate for fitness in *Drosophila*. *Nature* 359:58–60.
- Kimura, M. 1955. Solution of a process of random genetic drift with a continuous model. *Proceedings of the National Academy of Sciences, USA* 41:144–50.
- . 1988. Thirty years of population genetics with Dr. Crow. *Japanese Journal of Genetics* 63:1–10.
- Koenig, W. 1989. Sex-biased dispersal in the contemporary United States. *Ethology and Sociobiology* 10:263–78.
- Koenig, W., S. Albano, and J. Dickson. 1991. A comparison of methods to partition selection acting via components of fitness: Do larger male bullfrogs have greater hatching success? *Journal of Evolutionary Biology* 4:309–20.
- Lewontin, R. 1985. Population genetics. *Annual Review of Genetics* 19:81–102.
- Lummaa, V., et al. 1998. Natural selection on human twinning. *Nature* 394:533–34.
- Lyttle, T. 1991. Segregation distorters. *Annual Review of Genetics* 25:511–57.
- Mallett, J., et al. 1990. Estimates of selection and gene flow from measures of cline width and linkage disequilibrium in *Heliconius* hybrid zones. *Genetics* 124:921–36.
- Maynard Smith, J. 1991. The population genetics of bacteria. *Proceedings of the Royal Society of London, series B* 245:37–41.
- McCauley, D. 1991. Genetic consequences of local population extinction and recolonization. *Trends in Ecology and Evolution* 6:5–8.
- Menotti-Raymond, M., and S. O'Brien. 1993. Dating the genetic bottleneck of the African cheetah. *Proceedings of the National Academy of Sciences, USA* 90:3172–76.
- Pomiankowski, A., and L. Hurst. 1993. Siberian mice upset Mendel. *Nature* 363:396–97.
- Powers, D., et al. 1991. Genetic mechanisms for adapting to a changing environment. *Annual Review of Genetics* 25:629–59.
- Pyke, G. 1991. What does it cost a plant to produce floral nectar? *Nature* 350:58–59.
- Raymond, M., et al. 1996. Frequency-dependent maintenance of left handedness in humans. *Proceedings of the Royal Society of London, series B* 263:1627–33.
- Reznick, D., et al. 1997. Evaluation of the rate of evolution in natural populations of guppies (*Poecilia reticulata*). *Science* 275:1934–37.
- Rood, S., et al. 1988. Gibberellins: A phytohormonal basis for heterosis in maize. *Science* 241:1216–18.
- Schaeffer, S., and E. Miller. 1992. Estimates of gene flow in *Drosophila pseudoobscura* determined from nucleotide sequence analysis of the alcohol dehydrogenase region. *Genetics* 132:471–80.
- Smith, T. 1993. Disruptive selection and the genetic basis of bill size polymorphism in the African finch *Pyrenestes*. *Nature* 363:618–20.
- Spieess, E. 1989. *Genes in Populations*, 2d ed. New York: Wiley-Liss.
- Tamarin, R., and T. Kunz. 1974. *Microtus breweri*. *Mammalian Species* 45:1–3.
- Thoday, J., and J. Gibson. 1962. Isolation by disruptive selection. *Nature* 193:1164–66.
- Wallace, B. 1991. *Fifty Years of Genetic Load. An Odyssey*. Ithaca, N.Y.: Cornell University Press.
- Weber, K. 1992. How small are the smallest selectable domains of form? *Genetics* 130:345–53.
- Williams, G. 1992. *Natural Selection: Domains, Levels, and Challenges*. New York: Oxford University Press.
- Chapter 21 Genetics of the Evolutionary Process**
- Ayala, F. 1997. Vagaries of the molecular clock. *Proceedings of the National Academy of Sciences, USA* 94:7776–83.
- Barbadilla, A., L. King, and R. Lewontin. 1996. What does electrophoretic variation tell us about protein variation? *Molecular Biology and Evolution* 13:427–32.
- Barlow, G. 1991. Nature-nurture and the debates surrounding ethology and sociobiology. *American Zoologist* 31:286–96.
- Bethell, T. 1976. Darwin's mistake. *Harper's*, February, 70–75.
- Bowcock, A., et al. 1991. Drift, admixture, and selection in human evolution: A study with DNA polymorphisms. *Proceedings of the National Academy of Sciences, USA* 88:839–43.
- Brakefield, P. 1987. Industrial melanism: Do we have all the answers? *Trends in Ecology and Evolution* 2:117–22.
- Britton-Davidian, J., et al. 2000. Rapid chromosomal evolution in island mice. *Nature* 403:158.
- Bush, G. 1975. Modes of animal speciation. *Annual Review of Ecology and Systematics* 6:339–64.
- Buss, D. 1994. The strategies of human mating. *American Scientist* 82:238–49.
- Carson, H. 1987. The genetic system, the deme, and the origin of species. *Annual Review of Genetics* 21:405–23.
- Cavalli-Sforza, L., P. Menozzi, and A. Piazza. 1994. *The History and Geography of Human Genes*. Princeton, N.J.: Princeton University Press.
- Cohn, J. 1992. Naked mole-rats. *BioScience* 42:86–89.
- Conover, D., and D. Van Voorhees. 1990. Evolution of a balanced sex ratio by frequency-dependent selection in a fish. *Science* 250:1556–58.
- Coppens, Y. 1994. East side story: The origin of humankind. *Scientific American*, May, 88–95.
- Coyne, J. 1992. Genetics and speciation. *Nature* 355:511–15.

Appendix B Suggestions for Further Reading

B-21

- Crow, J. 1987. Population genetics history: A personal view. *Annual Review of Genetics* 21:1–22.
- Darwin, C. 1845. *The Voyage of the Beagle*. 1962 ed. Garden City, N.Y.: Doubleday.
- . 1859. *The Origin of Species by Means of Natural Selection or the Preservation of Favored Races in the Struggle for Life*. London: John Murray.
- Eldredge, N., and S. Gould. 1972. Punctuated equilibria: An alternative to phyletic gradualism. In *Models in Paleobiology*, edited by T. Schopf, pp. 85–115. San Francisco: Freeman.
- Elena, S., V. Cooper, and R. Lenski. 1996. Punctuated evolution caused by selection of rare beneficial mutations. *Science* 272:1802–4.
- Emlen, S., P. Wrege, and N. Demong. 1995. Making decisions in the family: An evolutionary perspective. *American Scientist* 83:148–57.
- Ford, E. 1979. *Ecological Genetics*, 4th ed. London: Chapman and Hall.
- Futuyma, D. 1997. *Evolutionary Biology*, 3d ed. Sunderland, Mass.: Sinauer Associates.
- Gibbons, A. 1996. On the many origins of species. *Science* 273:1496–99.
- . 1998. Calibrating the mitochondrial clock. *Science* 279:28–29.
- Gillespie, J. 1994. *The Causes of Molecular Evolution*. Oxford: Oxford University Press. (Reprint)
- Gojobori, T., E. Moriyama, and M. Kimura. 1990. Molecular clock of viral evolution, and the neutral theory. *Proceedings of the National Academy of Sciences, USA* 87:10015–18.
- Gould, S. 1994. The evolution of life on earth. *Scientific American*, October, 84–91.
- Gould, S., and N. Eldredge. 1993. Punctuated equilibrium comes of age. *Nature* 366:223–27.
- Grant, B., and P. Grant. 1993. Evolution of Darwin's finches caused by a rare climatic event. *Proceedings of the Royal Society of London, series B* 251:111–17.
- Grant, B., et al. 1998. Geographic and temporal variation in the incidence of melanism in peppered moth populations in America and Britain. *The Journal of Heredity* 89:465–71.
- Grant, P. 1999. *Ecology and Evolution of Darwin's Finches*. Princeton, N.J.: Princeton University Press.
- Hamilton, W. 1964. The genetical evolution of social behavior, I, II. *Journal of Theoretical Biology* 7:1–52.
- Hanson, B., et al., eds. 1999. The diversity of evolution. *Science* 284:2105–48. (Nine articles and viewpoints.)
- Hays, B. 1999. Experimental Lamarckism. *American Scientist* 87:494–98.
- Hutter, P., J. Roote, and M. Ashburner. 1990. A genetic basis for the inviability of hybrids between sibling species of *Drosophila*. *Genetics* 124:909–20.
- Johnson, T., et al. 1996. Late Pleistocene desiccation of Lake Victoria and rapid evolution of cichlid fishes. *Science* 273:1091–93.
- Kimura, M. 1985. *The Neutral Theory of Molecular Evolution*. New York: Cambridge University Press.
- . 1987. Molecular evolutionary clock and the neutral theory. *Journal of Molecular Evolution* 26:24–33.
- Koehn, R. 1970. Functional and evolutionary dynamics of polymorphic esterases in catostomid fishes. *American Fisheries Society, Transactions* 99:219–28.
- Kuhn, T. 1996. *The Structure of Scientific Revolutions*, 3d ed. Chicago: University of Chicago Press.
- Lewin, R. 1988. Molecular clocks turn a quarter century. *Science* 239:561–63.
- Lewontin, R. 1985. Population genetics. *Annual Review of Genetics* 19:81–102.
- . 1991. Twenty-five years ago in GENETICS: Electrophoresis in the development of evolutionary genetics: Milestone or millstone? *Genetics* 128:657–62.
- Lewontin, R., and J. Hubby. 1966. A molecular approach to the study of genic heterozygosity in natural populations. II. *Genetics* 54:595–609.
- Lindström, L., R. Alatalo, and J. Mappes. 1997. Imperfect Batesian mimicry—The effects of the frequency and the distastefulness of the model. *Proceedings of the Royal Society of London, Series B* 264:149–53.
- Losey, J., et al. 1997. A polymorphism maintained by opposite patterns of parasitism and predation. *Nature* 388:269–72.
- Losos, J., K. Warheit, and T. Schoener. 1997. Adaptive differentiation following experimental island colonization in *Anolis* lizards. *Nature* 387:70–73.
- Lynch, M., and P. Jarrell. 1993. A method for calibrating molecular clocks and its application to animal mitochondrial DNA. *Genetics* 135:1197–208.
- May, R. 1992. How many species inhabit the earth? *Scientific American*, October, 42–48.
- Mayr, E. 1997. The objects of selection. *Proceedings of the National Academy of Sciences, USA* 94:2091–94.
- Mogie, M. 1996. Malthus and Darwin: World views apart. *Evolution* 50:2086–88.
- National Academy of Sciences. 1999. *Science and Creationism—A View from the National Academy of Sciences*, 2d ed. Washington: National Academy Press.
- Niemälä, P., and J. Tuomi. 1987. Does the leaf morphology of some plants mimic caterpillar damage? *Oikos* 50:256–57.
- Nijhout, H. 1994. Developmental perspectives on evolution of butterfly mimicry. *BioScience* 44:148–57.
- Nowak, M., R. May, and K. Sigmund. 1995. The arithmetics of mutual help. *Scientific American*, June, 76–81.
- Packer, C., et al. 1991. A molecular genetic analysis of kinship and cooperation in African lions. *Nature* 351:562–65.
- Pfennig, D., and P. Sherman. 1995. Kin recognition. *Scientific American*, June, 98–103.
- Ritland, D., and L. Brower. 1991. The viceroxy butterfly is not a Batesian mimic. *Nature* 350:497–98.
- Sato, A., et al. 1999. Phylogeny of Darwin's finches as revealed by mtDNA sequences. *Proceedings of the National Academy of Sciences, USA* 96:5101–6.
- Scherer, S. 1990. The protein molecular clock: Time for a reevaluation. *Evolutionary Biology* 24:83–106.
- Schlieffen, U., D. Tautz, and S. Pääbo. 1994. Sympatric speciation suggested by monophyly of crater lake cichlids. *Nature* 368:629–32.
- Schmidt, K. 1996. Creationists evolve new strategy. *Science* 273:420–22.
- Sherman, P., J. Jarvis, and S. Braude. 1992. Naked mole rats. *Scientific American*, August, 72–78.
- Skibinski, D., M. Woodward, and R. Ward. 1993. A quantitative test of the neutral theory using pooled allozyme data. *Genetics* 135:233–48.
- Slatkin, M., ed. 1995. *Exploring Evolutionary Biology*. Sunderland, Mass.: Sinauer Associates.
- Stiassny, M., and A. Meyer. 1999. Cichlids of the rift lakes. *Scientific American*, February, 64–69.
- Tamarin, R., and M. Sheridan. 1987. Behavior-genetic mechanisms of population regulation in microtine rodents. *American Zoologist* 27:921–27.
- Thompson, K. 1998. 1798: Darwin and Malthus. *American Scientist* 86:226–29.
- Tregenza, T., and R. Butlin. 1999. Speciation without isolation. *Nature* 400:311–12. (With two accompanying research articles.)
- Vincek, V., et al. 1997. How large was the founding population of Darwin's finches? *Proceedings of the Royal Society of London, series B* 264:111–18.
- Williams, G. 1992. *Natural Selection: Domains, Levels, and Challenges*. New York: Oxford University Press.
- Wilson, E. 1975. *Sociobiology: The New Synthesis*. Cambridge, Mass.: Harvard University Press.
- Young, W. 1985. *Fallacies of Creationism*. Calgary, Alberta: Detselig Enterprises, Ltd.
- Zuckerandl, E., and L. Pauling. 1965. Evolutionary divergence and convergence in proteins. In *Evolving Genes and Proteins*, edited by V. Bryson and H. Vogel, pp. 97–166. New York: Academic Press.

GLOSSARY

acentric fragment A chromosomal piece without a centromere.

acrocentric chromosome A chromosome whose centromere lies very near one end.

acron The anterior end of the arthropod embryo from which eyes and antennae develop.

activation energy See free energy of activation ($\Delta G^\ddagger$).

activator A eukaryotic specific transcription factor that binds to an enhancer, often far upstream of a promoter.

active site The part of an enzyme where the actual enzymatic function is performed.

adaptive mutation See directed mutation.

adaptive value See fitness.

additive model A mechanism of quantitative inheritance in which alleles at different loci either add an amount to the phenotype or add nothing.

adenine See purines.

adjacent-1 segregation A separation of centromeres during meiosis in a reciprocal translocation heterozygote so that unbalanced zygotes are produced.

adjacent-2 segregation Separation of centromeres during meiosis in a translocation heterozygote so that homologous centromeres are pulled to the same pole.

A DNA The form of DNA with high water content; it has tilted base pairs and more base pairs per turn than does B DNA.

affected Individuals in a pedigree that exhibit the specific phenotype under study.

allele Alternative form of a gene.

allelic exclusion A process whereby only one immunoglobulin light chain and one heavy chain gene are transcribed in any one cell; the other genes are repressed.

allopatric speciation Speciation in which the evolution of reproductive isolating mechanisms occurs during the physical separation of the populations.

allopolyploidy Polyploidy produced by the hybridization of two species.

allosteric protein A protein whose shape is changed when it binds a particular molecule. In the new shape, the protein's ability to react to a second molecule is altered.

allotype Mutant of a nonvariant part of an immunoglobulin gene that follows the rules of simple Mendelian inheritance.

allozygosity Homozygosity in which the two alleles are alike but unrelated. See autozygosity.

allozymes Forms of an enzyme, controlled by alleles of the same locus, that differ in electrophoretic mobility. See isozymes.

alternate segregation A separation of centromeres during meiosis in a reciprocal translocation heterozygote so that balanced gametes are produced.

alternative splicing Various ways of splicing out introns in eukaryotic premessenger RNAs so that one gene produces several different messenger RNA and protein products.

altruism A form of behavior in which an individual risks lowering its fitness for the benefit of another.

Alu family A dispersed, intermediately repetitive DNA sequence found in the human genome about 300,000 times. The sequence is about 300 bp long. The name *Alu* comes from the restriction endonuclease that cleaves it.

aminoacyl-tRNA synthetases Enzymes that attach amino acids to their proper transfer RNAs.

amphidiploid An organism produced by hybridization of two species, followed by somatic doubling. It is an allotetraploid that appears to be a normal diploid.

anagenesis The evolutionary process whereby one species evolves into another without any splitting of the phylogenetic tree. See cladogenesis.

anaphase The stage of mitosis and meiosis in which sister chromatids or homologous chromosomes are separated by spindle fibers.

anaphase A The stage of anaphase in which chromatids are separated by the shortening of kinetochore microtubules.

anaphase B The stage of anaphase in which chromatids are separated by the general elongation of the spindle.

anaphase-promoting complex (APC) Protein complex that breaks down cyclin B and promotes anaphase among its various roles in controlling the cell cycle. (Also called the cyclosome.)

aneuploids Individuals or cells exhibiting aneuploidy.

aneuploidy The condition of a cell or of an organism that has additions or deletions of whole chromosomes.

angiosperms Plants whose seeds are enclosed within an ovary. Flowering plants.

antibody A protein produced by a B lymphocyte that protects the organism against antigens.

anticoding strand The DNA strand that forms the template for both the transcribed messenger RNA and the coding strand.

anticodon The three-base sequence on transfer RNA complementary to a codon on messenger RNA.

antigen A foreign substance capable of triggering an immune response in an organism.

antimutator mutations Mutations of DNA polymerase that decrease the overall mutation rate of a cell or of an organism.

anti-oncogene A gene that represses malignant growth and whose absence results in malignancy (e.g., *retinoblastoma*).

antiparallel strands Strands, as in DNA, that run in opposite directions with respect to their 3' and 5' ends.

antisense RNA RNA product of *mic* (*mRNA-interfering complementary RNA*) genes that regulates another gene by base pairing with, and thus blocking, its messenger RNA.

antisense strand See anticoding strand.

anti-sigma factor A protein that interferes with the action of a sigma factor.

antiterminator protein A protein that, when bound at its normal attachment sites, lets RNA polymerase read through normal terminator sequences (e.g., the N- and Q-gene products of phage λ).

AP endonucleases Endonucleases that initiate excision repair at apurinic and apyrimidinic sites on DNA.

apoptosis Programmed cell death.

archaea Highly specialized, bacterial-like organisms that make up the third kingdom of life on earth along with the bacteria and the eukaryotes. Identified by Carl Woese in 1977 based on ribosomal RNA sequences. Most are thermophilic, halophilic, or methanogenic.

ascospores Haploid spores found in the asci of Ascomycete fungi.

ascus The sac in Ascomycete fungi that holds the ascospores.

A (aminoacyl) site The site on the ribosome occupied by an aminoacyl-tRNA just prior to peptide bond formation.

assignment test A test that determines whether a locus is on a specific chromosome by observing the concordance of the locus and the specific chromosome in hybrid cell lines.

assortative mating The mating of individuals with similar phenotypes.

G-2 *Glossary*

- aster** Configuration at the centrosome with microtubules radiating out in all directions.
- ataxia-telangiectasia** A disease in human beings caused by a defect in X-ray-induced repair mechanisms.
- attenuator region** A control region at the promoter end of repressible amino acid operons that exerts transcriptional control based on the translation of a small leader peptide gene.
- attenuator stem** See terminator stem.
- autogamy** Nuclear reorganization in a single *Paramecium* cell similar to the changes that occur during conjugation.
- autonomously replicating sequence (ARS)** Eukaryotic site of the initiation of DNA replication consisting of an 11 base-pair consensus sequence and several other sequences covering 100–200 base pairs.
- autopolyploidy** Polyploidy in which all the chromosomes come from the same species.
- autoradiography** A technique in which radioactive molecules make their locations known by exposing photographic plates.
- autosomal set** A combination of nonsex chromosomes consisting of one from each homologous pair in a diploid species.
- autosomes** The nonsex chromosomes.
- autotrophs** Organisms that can utilize carbon dioxide as a carbon source.
- autozygosity** Homozygosity in which the two alleles are identical by descent (i.e., they are copies of an ancestral gene).
- auxotrophs** Organisms that have specific nutritional requirements.
- bacillus** A rod-shaped bacterium.
- backcross** The cross of an individual with one of its parents or with an organism with the same genotype as a parent.
- back mutation** The process that causes reversion. A change in a nucleotide pair in a mutant gene that restores the original sequence and hence the original phenotype.
- bacterial artificial chromosomes (BACs)** Artificial chromosomes used for sequencing that are derived from bacterial fertility factors (F plasmids).
- bacterial lawn** A continuous cover of bacteria on the surface of a growth medium.
- bacteriophages** Bacterial viruses.
- Balbani rings** The larger polytene chromosomal puffs. Generally synonymous with *puffs*. See chromosome puffs.
- Barr body** Heterochromatic body (X chromosome) found in the nuclei of normal female mammals but absent in the nuclei of normal males.
- basal bodies** Microtubule organizing center for cilia and flagella, composed of centrioles.
- base excision repair** The DNA excision repair mechanism that replaces a nucleotide lacking its base with a complete nucleotide. Glycosylases or the environment create the AP (apurinic and apyrimidinic) nucleotides.
- base flipping** A process whereby enzymes gain access to bases within the DNA double helix by first flipping the bases out of the interior to the outside.
- basic/helix-loop-helix/leucine zipper** A motif of proteins that bind DNA that consists of a series of basic amino acids followed by a helix-loop-helix domain and then a leucine zipper. See helix-turn-helix motif; leucine zipper.
- Batesian mimicry** Form of mimicry in which an innocuous model gains protection by resembling a noxious or dangerous host.
- B DNA** The right-handed, double-helical form of DNA described by Watson and Crick.
- β -galactosidase** The enzyme that splits lactose into glucose and galactose (coded by a gene in the *lac* operon).
- β -galactoside acetyltransferase** An enzyme that is involved in lactose metabolism and encoded by a gene in the *lac* operon.
- β -galactoside permease** An enzyme involved in concentrating lactose in the cell (coded by a gene in the *lac* operon).
- binary fission** Simple cell division in single-celled organisms.
- binomial expansion** The terms generated when a binomial expression is raised to a particular power.
- binomial theorem** The theorem that gives the terms of the expansion of a binomial expression raised to a particular power.
- biochemical genetics** The study of the relationships between genes and enzymes, specifically the role of genes in controlling the steps in biochemical pathways.
- bioinformatics** The science of storing, retrieving, and analyzing genomic data.
- biolistic** A method (*biological ballistic*) of transfecting cells by bombarding them with microprojectiles coated with DNA.
- biological species concept** The idea that organisms are classified in the same species if they are potentially capable of interbreeding and producing fertile offspring.
- bivalents** Structures, formed during prophase of meiosis I, consisting of the synapsed homologous chromosomes. Equivalent to a tetrad of chromatids.
- blastoderm** The outer layer of cells in an insect embryo after cleavage but before gastrulation. A syncytial blastoderm gives way to a cellular blastoderm when cell walls form.
- blunt-end ligation** The ligating or attaching of blunt-ended pieces of DNA by, for example, T₄ DNA ligase. Used in creating hybrid vectors.
- bottleneck** A brief reduction in the size of a population, which usually leads to random genetic drift.
- bouquet stage** A stage during zygonema in which chromosome ends, attached to the nuclear membrane, come to lie near each other.
- branch migration** The process in which a crossover point between two duplexes slides along the duplexes.
- breakage and reunion** The general mode by which recombination occurs. DNA duplexes are broken and reunited in a crosswise fashion according to the Holliday model.
- breakage-fusion-bridge cycle** Damage that a dicentric chromosome goes through during each cell cycle.
- buoyant density of DNA** A measure of the density or size of DNA determined by the equilibrium point reached by DNA after density gradient centrifugation.
- calculus of the genes** Apparent calculation by the genes to determine when a particular altruistic behavior is beneficial to inclusive fitness and hence worth doing.
- cancer** An informal term for a diverse class of diseases marked by abnormal cell proliferation.
- cancer-family syndromes** Pedigree patterns in which unusually large numbers of blood relatives develop certain kinds of cancers.
- cap** A methylated guanosine added to the 5' end of eukaryotic messenger RNA.
- capsid** The protein shell of a virus.
- capsomere** Protein clusters making up discrete subunits of a viral protein shell.
- carcinoma** Tumor arising from epithelial tissue (e.g., glands, breasts, skin, linings of the urogenital and respiratory systems).
- cassette mechanism** The mechanism by which homothallic yeast cells alternate mating types. The mechanism involves two silent transposons (cassettes) and a region where these cassettes can be expressed (cassette player).
- catabolite activator protein (CAP)** A protein that, when bound with cyclic AMP, can attach to sites on sugar-metabolizing operons to enhance transcription of these operons.
- catabolite repression** Repression of certain sugar-metabolizing operons in favor of glucose utilization when glucose is present in the environment of the cell.
- cdNA** See complementary DNA.
- cell cycle** The cycle of cell growth, replication of the genetic material, and nuclear and cytoplasmic division.
- cell-free system** A mixture of cytoplasmic components from cells, lacking nucleic acids and membranes. Used for in vitro protein synthesis and other purposes.
- cellular immunity** Immunity controlled by killer and helper T cells, which recognize infected body cells and either cause the infected cells to destroy internal invaders or destroy the infected cells directly.
- centimorgan** A chromosome-mapping unit. One centimorgan equals 1% recombinant offspring.

central dogma The original postulate of the way that information can be transferred between DNA, RNA, and protein, barring any transfer originating in protein.

centric fragment A piece of chromosome containing a centromere.

centrioles Cylindrical organelles, found in eukaryotes (except in higher plants), that reside in the centrosome. Also called basal bodies when they organize flagella or cilia.

centromere Constrictions in eukaryotic chromosomes on which the kinetochore lies. Also, the DNA sequence within the constriction that is responsible for appropriate function.

centromeric fission Creation of two chromosomes from one by splitting at the centromere.

centrosome The spindle-microtubule organizing center in eukaryotes except for those, such as fungi, that use spindle pole bodies to organize the spindles.

chaperone See molecular chaperone.

Chargaff's rule Chargaff's observation that in the base composition of DNA, the quantity of adenine equals the quantity of thymine, and the quantity of guanine equals the quantity of cytosine (equal purine and pyrimidine content).

Charon phages Phage lambda derivatives used as vehicles in DNA cloning.

checkpoint Used to describe points in the cell cycle that can be stopped if certain conditions are not met.

chemiluminescent techniques Techniques in which various molecules are made visible when exposed to ultraviolet or laser light if the molecules have a fluorescent tag.

chiasmata X-shaped configurations seen in tetrads during the later stages of prophase I of meiosis. They represent physical crossovers (singular: *chiasma*).

chimeras Individuals made up of two or more cell lines in which the cells originated in different zygotes. See mosaics.

chimeric plasmid Hybrid, or genetically mixed, plasmid used in DNA cloning.

chi site Sequence of DNA at which the RecBCD protein cleaves one of the DNA strands during recombination.

chi-square distribution The sampling distribution of the chi-square statistic. A family of curves whose shapes depend on degrees of freedom.

chloroplast The organelle that carries out photosynthesis and starch grain formation.

chromatids The subunits of a chromosome prior to anaphase of meiosis or mitosis. At anaphase of meiosis II or mitosis, when the sister chromatids separate, each chromatid becomes a chromosome.

chromatin The nucleoprotein material of the eukaryotic chromosome.

chromatin assembly factors Proteins involved in the construction of nucleosomes.

chromatin remodeling The change in the structure or positioning of nucleosomes, usually to allow transcription.

chromatosome The core nucleosome plus the H1 protein, a unit that includes approximately 168 base pairs of DNA.

chromomeres Dark regions of chromatin condensation in eukaryotic chromosomes at meiosis, mitosis, or endomitosis.

chromosomal painting A variant of the technique known as fluorescent in situ hybridization. Fluorescent dyes, attached to numerous nucleotide probes, give each human chromosome a different fluorescent signature.

chromosomal theory of inheritance

The theory that chromosomes are linear sequences of genes.

chromosome The form of the genetic material in viruses and cells. A circle of DNA in most prokaryotes; a DNA or an RNA molecule in viruses; a linear nucleoprotein complex in eukaryotes.

chromosome jumping A technique for isolating clones from a genomic library that are not contiguous but skip a region between known points on the chromosome. This is usually done to bypass regions that are difficult or impossible to "walk" through or regions known not to be of interest.

chromosome puffs Diffuse, uncoiled regions in polytene chromosomes where transcription is actively taking place.

chromosome walking A technique for studying segments of DNA, larger than can be individually cloned, by using overlapping cloned DNA.

cis Meaning "on the near side of"; refers to geometric configurations of atoms or mutants on the same chromosome.

cis-dominant Mutants (e.g., of an operator) that control the functioning of genes on the same piece of DNA.

cis-trans complementation test A mating test to determine whether two mutants on opposite chromosomes complement each other; a test for allelism.

cistron Term Benzer coined for the smallest genetic unit that exhibits the *cis-trans* position effect; synonymous with *gene*.

cladogenesis The evolutionary process whereby one species splits into two or more species. See anagenesis.

classical linkage map A chromosomal map, measured in centimorgans, based on genetic crosses to locate the relative distances between genes and their relative locations on chromosomes.

clinal selection Selection that changes gradually along a geographic gradient.

clonal evolution theory The theory that cancer develops from sequential changes (mutations) in the genome of a single cell.

clone A group of cells arising from a single ancestor.

coccus A spherical bacterium.

coding strand The DNA strand with the same sequence as the transcribed messenger RNA (given U in RNA and T in DNA). Compare with "anticoding strand."

codominance The relationship of alleles in a heterozygote that shows the individual expression of each allele in the phenotype.

codon preference The idea that for amino acids with several codons, one or a few are preferred and are used disproportionately. They would correspond with abundant transfer RNAs.

codons The sequences of three RNA or DNA nucleotides that specify either an amino acid or termination of translation.

coefficient of coincidence The number of observed double crossovers, divided by the number expected based on the independent occurrence of crossovers.

coefficient of relationship, *r* The proportion of alleles held in common by two related individuals.

cohesin Proteinaceous complex that holds sister chromatids together until anaphase of mitosis.

cointegrate A fusion of two elements. An intermediate structure in transposition.

colicinogenic factors See col plasmids.

col plasmids Plasmids that produce antibiotics (colicinogens) that the host uses to kill other strains of bacteria.

combinatorial control Transcriptional control in eukaryotes, which involves a large number of polypeptides, many of which recognize specific DNA sequences.

common ancestry The shared genetic inheritance of two individuals who are blood relatives. When two parents have a common ancestor, their offspring will be inbred.

compensasome The multisubunit dosage compensation complex in *Drosophila* with attendant RNAs.

competence factor A surface protein that binds extracellular DNA and enables the bacterial cell to be transformed.

complementarity The correspondence of DNA bases in the double helix so that adenine in one strand is opposite thymine in the other strand and cytosine in one strand is opposite guanine in the other. This relationship explains Chargaff's rule.

complementary DNA (cDNA) DNA synthesized by reverse transcriptase using RNA as a template.

complementation The production of the wild-type phenotype by a cell or an organism that contains two mutant genes. If complementation occurs, the mutants are almost certainly nonallelic.

complementation group Cistron (determined by the *cis-trans* complementation test).

complete medium A culture medium that is enriched to contain all of the growth requirements of a strain of organisms.

G-4

Glossary

- component of fitness** A particular aspect in the life cycle of an organism upon which natural selection acts.
- composite transposon** A transposon constructed of two IS elements flanking a control region that frequently contains host genes.
- concordance** The amount of phenotypic similarity between individuals.
- condensin** A protein complex including SMC (structural maintenance of chromosomes) proteins needed for the condensation of interphase chromosomes to mitotic chromosomes.
- conditional-lethal mutant** A mutant that is lethal under one condition but not lethal under another.
- confidence limits** A statistical term for a pair of numbers that predicts the range of values within which a particular parameter lies.
- conjugation** A process whereby two cells come in contact and exchange genetic material. In prokaryotes, the transfer is a one-way process.
- consanguineous** Meaning “between blood relatives”; usually refers to inbreeding or incestuous matings.
- consensus sequence** A sequence of the common nucleotides found in many different DNA or RNA samples of homologous regions (e.g., promoters).
- conservative replication** A postulated mode of DNA replication in which an intact double helix acts as a template for a new double helix; known to be incorrect.
- conserved sequence** A sequence found in many different DNA or RNA samples (e.g., promoters) that is invariant in the sample.
- constitutive heterochromatin** Heterochromatin that surrounds the centromere. *See* satellite DNA.
- constitutive mutant** A mutant whose transcription is no longer under regulatory control.
- contigs** Genomic libraries of overlapping, contiguous clones that cover complete regions of a chromosome.
- continuous replication** In DNA, uninterrupted replication in the 5' to 3' direction using a 3' to 5' template.
- continuous variation** Variation measured on a continuum rather than in discrete units or categories (e.g., height in human beings).
- corepressor** The metabolite that when bound to the repressor (of a repressible operon) forms a functional unit that can bind to its operator and block transcription.
- correlation coefficient** A statistic that gives a measure of how closely two variables are related.
- cosmid** A hybrid plasmid that contains *cos* sites at each end. *Cos* sites are recognized during head filling of lambda phages. Cosmids are useful for cloning segments of foreign DNA up to 50 kb.
- cotransduction** The simultaneous transduction of two or more genes.
- coupling** Allelic arrangement in which mutants are on the same chromosome and wild-type alleles are on the homologue.
- covariance** A statistical value measuring the simultaneous deviations of x and y variables from their means.
- CpG islands** Stretches of CG repeats (in which CpG indicates sequential bases on the same strand of DNA, rather than a C-G base pair). These repeats, found in imprinting centers, are important in regulation.
- crisscross pattern of inheritance** The phenotypic pattern of inheritance controlled by X-linked recessive alleles.
- critical chi-square** A chi-square value for a given degree of freedom and probability level, to which we can compare an experimental chi-square.
- crossbreed** To facilitate fertilization between separate individuals.
- cross-fertilization** *See* crossbreed.
- crossing over** A process in which homologous chromosomes exchange parts by a breakage-and-reunion process.
- crossovers** *See* chiasmata.
- crossover suppression** The apparent lack of crossing over within an inversion loop in heterozygotes. Usually due to mortality of zygotes carrying defective crossover chromosomes rather than to actual suppression.
- C-value paradox** Structural and junk DNA create large eukaryotic genomes and large differences in DNA content between eukaryotic species.
- cyclic AMP** A form of AMP (adenosine monophosphate) used frequently as a second messenger in eukaryotic hormone nets and in catabolite repression in prokaryotes.
- cyclin** Family of proteins involved in cell cycle control.
- cyclin-dependent kinase (CDK)** Family of kinases (phosphorylating enzymes) that, when combined with cyclin, are active in controlling checkpoints in the cell cycle.
- cyclosome** *See* anaphase-promoting complex (APC).
- cytogenetics** The study of cells from the perspective of genetics. In practice, the study of changes in the gross structure and number of chromosomes in cells.
- cytokinesis** The division of the cytoplasm of a cell into two daughter cells. *See* karyokinesis.
- cytoplasmic inheritance** Extrachromosomal inheritance controlled by nonnuclear genomes.
- cytosine** *See* pyrimidines.
- cytotoxic T lymphocytes** T cells responsible for attacking host cells that have been infected with an invading bacterium or virus.
- dauermodification** The persistence for several generations of an environmentally induced trait.
- degenerate code** A code in which several code words have the same meaning. The genetic code is degenerate because many different codons may specify the same amino acid.
- degrees of freedom** An estimate of the number of independent categories in a particular statistical test or experiment.
- deletion chromosome** A chromosome with part deleted.
- deme** A locally interbreeding population.
- denatured** Loss of natural configuration (of a molecule) through heat or other treatment. Denatured DNA is single-stranded.
- denominator elements** Genes on the autosomes of *Drosophila* that regulate the sex switch (*sxl*) to the off condition (maleness). Refers to the denominator of the X/A genic balance equation.
- density-gradient centrifugation** A method of separating molecular entities by their differential sedimentation in a centrifugal gradient.
- depauperate fauna** A fauna, especially common on islands, lacking many species found in similar habitats.
- derepressed** The condition of an operon that is transcribing because repressor control has been lifted.
- deterministic** Referring to events that have no random or probabilistic aspects but proceed in a fixed, predictable fashion.
- development** The process of orderly change an individual goes through in the formation of structure.
- diakinesis** The final stage of prophase I of meiosis.
- dicentric chromosome** A chromosome with two centromeres.
- dictyotene** A prolonged diplotene of primary oocytes that can last many years.
- dideoxy method** A method of DNA sequencing that uses chain-terminating (dideoxy) nucleotides.
- dihybrid** An organism heterozygous at two loci.
- dimerization** The chemical union of two similar molecules.
- diploid** Having each chromosome in two copies per nucleus or cell.
- diplotene (diplotene stage)** The stage of prophase of meiosis I in which chromatids appear to repel each other.
- directed mutation** A form of mutation that appears to respond to the needs of the cell but may, in fact, be due to the cell's hypermutable state under duress.
- directional selection** A type of selection that removes individuals from one end of a phenotypic distribution and thus causes a shift in the distribution.
- disassortative mating** The mating of two individuals with dissimilar phenotypes.

discontinuous replication In DNA, the replication in short 5' to 3' segments, using the 5' to 3' strand as a template while going backward, away from the replication fork.

discontinuous variation Variation that falls into discrete categories (e.g., the green and yellow color of garden peas).

discrete generations Generations that have no overlapping reproduction. All reproduction takes place between individuals of the same cohort.

dispersive replication A postulated mode of DNA replication combining aspects of conservative and semiconservative replication; known to be incorrect.

disruptive selection A type of selection that removes individuals from the center of a phenotypic distribution and thus causes the distribution to become bimodal.

D-loop Configuration found during DNA replication of chloroplast and mitochondrial chromosomes wherein the origin of replication is different on the two strands. The first structure formed is a displacement loop, or D-loop.

DNA cloning See gene cloning.

DNA-DNA hybridization The process of taking DNA from the same or different sources and heating and then cooling it, causing double helices to re-form at homologous regions. This technique is useful for determining sequence similarities and degrees of repetitiveness among DNAs.

DNA fingerprint A pattern of bands created on an electrophoretic gel of a DNA digest probed for a variable locus.

DNA glycosylases Endonucleases that initiate excision repair at the sites of various damaged or improper bases in DNA.

DNA gyrase A topoisomerase that relieves supercoiling in DNA by creating a transient break in the double helix.

DNA ligase An enzyme that closes nicks or discontinuities in one strand of double-stranded DNA by creating an ester bond between adjacent 3'-OH and 5'-PO₄ ends on the same strand.

DNA polymerase One of several classes of enzymes that polymerize DNA nucleotides using single-stranded DNA as a template.

DNA-RNA hybridization The process of heating and then cooling a mixture of DNA and RNA so that the RNA can hybridize (form a double helix) with DNA with a complementary nucleotide sequence.

docking protein Responsible for attaching (docking) a ribosome to a membrane by interacting with a signal particle attached to a ribosome destined to be membrane bound.

dominant An allele that expresses itself even when heterozygous. Also, the trait controlled by that allele.

dosage compensation A mechanism by which species with sex chromosomes ensure that one sex does not have differential activity of alleles on the sex chromosomes.

dot blotting A blotting technique, used on DNA already cloned, that eliminates the electrophoretic separation step.

Autoradiographs reveal dots rather than bands on a gel, indicating a probed sequence.

double digest The product formed when two different restriction endonucleases act on the same segment of DNA.

double helix The normal structural configuration of DNA consisting of two helices rotating about the same axis.

downstream A convention on DNA related to the position and direction of transcription by RNA polymerase (5'→3'). Downstream (in the 3' direction) is in the direction of transcription, whereas upstream (in the 5' direction) is in the direction from which the polymerase has come.

downstream promoter element (DPE)

A consensus sequence at about +28 to +34 of RNA polymerase II promoters that have initiator elements but not TATA boxes.

dyad Two sister chromatids attached to the same centromere.

dynein Microtubule motor protein.

dysplasia Excessive cell growth that involves pathological changes to the cells and their nuclei.

electrophoresis The separation of molecular entities by electric current.

electroporation A technique for transfecting cells by applying a high-voltage electric pulse.

elongation complex The form of RNA polymerase II that actively carries out basal transcription.

elongation factors (EF-Ts, EF-Tu, EF-G)

Proteins necessary for the proper elongation and translocation processes during translation at the ribosome in prokaryotes.

Replaced by eEF1 α and eEF1 β in eukaryotes.

endogenote Bacterial host chromosome.

endomitosis Chromosomal replication without nuclear or cellular division that results in cells with many copies of the same chromosome, such as in the salivary glands of *Drosophila*.

endonucleases Enzymes that make nicks internally in the backbone of a polynucleotide. They hydrolyze internal phosphodiester bonds.

enhancer A eukaryotic DNA sequence that increases transcription of a gene by binding specific transcription factors.

enriched medium See complete medium.

enzyme Protein catalyst.

epigenetic effect An environmentally induced change in the genetic material that does not cause a change in base pairs.

Generally, a phenomenon of differential expression of alleles of a locus depending on the parent of origin. Also applied to an effect in proteins.

epistasis The masking of the action of alleles of one gene by allelic combinations of another gene.

equational division A division, such as the second meiotic division, that does not reduce chromosomal numbers.

E (exit) site Site on the ribosome that depleted transfer RNAs pass through during ejection.

euchromatin Regions of eukaryotic chromosomes that are diffuse during interphase. Presumably the actively transcribing DNA of the chromosomes.

eugenics A social movement designed to improve humanity by encouraging those with beneficial traits to breed and discouraging those with undesirable traits from breeding.

eukaryotes Organisms with true nuclei.

euploidy The condition of a cell or organism that has one or more complete sets of chromosomes.

evolution A change in phenotypic frequencies in a population.

evolutionary rates The rate of divergence between taxonomic groups, measurable as number of amino acid substitutions per million years.

excision repair A process whereby cells remove part of a damaged DNA strand and replace it through DNA synthesis, using the undamaged strand as a template.

exconjugant Each of the two cells that separate after conjugation has taken place.

exogenote DNA that a bacterial cell has taken up through one of its sexual processes.

exon In a gene that has intervening sequences (introns), a region that is actually exported from the nucleus to be expressed or become part of a transfer or ribosomal RNA.

exon shuffling The hypothesis put forward by Walter Gilbert that exons code for the functional units of a protein, and that the evolution of new genes proceeds by recombination or the exclusion of exons.

exonucleases Enzymes that digest nucleotides from the ends of polynucleotide molecules. They hydrolyze phosphodiester bonds of terminal nucleotides.

experimental design A branch of statistics that attempts to outline the way in which experiments should be carried out so the data gathered has statistical value.

expression vector A hybrid vector (plasmid) that expresses its cloned genes.

expressivity The degree of expression of a genetically controlled trait.

F₁ See filial generation.

factorial The product of all integers from the specified number down to one (unity).

Fanconi's anemia A disease in human beings with a syndrome of congenital malformations; associated with various cancers.

fate map A map of the developmental fate of a zygote or early embryo showing the adult organs that will develop from a given position on the zygote or early embryo.

G-6

Glossary

F-duction See sexduction.

fecundity selection The forces causing one genotype to be more fertile than another genotype.

feedback inhibition A posttranslational control mechanism in which the end product of a biochemical pathway inhibits the activity of the first enzyme of the same pathway.

fertility factor The plasmid that allows a prokaryote to engage in conjugation with, and pass DNA into, an F^- cell.

F factor See fertility factor.

filial generation Offspring generation. F_1 is the first offspring, or filial, generation; F_2 is the second; and so on.

fimbriae See pili.

first-division segregation (FDS) The allelic arrangement of ordered spores that indicates the lack of recombination between a locus and its centromere.

fitness, W The relative reproductive success of a genotype as measured by survival, fecundity, or other life-history parameters.

5' untranslated region (5' UTR) See leader.

floral meristem The shoot apical meristem sets aside this tissue that gives rise to flowers.

floral-meristem identity genes At least five genes known to establish the identity of the floral meristem.

fluctuation test An experiment by Luria and Delbrück that compared the variance in number of mutations among small cultures with that among subsamples of a large culture to determine the mechanism of inherited change in bacteria.

fluorescent in situ hybridization (FISH) A technique in which a fluorescent dye is attached to a nucleotide probe that then binds to a specific site on a chromosome and makes itself visible by its fluorescence.

Fokker-Planck equation An equation that describes diffusion processes. It is used by population geneticists to describe random genetic drift.

footprinting A technique to determine the length of nucleic acid in contact with a protein. While in contact, the free DNA is digested. The remaining DNA is then isolated and characterized.

founder effect Genetic drift observed in a population founded by a small, nonrepresentative sample of a larger population.

F-pili Sex pili. Hairlike projections on an F^+ or Hfr bacterium involved in anchorage during conjugation.

fragile site A chromosomal region that has a tendency to break.

fragile-X syndrome The most common form of inherited mental retardation. Named for its association with an X chromosome with a tip that breaks or appears uncondensed. Inheritance involves imprinting.

frameshift A mutation in which there is an addition or deletion of nucleotides that causes the codon reading frame to shift.

free energy of activation ($\Delta G^\ddagger$) Energy needed to initiate a chemical reaction.

frequency-dependent selection A selection whereby a genotype is at an advantage when rare and at a disadvantage when common.

functional alleles Mutations that fail to complement each other in a *cis-trans* complementation test.

fundamental number (NF) The number of chromosome arms in a somatic cell of a particular species.

gamete A germ cell having a haploid chromosomal complement. Gametes from parents of opposite sexes fuse to form zygotes.

gametic selection The forces acting to cause differential reproductive success of one allele over another in a heterozygote.

gametophyte The haploid stage of a plant life cycle that produces gametes (by mitosis). It alternates with a diploid, sporophyte generation.

G-bands Eukaryotic chromosomal bands produced by treatment with Giemsa stain.

gene Inherited determinant of the phenotype. See *cistron*; *locus*.

gene amplification A process or processes by which the cell increases the number of repeats of a particular gene within the genome.

gene cloning Production of large numbers of a piece of DNA after that piece of DNA is inserted into a vector and taken up by a cell. Cloning occurs as the vector replicates.

gene conversion In Ascomycete fungi, a 2:2 ratio of alleles is expected after meiosis, yet a 3:1 ratio is sometimes observed. The gene conversion mechanism is explained by repair of heteroduplex DNA produced by recombination.

gene family A group of genes that has arisen by duplication of an ancestral gene. The genes in the family may or may not have diverged.

gene flow The movement of genes from one population to another by interbreeding between individuals in the two populations.

gene pool All of the alleles available among the reproductive members of a population from which gametes can be drawn.

generalized transduction Form of transduction in which any region of the host genome can be transduced. See *specialized transduction*.

general transcription factors Eukaryotic proteins that form part of the RNA polymerase holoenzymes.

genetic code The linear sequences of nucleotides that specify the amino acids during the process of translation at the ribosome.

genetic engineering Popular term for recombinant DNA technology. See *recombinant DNA technology*.

genetic load The relative decrease in the mean fitness of a population due to the presence of genotypes that have less than the highest fitness.

genetic polymorphism The occurrence together in the same population of more than one allele at the same locus, with the least frequent allele occurring more frequently than can be accounted for by mutation.

genic balance theory Bridges's theory that the sex of a fruit fly is determined by the relative number of X chromosomes and autosomal sets.

genome The entire genetic complement of a prokaryote or virus or the haploid genetic complement of a eukaryote.

genomic equivalence The concept that differentiated cells in a eukaryotic organism have identical genetic contents.

genomic library A set of cloned fragments making up the entire genome of an organism or species.

genomics The study of the mapping and sequencing of genomes. Bioinformatics is the science of mining the data from these DNA sequences obtained from sequencing.

genophore The chromosome (genetic material) of prokaryotes and viruses.

genotype The genes that an organism possesses.

Giemsa stain A complex of stains specific for the phosphate groups of DNA.

Goldstein-Hogness box See *TATA box*.

green fluorescent protein A reporter system that uses the gene from a jellyfish that specifies a protein that fluoresces green when ultraviolet light is shined on it, indicating the success of a transfection experiment.

group I introns Self-splicing introns that require a guanine-containing nucleotide for splicing; the intron is released in a linear form.

group II introns Self-splicing introns that do not require an external nucleotide for splicing; the intron is released in a lariat form.

group selection Selection for traits that would be beneficial to a population at the expense of the individual possessing the trait.

G-tetraplex A structure of four guanines that can base pair to form a planar structure that may be involved in novel structures at the end of eukaryotic chromosomes.

guanine See *purines*.

guide RNA (gRNA) RNA that guides the insertion of uridines (RNA editing) into messenger RNAs in trypanosomes. Found in transcripts from minicircles and maxicircles of DNA in kinetoplasts.

gynandromorphs Mosaic individuals having simultaneous aspects of both the male and the female phenotype.

hammerhead ribozyme A catalytic RNA, shaped like a hammerhead, capable of splitting other RNA molecules with appropriate complementary sequences.

haplodiploidy The sex-determining mechanism found in some insect groups among which males are haploid and females are diploid.

haploid The state of having one copy of each chromosome per nucleus or cell.

HAT medium A selection medium for hybrid cell lines; contains hypoxanthine, aminopterin, and thymidine. HPRT⁺ TK⁺ cell lines can survive in this medium.

heat shock proteins Proteins that appear in a cell after the cell has been subjected to elevated temperatures.

helicase A protein that unwinds DNA, usually at replicating Yjunctions.

helix-turn-helix motif Configuration found in DNA-binding proteins consisting of a recognition helix and a stabilizing helix, separated by a short turn.

hemizygous The condition of loci present in only one copy in a diploid organism, such as loci on the X chromosome of the heterogametic sex of a diploid species.

heritability A measure of the degree to which the variance in the distribution of a phenotype is due to genetic causes. In the broad sense, it is measured by the total genetic variance divided by the total phenotypic variance. In the narrow sense, it is measured by the genetic variance due to additive genes divided by the total phenotypic variance.

hermaphrodite An individual with both male and female genitalia.

heterochromatin Chromatin that remains tightly coiled (and darkly staining) throughout the cell cycle.

heteroduplex DNA See hybrid DNA.

heterogametic The sex with heteromorphic sex chromosomes; during meiosis, it produces different kinds of gametes in accordance with these sex chromosomes.

heterogeneous nuclear mRNA (hnRNA) The original RNA transcripts found in eukaryotic nuclei before posttranscriptional modifications.

heterokaryon A cell that contains two or more nuclei from different origins.

heteromorphic chromosome pair Members of a homologous pair of chromosomes that are not morphologically identical (e.g., the sex chromosomes).

heteroplasmy The existence within an organism of genetic heterogeneity within the populations of mitochondria or chloroplasts.

heterothallic A botanical term used for organisms in which the two sexes reside in different individuals.

heterotrophs Organisms that require an organic form of carbon as a carbon source.

heterozygote A diploid or polyploid with different alleles at a particular locus.

heterozygote advantage A selection model in which heterozygotes have the highest fitness.

heterozygous DNA See hybrid DNA.

Hfr High frequency of recombination. A strain of bacteria that has incorporated an F factor into its chromosome and can then transfer the chromosome during conjugation.

histone acetyl transferases (HATs)

Proteins that remodel chromatin by acetylating histones.

histones Arginine- and lysine-rich basic proteins making up a substantial portion of eukaryotic nucleoprotein.

hnRNA See heterogeneous nuclear mRNA.

Hogness box See TATA box.

holandric trait Trait controlled by a locus found only on the Y chromosome. Involves father-to-son transmission.

Holliday junction A junction point between two cross-linked DNA double helices. It is an intermediate stage in DNA recombination.

holoenzyme The complete enzyme, including all subunits. Often used in reference to RNA and DNA polymerases.

homeo box A consensus sequence of about 180 base pairs discovered in homeotic genes in *Drosophila*. Also found in other developmentally important genes from yeast to human beings.

homeo domain The sixty amino acid polypeptide translated from the homeo box.

homeotic gene Gene that controls the developmental fate of a cell type; mutations of the homeotic gene cause one cell type to follow the developmental pathway of another cell type.

homogametic The sex with homomorphic sex chromosomes; it produces only one kind of gamete in regard to the sex chromosomes.

homologous chromosomes Members of a pair of essentially identical chromosomes that synapse during meiosis.

homologous recombination Breakage and reunion between homologous lengths of DNA mediated by RecA and RecBCD.

homomorphic chromosome pairs Morphologically identical members of a homologous pair of chromosomes.

homoplasmy The existence within an organism of only one type of plastid; usually referring to the genetic identity of mitochondria or chloroplasts.

homothallic A botanical term used for groups whose individuals are not of different sexes.

homozygote A diploid or a polyploid with identical alleles at a locus.

humoral immunity Immunity due to antibodies in the serum and lymph.

H-Y antigen The histocompatibility Yantigen, a protein found on the cell surfaces of male mammals.

hybrid Offspring of unlike parents.

hybrid DNA DNA whose two strands have different origins.

hybridoma A cell resulting from the fusion of a spleen cell and a multiple myeloma cell. These cells can be maintained indefinitely in cell culture, in which they produce monoclonal antibodies.

hybrid plasmid A plasmid that contains an inserted piece of foreign DNA.

hybrid vector See hybrid vehicle.

hybrid vehicle A plasmid or phage containing an inserted piece of foreign DNA.

hybrid zone Geographical region in which previously isolated populations that have evolved differences come into contact and form hybrids.

hyperplasia Excessive cell growth that does not involve pathological changes to the cells.

hypervariable loci Loci with many alleles; especially those whose variation is due to variable numbers of tandem repeats.

hypostatic gene A gene whose expression is masked by an epistatic gene.

identity by descent The state of two alleles when they are identical copies of the same ancestral allele (autozygous).

idiogram A photograph or diagram of the chromosomes of a cell arranged in an orderly fashion. See karyotype.

idiotypic variation Variation in the variable parts of immunoglobulin genes.

idling reaction The production of guanosine tetraphosphate (3'-ppGpp-5') by the stringent factor when a ribosome encounters an uncharged transfer RNA in the A site.

immunity The ability of an organism to resist infection.

immunoglobulins (Igs) Specific proteins produced by derivatives of B lymphocytes that protect an organism from antigens.

imprinting See molecular imprinting.

imprinting center (IC) A region responsible for the control of imprinting. The imprinting mark is almost certainly DNA methylation, which is able to turn off gene transcription.

inbreeding The mating of genetically related individuals.

inbreeding coefficient, F The probability of autozygosity.

inbreeding depression A depression of vigor or yield due to inbreeding.

incestuous A mating between blood relatives who are more closely related than the law of the land allows.

inclusive fitness The expansion of the concept of the fitness of a genotype to include benefits accrued to relatives of an individual since relatives share parts of their genomes. An apparently altruistic act toward a relative may thus enhance the fitness of the individual performing the act.

G-8

Glossary

incomplete dominance The situation in which both alleles of the heterozygote influence the phenotype. The phenotype is usually intermediate between the two homozygous forms.

independent assortment, rule of Mendel's second rule, describing the independent segregation of alleles of different loci.

inducible system A system (a coordinated group of enzymes involved in a catabolic pathway) is inducible if the metabolite it works upon causes transcription of the genes controlling these enzymes. These systems are primarily prokaryotic operons.

induction In regard to temperate phages, the process of causing a prophage to become virulent.

informatics See bioinformatics.

initiation codon The messenger RNA sequence AUG, which specifies methionine, the first amino acid used in the translation process. (Occasionally, GUG is recognized as an initiation codon.)

initiation complex The complex formed for initiation of translation. It consists of the 30S ribosomal subunit, messenger RNA, N-formyl-methionine transfer RNA, and three initiation factors.

initiation factors (IF1, IF2, IF3) Proteins (prokaryotic with eukaryotic analogues) required for the proper initiation of translation.

initiator element (inr) A CT-rich area found in RNA polymerase II promoters without TATA boxes.

initiator proteins Proteins that recognize the origin of replication on a replicon and take part in primosome construction.

insertion mutagenesis Change in gene action due to an insertion event that either changes a gene directly or disrupts control mechanisms.

insertion sequences (IS) Small, simple transposons. See transposable genetic element.

inside marker The middle locus of three linked loci.

intercalary heterochromatin

Heterochromatin, other than centromeric heterochromatin, dispersed throughout eukaryotic chromosomes.

intergenic suppression A mutation at a second locus that apparently restores the wild-type phenotype to a mutant at a first locus.

interkinesis The abbreviated interphase that occurs between meiosis I and II. No DNA replication occurs here.

internal ribosome entry site Sequence in eukaryotic messenger RNAs that allows ribosomes to initiate translation at a point other than the 5' cap.

interphase The metabolically active, nondividing stage of the cell cycle.

interpolar microtubules Microtubules extending from one pole of the spindle and overlapping spindle fibers from the other pole, but not in contact with kinetochores.

interrupted mating A mapping technique in which bacterial conjugation is disrupted after specified time intervals.

intersex An organism with external sexual characteristics of both sexes.

intervening sequences (introns) DNA sequences within a gene that are transcribed but removed prior to translation.

intra-allelic complementation The restoration of activity in an enzyme made of subunits in a heterozygote of two mutants that, when homozygous, are not active. Caused by the interaction of the subunits in the protein.

intragenic suppression A second change within a mutant gene that results in an apparent restoration of the original phenotype.

intron See intervening sequences.

inversion The replacement of a section of a chromosome in the reverse orientation.

inverted repeat sequence A nucleotide sequence read in opposite orientations on the same double helix.

in vitro Biological or chemical work done in the test tube (literally, "in glass") rather than in living systems.

IS elements See insertion sequences.

isochromosome A chromosome with two genetically and morphologically identical arms.

isozymes Different electrophoretic forms of the same enzyme. Unlike allozymes, isozymes are due to differing subunit configurations rather than allelic differences.

junctional diversity Variability in immunoglobulins caused by variation in the exact crossover point during V-J, V-D, and D-J joining.

kappa particles The bacteriophage particles that give a *Paramecium* the killer phenotype.

karyokinesis The process of nuclear division. See cytokinesis.

karyotype The chromosome complement of a cell. See idiogram.

kinesin Microtubule motor protein.

kinetochore The chromosomal attachment point for the spindle fibers, located on the centromere.

kinetochore microtubules Microtubules radiating from the centrosome and attached to kinetochores of chromosomes during mitosis and meiosis.

kin selection The mode of natural selection that acts on an individual's inclusive fitness.

Klenow fragment Proteolytic fragment—obtained by treatment with a protease, or protein-cleaving enzyme—of *E. coli* DNA polymerase I with both 5'→3' polymerase activity and 3'→5' exonuclease activity. (It has been studied extensively because it has been easier for X-ray crystallographers and biochemists to work with this fragment than with the whole enzyme.)

knockout mice Transgenic mice that have been made homozygous for a nonfunctioning allele at a particular locus.

lac operon The inducible operon, including three loci involved in the uptake and breakdown of lactose.

ladder gel See stepladder gel.

lagging strand Strand of DNA being replicated discontinuously.

lampbrush chromosomes Chromosomes of amphibian oocytes having loops that are suggestive of a lampbrush.

leader The length of messenger RNA from the 5' end to the initiation codon, AUG.

leader peptide gene A small gene within the attenuator control region of a repressible amino acid operon. Translation of the gene tests the concentration of amino acids in the cell.

leader transcript The messenger RNA transcribed by the attenuator region of a repressible amino acid operon. The transcript is capable of several alternative stem-loop structures, depending on the translation of a short leader peptide gene.

leading strand Strand of DNA being replicated continuously.

leptonema (leptotene stage) The first stage of prophase I of meiosis, in which chromosomes become distinct.

lethal-equivalent alleles Alleles whose summed effect is that of lethality—for example, four alleles, each of which would be lethal 25% of the time (or to 25% of their bearers), are equivalent to one lethal allele.

leucine zipper Configuration of a DNA-binding protein in which leucine residues on two helices interdigitate, in zipper fashion, to stabilize the protein.

leukemia Cancer of the bone marrow resulting in excess production of leukocytes.

level of significance The probability value in statistics used to reject the null hypothesis.

linkage The association of loci on the same chromosome.

linkage disequilibrium The condition among alleles at different loci such that allelic combinations in a gamete do not follow the product rule of probability.

linkage equilibrium The condition among alleles at different loci such that any allelic combination in a gamete occurs as the product of the frequencies of each allele at its own locus.

linkage groups Associations of loci on the same chromosome. In a species, there are as many linkage groups as there are homologous pairs of chromosomes.

linkage number The number of times one strand of a helix coils about the other.

linker A small segment of DNA that contains a restriction site. It can be added to blunt-ended DNA to give that DNA a particular restriction site for cloning.

liposomes Transfecting DNA is delivered to target cells by way of these membrane-bound vesicles.

locus The position of a gene on a chromosome (plural: *loci*).

lod score method A technique (*logarithmic odds*) for determining the most likely recombination frequency between two loci from pedigree data.

long interspersed elements (LINEs) Sequences of DNA, up to seven thousand base pairs in length, interspersed in eukaryotic chromosomes in many copies.

lymphoma Cancer of the lymph nodes and spleen that causes excessive production of lymphocytes.

Lyon hypothesis The hypothesis that suggests that the Barr body is an inactivated X chromosome.

lysate The contents released from a lysed cell.

lysis The breaking open of a cell by the destruction of its wall or membrane.

lysogenic The state of a bacterial cell that has an integrated phage (prophage) in its chromosome.

major histocompatibility complex A group of highly polymorphic genes whose products appear on the surfaces of cells, imparting to them the property of "self" (belonging to that organism). Some other functions are also involved.

mapping The process of locating the positions of genes on chromosomes.

mapping function The mathematical relationship between measured map distance in a given experiment and the actual recombination frequency.

map unit The distance equal to 1% recombination between two loci.

mate-killer A *Paramecium* phenotype induced by intracellular bacterial-like mu particles.

maternal effect The effect of the maternal parent's genotype on the phenotype of her offspring.

maternal-effect gene A gene expressed in maternal tissue that influences a developing embryo.

mating type In many species of microorganisms, individuals can be divided into two mating types. Mating can take place only between individuals of opposite mating types due to the interaction of cell surface components.

maturation-promoting factor (MPF)

A protein complex of cyclin B and p34^{cdc2} that initiates mitosis during the cell cycle.

Also called the mitosis-promoting factor.

mean The arithmetic average; the sum of the data divided by the sample size.

mean fitness of the population, $\bar{W}$ The sum of the fitnesses of the genotypes of a population weighted by their proportions; hence, a weighted mean fitness.

meiosis The nuclear process in diploid eukaryotes that results in gametes or spores with only one member of each original homologous pair of chromosomes per nucleus.

meiotic drive See gametic selection.

merozygote A bacterial cell having a second copy of a particular chromosomal region in the form of an exogenote.

messenger RNA (mRNA) A complementary copy of a gene that is translated into a polypeptide at the ribosome.

metacentric chromosome A chromosome whose centromere is located in the middle.

metafemale A fruit fly with an X/A ratio greater than unity.

metagon An RNA necessary for the maintenance of mu particles in *Paramecium*.

metamale A fruit fly with X/A ratio below 0.5.

metaphase The stage of mitosis or meiosis in which spindle fibers are attached to kinetochores, and the chromosomes are positioned in the equatorial plane of the cell.

metaphase plate The plane of the equator of the spindle into which chromosomes are positioned during metaphase.

metastasis The migration of cancerous cells to other parts of the body.

metrical variation See continuous variation.

microsatellite DNA Repeats of very short sequences of DNA, such as CACACACA, dispersed throughout the eukaryotic genome. The loci can be studied by polymerase chain reaction amplification.

microtubule organizing center Active center from which microtubules are organized. The spindle is organized by the centrosome, which may or may not contain a centriole.

microtubules Hollow cylinders made of the protein tubulin (α and β subunits) that make up, among other things, the spindle fibers.

mimicry A phenomenon in which an individual gains an advantage by looking like the individuals of a different species.

minimal medium A culture medium for microorganisms that contains the minimal necessities for growth of the wild-type.

mismatch repair A form of excision repair initiated at the sites of mismatched bases in DNA.

missense mutation Mutations that change a codon for one amino acid into a codon for a different amino acid.

mitochondrion The eukaryotic cellular organelle in which the Krebs cycle and electron transport reactions take place.

mitosis The nuclear division producing two daughter nuclei identical to the original nucleus.

mitosis-promoting factor See maturation-promoting factor (MPF).

mitotic apparatus See spindle.

mixed families Groups of four codons sharing their first two bases and coding for more than one amino acid.

modern linkage map A chromosomal map based on the positions of RFLP markers along its length.

molecular chaperone A protein that aids in the folding of a second protein. The chaperone prevents proteins from forming structures that would be inactive.

molecular evolutionary clock

A measurement of evolutionary time in nucleotide substitutions per year.

molecular imprinting The phenomenon in which there is differential expression of a gene depending on whether it was maternally or paternally inherited.

molecular mimicry The situation in which one type of molecule resembles another type in order to function. For example, the prokaryotic ribosomal release factors, RF1 and RF2, mimic the structure of a transfer RNA.

monocistronic Usually referring to a messenger RNA that carries the information for only one gene (cistron).

monoclonal antibody The antibody from a clone of cells producing the same antibody. An individual with multiple myeloma usually produces monoclonal antibodies.

monohybrids Offspring of parents that differ in only one genetic characteristic. Usually implies heterozygosity at a single locus under study.

monosomic A diploid cell missing a single chromosome.

monovalent A single chromosome composed of two sister chromatids. Equivalent to a dyad.

morphogen A substance transported into or produced in a developing embryo that diffuses to form a gradient that helps determine cell differentiation.

morphological species concept The idea that organisms are classified in the same species if they appear similar.

mosaicism The condition of being a mosaic. See mosaics.

mosaics Individuals made up of two or more cell lines in which the cells originated in the same zygote.

mRNA See messenger RNA (mRNA).

Müllerian mimicry A form of mimicry in which noxious species evolve to resemble each other.

multihybrid An organism heterozygous at numerous loci.

G-10

Glossary

- multinomial expansion** The terms generated when a multinomial is raised to a power.
- mu particles** Bacteriophage particles found in the cytoplasm of *Paramecium* that impart the mate-killer phenotype.
- mutability** The ability to change.
- mutants** Alternative phenotypes to the wild-type; the phenotypes produced by alternative alleles.
- mutation** The process by which a gene or chromosome changes structurally; the end result of this process.
- mutation rate** The proportion of mutations per cell division in bacteria or single-celled organisms or the proportion of mutations per gamete in higher organisms.
- mutator mutations** Mutations of DNA polymerase that increase the overall mutation rate of a cell or of an organism.
- muton** A term Benzer coined for the smallest mutable site within a cistron.
- natural selection** The process in nature whereby one genotype leaves more offspring than another genotype because of superior life history attributes such as survival or fecundity.
- negative interference** The phenomenon whereby a crossover in a particular region facilitates the occurrence of other apparent crossovers in the same region of the chromosome.
- N-end rule** The life span of a protein is determined by its amino-terminal (N-terminal) amino acid.
- neo-Darwinism** The merger of classical Darwinian evolution with population genetics.
- neoplasm** New growth of abnormal tissue.
- neutral gene hypothesis** The hypothesis that most genetic variation in natural populations is not maintained by natural selection.
- NF** See fundamental number.
- nickase** See DNA gyrase.
- noncoding strand** See anticoding strand.
- nondisjunction** The failure of a pair of homologous chromosomes to separate properly during meiosis.
- nonhistone proteins** The proteins remaining in chromatin after the histones are removed.
- nonparental ditype (NPD)** A spore arrangement in Ascomycete fungi that contains only the two recombinant-type ascospores (assuming two segregating loci).
- nonparentals** See recombinants.
- nonrecombinants** In mapping studies, offspring that have alleles arranged as in the original parents.
- nonsense codon** One of the messenger RNA sequences (UAA, UAG, UGA) that signals the termination of translation.
- nonsense mutations** Mutations that change a codon for an amino acid into a nonsense codon.
- normal distribution** Any of a family of bell-shaped frequency curves whose relative positions and shapes are defined on the basis of the mean and standard deviation.
- northern blotting** A gel transfer technique used for RNA. See Southern blotting.
- N segments** Sequences of nucleotides added in a template-free fashion at the joining junctions of heavy-chain antibody genes.
- nuclease-hypersensitive site** A region of a eukaryotic chromosome that is specifically vulnerable to nuclease attack because it is not wrapped as nucleosomes.
- nucleolar organizer** The chromosomal region around which the nucleolus forms; site of tandem repeats of the major ribosomal RNA gene.
- nucleolus** The globular, nuclear organelle formed at the nucleolar organizer. Site of ribosome construction.
- nucleoprotein** The substance of eukaryotic chromosomes consisting of proteins and nucleic acids.
- nucleoside** A sugar-base compound that is a nucleotide precursor. Nucleotides are nucleoside phosphates.
- nucleosomes** Arrangements of DNA and histones forming regular spherical structures in eukaryotic chromatin.
- nucleotide** Subunits that polymerize into nucleic acids (DNA or RNA). Each nucleotide consists of a nitrogenous base, a sugar, and one or more phosphate groups.
- nucleotide excision repair** The DNA excision repair mechanism responsible for repairing thymine dimers and other lesions. Enzymes excise a short segment of one of the DNA strands and then repair and ligate the DNA.
- null hypothesis** The statistical hypothesis that there are no differences between observed data and those data expected based on the assumption of no experimental effect.
- nullisomic** A diploid cell missing both copies of the same chromosome.
- numerator elements** Genes on the X chromosome in *Drosophila* that regulate the sex switch gene (*sex*) to the on condition (femaleness). Refers to the numerator of the X/A genic balance equation.
- nutritional-requirement mutants** See auxotrophs.
- Okazaki fragments** Segments of newly replicated DNA produced during discontinuous DNA replication.
- oncogene** Genes capable of transforming a cell. They are found in the active state in retroviruses and transformed cells and in the inactive state in nontransformed cells, in which they are called proto-oncogenes.
- one-gene-one-enzyme hypothesis** Hypothesis of Beadle and Tatum that states that one gene controls the production of one enzyme. Later modified to the concept that one cistron controls the production of one polypeptide.
- oogenesis** The process of ovum formation in female animals.
- oogonia** Cells in females that produce primary oocytes by mitosis.
- open reading frames (ORFs)** Sequence of codons between the initiation and termination codons in a gene.
- operator** A DNA sequence recognized by a repressor protein or repressor-corepressor complex. When the operator is complexed with the repressor, transcription is prevented.
- operon** A sequence of adjacent genes all under the transcriptional control of the same operator.
- origin recognition complex (ORC)** A complex of six proteins that bind to the eukaryotic autonomously replicating sequences (ARS). Needed for the initiation of DNA replication in concert with other proteins.
- outbreeding** The mating of genetically unrelated individuals.
- ovum** Egg. The one functional product of each meiosis in female animals.
- pachynema (pachytene stage)** The stage of prophase I of meiosis in which chromatids are first distinctly visible.
- palindrome** A sequence of words, phrases, or nucleotides that reads the same regardless of the direction from which one starts; the sites of recognition of type II restriction endonucleases.
- panmictic** Referring to unstructured (random mating) populations.
- paracentric inversion** A chromosomal inversion that does not include the centromere.
- paramycin** A toxin liberated by a "killer" *Paramecium*.
- parameters** Measurements of attributes of a population; denoted by Greek letters.
- parapatric speciation** Speciation in which reproductive isolating mechanisms evolve when a population enters a new niche or habitat within the range of the parent species.
- parasegment** The first series of segments that form in a developing insect embryo; they form after about 5.5 hours in the developing *Drosophila* embryo.
- parental ditype (PD)** A spore arrangement in Ascomycete fungi that contains only the two nonrecombinant-type ascospores.
- parental imprinting** See molecular imprinting.
- parentals** See nonrecombinants.
- parthenogenesis** The development of an individual from an unfertilized egg that did not arise by meiotic chromosomal reduction.
- partial digest** A restriction digest that has not been allowed to go to completion, and thus contains pieces of DNA that have restriction endonuclease sites that have not been cleaved.

partial dominance See incomplete dominance.

Pascal's triangle A triangular array of numbers made up of the coefficients of the binomial expansion.

path diagram A modified pedigree showing only the direct line of descent from common ancestors.

pedigree A representation of the ancestry of an individual or family; a family tree.

penetrance The normal appearance of genetically controlled traits in the phenotype.

peptidyl transferase The enzymatic center responsible for peptide bond formation during translation at the ribosome.

pericentric inversion A chromosomal inversion that includes the centromere.

permissive temperature A temperature at which temperature-sensitive mutants are normal.

PEST hypothesis Degradation of a protein in less than two hours is signaled by a region within the protein rich in proline (P), glutamic acid (E), serine (S), or threonine (T).

petite mutations Mutations of yeast that produce small colonies, like those grown under anaerobic conditions.

phages See bacteriophages.

phenocopy A phenotype that is not genetically controlled but looks like a genetically controlled phenotype.

phenotype The observable attributes of an organism.

pheromone A chemical signal, analogous to a hormone, that passes information between individuals.

phosphodiester bond A diester bond linking two nucleotides together (between phosphoric acid and sugars) to form the nucleotide polymers DNA and RNA.

photocrosslinking A technique used to determine which moieties (proteins, DNA) are in close proximity during a particular process.

photoreactivation The process whereby dimerized pyrimidines (usually thymines) in DNA are restored by an enzyme (deoxyribodipyrimidine photolyase) that requires light energy.

phyletic evolution See anagenesis.

phyletic gradualism The process of gradual evolutionary change over time.

phylogenetic tree A diagram showing evolutionary lineages of organisms.

physical map Chromosomal map in which distances are in physical units of base pairs. These maps can be of microsatellite markers or of sequence-tagged sites.

pili (fimbriae) Hairlike projections on the surface of bacteria; Latin for "hair."

plaques Clear areas on a bacterial lawn caused by cell lysis due to viral attack.

plasmid An autonomous, self-replicating genetic particle, usually of double-stranded DNA.

plastid A chloroplast prior to the development of chlorophyll.

pleiotropy The phenomenon whereby a single mutant affects several apparently unrelated aspects of the phenotype.

point centromere The type of centromere, such as that found in *Saccharomyces cerevisiae*, that has defined sequences large enough to accommodate one spindle microtubule.

point mutations Small mutations that consist of a replacement, addition, or deletion of one or a few bases.

polar bodies The small cells that are the by-products of meiosis in female animals. One functional ovum and as many as three polar bodies result from meiosis of each primary oocyte.

polarity Meaning "directionality" and referring either to an effect seen in only one direction from a point of origin or to the fact that linear entities (such as a single strand of DNA) have ends that differ from each other.

polar mutant An organism with a mutation, usually within an operon, that prevents the expression of genes distal to itself.

pollen grain The male gametophyte in higher plants.

poly-A tail A sequence of adenosine nucleotides added to the 3' end of eukaryotic messenger RNAs.

polycistronic Referring to prokaryotic messenger RNAs that contain several genes within the same messenger RNA transcript.

polygenic inheritance See quantitative inheritance.

polymerase chain reaction (PCR) A method to amplify DNA segments rapidly in temperature-controlled cycles of denaturation, primer binding, and replication.

polymerase cycling The process by which a DNA polymerase III enzyme completes an Okazaki fragment, releases it, and begins synthesis of the next Okazaki fragment.

polymerized Formed into a complex compound by linking together smaller elements.

polynucleotide phosphorylase An enzyme that can polymerize diphosphate nucleotides without the need for a primer. The function of this enzyme *in vivo* is probably in its reverse role as an RNA exonuclease.

polyploids Organisms with greater than two chromosome sets.

polyribosome See polysome.

polysome The configuration of several ribosomes simultaneously translating the same messenger RNA. Shortened form of the term *polyribosome*.

polytene chromosome Large chromosome, seen, for example, in *Drosophila* salivary glands, consisting of many chromatids formed by rounds of endomitosis. Synapsis of homologous chromosomes occurs during the process.

population A group of organisms of the same species relatively isolated from other groups of the same species. See deme.

position effect An alteration of phenotype caused by a change in the relative arrangement of the genetic material.

positive interference When the occurrence of one crossover reduces the probability that a second will occur in the same region.

postreplicative repair A DNA repair process initiated when DNA polymerase bypasses a damaged area.

posttranscriptional modifications Changes in eukaryotic messenger RNA made after transcription has been completed. These changes include additions of caps and tails and removal of introns.

preemptor stem A configuration of leader transcript messenger RNA that does not terminate transcription in the attenuator-controlled amino acid operons.

pre-initiation complex (PIC) The form of the RNA polymerase II enzyme with general transcription factors bound equivalent to the *E. coli* holoenzyme. Phosphorylation of the enzyme then allows transcription to begin.

Pribnow box Consensus sequence of TATAAT in prokaryotic promoters centered at the position -10 .

primary oocytes The cells that undergo meiosis in female animals.

primary spermatocytes The cells that undergo meiosis in male animals.

primary structure The sequence of polymerized amino acids in a protein.

primary transcript The product of eukaryotic transcription before posttranscriptional modification takes place.

primase An enzyme that creates a messenger RNA primer for Okazaki fragment initiation.

primer In DNA replication, a length of double-stranded DNA that continues as a single-stranded template in the 3' to 5' direction.

primosome A complex of two proteins, a primase and helicase, that initiates RNA primers on the lagging DNA strand during DNA replication.

prion Infectious agent responsible for several neurological diseases (scrapie, kuru, Creutzfeld-Jakob syndrome, mad-cow disease). It is a protein that lacks DNA or RNA.

probability The expectation of the occurrence of a particular event.

probability theory The conceptual framework concerned with quantification of probabilities. See probability.

proband See propositus.

probe In recombinant DNA work, a radioactive nucleic acid complementary to a region being searched for in a restriction digest or genomic library.

processivity The ability of an enzyme to repetitively continue its catalytic function without dissociating from its substrate.

G-12

Glossary

product rule The rule that states that the probability that two independent events will both occur is the product of their separate probabilities.

progeny testing Breeding of offspring to determine their genotypes and those of their parents.

prokaryotes Organisms that lack true nuclei.

promoter A DNA region that RNA polymerase binds to in order to initiate transcription.

proofread Technically, to read for the purpose of detecting errors for later correction. DNA polymerase has 3' to 5' exonuclease activity, which it uses during polymerization to remove incorrect nucleotides it has recently added.

prophage A temperate phage replicating with the host that can later initiate the lytic cycle.

prophase The initial stage of mitosis or meiosis in which chromosomes become visible and the spindle apparatus forms.

proplastid Mutant plastids that do not grow and develop into chloroplasts.

propositus (proposita) The person through whom a pedigree was discovered.

proteasome A barrel-shaped cellular organelle for protein breakdown involving the ubiquitin pathway.

proteome From *proteins* of the *genome*; the complete set of proteins from a particular genome. It is the protein analogue to "genome."

proteomics The study of the complete set of proteins from a particular genome. It is the protein analogue to "genomics."

proto-oncogene A cellular oncogene in an untransformed cell.

prototrophs Strains of organisms that can survive on minimal medium.

pseudoalleles Genes that are functionally but not structurally allelic. Within gene families, pseudoalleles are alleles that are not expressed.

pseudoautosomal gene A gene that occurs on both sex-determining heteromorphic chromosomes.

pseudodominance The phenomenon in which a recessive allele shows itself in the phenotype when only one copy of the allele is present, as in hemizygous alleles or in deletion heterozygotes.

P (peptidyl) site The site on the ribosome occupied by the peptidyl-tRNA just before peptide bond formation.

punctuated equilibrium The evolutionary process involving long periods without change (stasis) punctuated by short periods of rapid speciation.

Punnett square A diagrammatic representation of a particular cross used to predict the progeny of the cross.

purines Nitrogenous bases of which guanine and adenine are found in DNA and RNA.

pyrimidines Nitrogenous bases of which thymine is found in DNA, uracil in RNA, and cytosine in both.

quantitative inheritance The mechanism of genetic control of traits showing continuous variation.

quantitative trait loci Chromosomal regions contributing to the inheritance of a quantitative trait. These regions may contain one or more polygenes that contribute to the phenotype.

quantitative variation *See* continuous variation.

quaternary structure The association of polypeptide subunits to form the final structure of a protein.

random genetic drift Changes in allelic frequency due to sampling error.

random mating The mating of individuals in a population such that the union of individuals with the trait under study occurs according to the product rule of probability.

random strand analysis Mapping studies in organisms that do not keep all the products of meiosis together.

read-through Transcription or translation beyond the normal termination signals in DNA or RNA, respectively.

realized heritability Heritability measured by a response to selection.

recessive An allele (or phenotype) that does not express itself in the heterozygous condition.

reciprocal cross A cross with the phenotype of each sex reversed as compared with the original cross. Made to test the role of parental sex on inheritance pattern.

reciprocal translocation A chromosomal configuration in which the ends of two nonhomologous chromosomes are broken off and become attached to the nonhomologues.

recombinant DNA technology Techniques of gene cloning. Recombinant DNA refers to the hybrid of foreign and vector DNA. *See* gene cloning.

recombinant plasmid A plasmid that contains an inserted piece of foreign DNA.

recombinants In mapping studies, offspring with allelic arrangements made up of a combination of the original parental alleles.

recombination The nonparental arrangement of alleles in progeny that can result from either independent assortment or crossing over.

recombination nodule Proteinaceous nodules found on bivalents during zygonema and pachynema associated with crossing over.

recon A term Benzer coined for the smallest recombinable unit within a cistron.

reductional division The first meiotic division. It reduces the number of chromosomes and centromeres to half that in the original cell.

regional centromere The type of centromere found in higher eukaryotes that can accommodate several spindle microtubules.

regulator gene A gene primarily involved in control of the production of another gene's product.

reinitiation The initiation of translation by a ribosome that has just completed translation of a region of the messenger RNA upstream of the current point of initiation.

relative Darwinian fitness *See* fitness.

relaxed mutant A mutant that does not exhibit the stringent response under amino acid starvation.

release factors (RF1 and RF2) Proteins in prokaryotes responsible for termination of translation and release of the newly synthesized polypeptide when a nonsense codon appears in the A site of the ribosome. Replaced by eRF in eukaryotes.

repetitive DNA DNA made up of copies of the same nucleotide sequence.

replica-planting A technique to rapidly transfer microorganism colonies to numerous petri plates.

replication-coupling assembly factor

A protein complex in fruit flies that assembles new nucleosomes.

replicons A replicating genetic unit including a length of DNA and its site for the initiation of replication.

replisome The DNA-replicating structure at the Y-junction, consisting of two DNA polymerase III enzymes and a primosome (primase and DNA helicase).

reporter systems Genetic constructs that allow an investigator to determine that a specific locus is active by measuring the phenotypic expression of an associated locus, such as the luciferase reporter, which glows if watered with luciferin.

repressible system A coordinated group of enzymes involved in a synthetic pathway (anabolic) is repressible if excess quantities of the end product of the pathway lead to the termination of transcription of the genes for the enzymes. These systems are primarily prokaryotic operons.

repressor The protein product of a regulator gene that acts to control transcription of inducible and repressible operons.

reproductive isolating mechanisms Environmental, behavioral, mechanical, and physiological barriers that prevent two individuals of different populations from producing viable progeny.

reproductive success The relative production of offspring by a particular genotype.

repulsion Allelic arrangement in which each homologous chromosome has mutant and wild-type alleles.

resistance transfer factor Infectious transfer part of R plasmids.

restricted transduction See specialized transduction.

restriction digest The results of the action of a restriction endonuclease on a DNA sample.

restriction endonucleases Endonucleases that recognize certain DNA sequences, then cleave them. They protect cells from viral infection; they are useful in recombinant DNA work.

restriction fragment length

polymorphism (RFLP) Variations (among individuals) in banding patterns of electrophoresed restriction digests.

restriction map A physical map of a piece of DNA showing recognition sites of specific restriction endonucleases separated by lengths marked in numbers of bases.

restriction site The sequence of DNA recognized by a restriction endonuclease.

restrictive temperature A temperature at which temperature-sensitive mutants display the mutant phenotype.

retinoblastoma A childhood cancer of retinoblast cells caused by the inactivation of an anti-oncogene.

retrotransposons Transposable genetic elements found in eukaryotic DNA that move through the reverse transcription of an RNA intermediate.

reverse transcriptase An enzyme that can use RNA as a template to synthesize DNA.

reversion The return of a mutant to the wild-type phenotype by way of a second mutational event.

R factors See R plasmids.

rho-dependent terminator A DNA sequence signaling the termination of transcription; termination requires the presence of the rho protein.

rho-independent terminator A DNA sequence signaling the termination of transcription; the rho protein is not required for termination.

rho protein A protein involved in the termination of transcription.

ribosomal RNA (rRNA) RNA components of the subunits of the ribosomes.

ribosome recycling factor (RRF) A protein needed to prepare ribosomal subunits that have just finished translating a messenger RNA for another cycle of translation.

ribosomes Organelles at which translation takes place. They are made up of two subunits consisting of RNA and proteins.

ribozyme Catalytic or autocatalytic RNA.

RNA editing The insertion of uridines into messenger RNAs after transcription is completed; controlled by guide RNA. May also involve insertion of cytidines in some organisms or possible deletions of bases.

RNA phages Phages whose genetic material is RNA. They are the simplest phages known.

RNA polymerase The enzyme that polymerizes RNA by using DNA as a template. (Also known as *transcriptase* or *RNA transcriptase*.)

RNA replicase A polymerase enzyme that catalyzes the self-replication of single-stranded RNA.

Robertsonian fusion Fusion of two acrocentric chromosomes at the centromere.

rolling-circle replication A model of DNA replication that accounts for a circular DNA molecule producing linear daughter double helices.

R plasmids Plasmids that carry genes that control resistance to various drugs.

rRNA See ribosomal RNA (rRNA).

rule of independent assortment See independent assortment, rule of.

rule of segregation See segregation, rule of.

sampling distribution The distribution of frequencies with which various possible events could occur, or a probability distribution defined by a particular mathematical expression.

sarcoma Tumor of tissue of mesodermal origin (e.g., muscle, bone, cartilage).

satellite DNA Highly repetitive eukaryotic DNA primarily located around centromeres. Satellite DNA usually has a different buoyant density than the rest of the cell's DNA.

scaffold The eukaryotic chromosomal structure that remains when DNA and histones have been removed.

scanning hypothesis Proposed mechanism by which the eukaryotic ribosome recognizes the initiation region of a messenger RNA after binding the 5' capped end of it. The ribosome scans the messenger RNA for the initiation codon.

scientific method A procedure scientists use to test hypotheses, making predictions about the outcome of an observation or experiment before the experiment is performed. The results provide support or refutation of the hypothesis.

screening technique A technique to isolate a specific genotype or phenotype of an organism.

secondary oocytes The cells formed by meiosis I in female animals.

secondary spermatocytes The products of the first meiotic division in male animals.

secondary structure The flat or helical configuration of the polypeptide backbone of a protein.

second-division segregation (SDS) The allelic arrangement in the spores of Ascomycete fungi with ordered spores that indicates a crossover between a locus and its centromere

securin An inhibitory protein that prevents separin from acting on cohesin to separate sister chromatids.

segmentation genes Genes of developing embryos that determine the number and fate of segments.

segregation, rule of Mendel's first principle, which describes how genes pass from one generation to the next.

segregational load Genetic load caused when a population is segregating less fit homozygotes because of heterozygote advantage.

segregation distortion See gametic selection.

selection See natural selection.

selection coefficients, *s*, *t* The sum of forces acting to lower the relative reproductive success of a genotype.

selection-mutation equilibrium

An equilibrium allelic frequency resulting from the balance between selection against an allele and mutation re-creating this allele.

selective medium A culture medium enriched with a particular substance to allow the growth of particular strains of organisms.

selfed See self-fertilization.

self-fertilization Fertilization in which the two gametes come from the same individual.

selfish DNA A segment of the genome with no apparent function, although it can control its own copy number.

semiconservative replication The mode by which DNA replicates. Each strand acts as a template for a new double helix. See template.

semisterility Nonviability of a proportion of gametes or zygotes.

sense strand See coding strand.

separin An enzyme that breaks down cohesin and allows sister chromatids to separate at the start of anaphase of mitosis.

sequence-tagged sites (STSs) DNA lengths of 100–500 base pairs that are unique in the genome. They are created by polymerase chain reaction amplification of primers that are then tested to be sure the sequence is unique.

sex chromosomes Heteromorphic chromosomes whose distribution in a zygote determines the sex of the organism.

sex-conditioned traits Traits that appear more often in one sex than in another.

sex-determining region Y (SRY) The sex switch, or testis-determining factor, in human beings, located on the Y chromosome (*Sry* in mice).

sexduction A process whereby a bacterium gains access to and incorporates foreign DNA brought in by a modified F factor during conjugation.

sex-influenced traits See sex-conditioned traits.

Sex-lethal A gene in *Drosophila*, located on the X chromosome, that is a sex switch, directing development toward femaleness when in the “on” state. It is regulated by numerator and denominator elements that act to influence the genic balance ratio (X/A).

sex-limited traits Traits expressed in only one sex. They may be controlled by sex-linked or autosomal loci.

G-14

Glossary

sex-linked The inheritance pattern of loci located on the sex chromosomes (usually the X chromosome in XY species); also refers to the loci themselves.

sex-ratio phenotype A trait in *Drosophila* whereby females produce mostly, if not only, daughters.

sex switch A gene in mammals, normally found on the Y chromosome, that directs the indeterminate gonads towards development as testes.

sexual selection The forces, determined by mate choice, that act to cause one genotype to mate more frequently than another genotype.

Shine-Dalgarno hypothesis A proposal that prokaryotic messenger RNA is aligned at the ribosome by complementarity between the messenger RNA upstream from the Initiation codon and the 3' end of the 16S ribosomal RNA.

shoot apical meristem The major meristematic tissue of the plant; surrounds the shoot.

short interspersed elements (SINEs) Sequences of DNA interspersed in eukaryotic chromosomes in many copies. Alu, a three-hundred base-pair sequence, is found about half a million times in human DNA.

shotgun cloning The random cloning of pieces of the DNA of an organism without regard to the genes or sequences present in the cloned DNA.

shunting Process in which the first initiation codon on a messenger RNA is bypassed for an initiation codon further down the messenger. The process is probably guided by secondary structure in the messenger.

siblings (sibs) Brothers and sisters.

sigma factor The protein that gives promoter-recognition specificity to the RNA polymerase core enzyme of bacteria.

signal hypothesis The major mechanism whereby proteins that must insert into or across a membrane are synthesized by a membrane-bound ribosome. The first thirteen to thirty-six amino acids synthesized, termed a *signal peptide*, are recognized by a *signal recognition particle* that draws the ribosome to the membrane surface. The signal peptide may be removed later from the protein.

signal peptide See signal hypothesis.

signal recognition particle See signal hypothesis.

signal transduction pathway A pathway in which the action of kinase enzymes that free transcription factors, or some other action, translates an environmental signal into some form of gene action.

single-nucleotide polymorphisms (SNPs) Differences between individuals involving single base pairs that are located about every 1,000 bases along the human genome. SNPs are useful for mapping disease genes.

single-strand binding proteins Proteins that attach to single-stranded DNA, usually near the replicating Y-junction, to stabilize the single strands.

sister chromatids See chromatids.

site-specific recombination A crossover event, such as the integration of phage λ , that requires homology of only a very short region and uses an enzyme specific for that recombination.

small nuclear ribonucleoproteins

(**snRNPs**) Components of the spliceosome, the intron-removing apparatus in eukaryotic nuclei. See snRNPs.

small nucleolar ribonucleoprotein

particles (snoRNPs) Particles composed of RNA and protein found in the nucleolus that modify ribosomal RNAs, particularly by converting some uridines to pseudouridines and methylating some ribose sugars.

small nucleolar RNAs (snoRNAs) RNAs found in small nucleolar ribonucleoprotein particles (snoRNPs) that take part in modifying ribosomal RNA in the nucleolus.

SMC proteins For structural maintenance of chromosomes; proteins that aid mitotic segregation, sister-chromatid adhesion, dosage compensation, recombination, and other chromosomal activities.

snRNPs See small nuclear ribonucleoproteins.

sociobiology The study of the evolution of social behavior in animals.

somatic doubling A disruption of the mitotic process that produces a cell with twice the normal chromosome number.

somatic hypermutation The occurrence of a high level of mutation in the variable regions of immunoglobulin genes.

SOS box The region of the promoters of various genes that the LexA repressor recognizes. Release of repression results in the induction of the SOS response.

SOS response Repair systems (RecA, Uvr) induced by the presence of single-stranded DNA that usually occurs from postreplicative gaps caused by various types of DNA damage. The RecA protein, stimulated by single-stranded DNA, is involved in the inactivation of the LexA repressor, thereby inducing the response.

Southern blotting A method, first devised by E. M. Southern, used to transfer DNA fragments from an agarose gel to a nitrocellulose gel for the purpose of DNA-DNA or DNA-RNA hybridization during recombinant DNA work.

specialized transduction Form of transduction based on faulty looping out by a temperate phage. Only neighboring loci to the attachment site can be transduced. See generalized transduction.

speciation A process whereby, over time, one species evolves into a different species (anagenesis) or one species diverges to become two or more species (cladogenesis).

species A group of organisms capable of interbreeding to produce fertile offspring.

specific transcription factors Proteins needed for activation of transcription at specific eukaryotic promoters. Also, negative factors that can inhibit transcription at a specific eukaryotic promoter.

spermatids The four products of meiosis in males that develop into sperm.

spermatogenesis The process of sperm production.

spermatogonium A cell type in the testes of male vertebrates that gives rise to primary spermatocytes by mitosis.

sperm cells The gametes of males.

spermiogenesis The process by which spermatids mature into sperm cells.

spindle The microtubule apparatus that controls chromosomal movement during mitosis and meiosis.

spindle pole body Spindle microtubule organizing center found in fungi.

spiral cleavage The cleavage process in mollusks and some other invertebrates whereby the spindle is tipped at mitosis in relation to the original egg axis.

spirillum A spiral bacterium.

spliceosome Protein-RNA complex that removes introns in eukaryotic nuclear RNAs.

sporophyte The stage of a plant life cycle that produces spores by meiosis and alternates with the gametophyte stage.

stabilizing selection A type of selection that removes individuals from both ends of a phenotypic distribution, thus maintaining the same distributional mean.

standard deviation The square root of the variance.

standard error of the mean The standard deviation divided by the square root of the sample size. It is the standard deviation of a sample of means.

statistics Measurements of attributes of a sample from a population; denoted by Roman letters. See parameters.

stem-loop structure A lollipop-shaped structure formed when a single-stranded nucleic acid molecule loops back on itself to form a complementary double helix (stem), topped by a loop.

stepladder gel A DNA-sequencing gel. The numerous bands in each lane give the appearance of a stepladder.

stochastic A process with an indeterminate or random element as opposed to a deterministic process with no random element.

stringent factor A protein that catalyzes the formation of an unusual nucleotide (guanosine tetraphosphate) during the stringent response under amino acid starvation.

stringent response A translational control mechanism in prokaryotes that represses transfer RNA and ribosomal RNA synthesis during amino acid starvation.

structural alleles Alleles whose alterations include some of the same base pairs.

submetacentric chromosome A chromosome whose centromere lies between its middle and its end, but closer to the middle.

subtelocentric chromosome

A chromosome whose centromere lies between its middle and its end, but closer to the end.

sum rule The rule that states that the probability that two mutually exclusive events will occur is the sum of the probabilities of the individual events.

supercoiling Negative or positive coiling of double-stranded DNA that differs from the relaxed state.

supergenes Several loci, which usually control related aspects of the phenotype, in close physical association.

suppressor gene A gene that, when mutated, apparently restores the wild-type phenotype to a mutant of another locus.

surveillance mechanism Used to describe mechanisms that oversee checkpoints in the cell cycle, points in which the process can be halted if certain conditions are not met.

survival of the fittest In evolutionary theory, survival of only those organisms best able to obtain and utilize resources (the fittest). This phenomenon is the cornerstone of Darwin's theory of evolution.

Svedberg unit A unit of sedimentation during centrifugation. Abbreviation is S, as in 50S.

swivelase See DNA gyrase.

sympatric speciation Speciation in which reproductive isolating mechanisms evolve within the range and habitat of the parent species. This type of speciation may be common in parasites.

synapsis The point-by-point pairing of homologous chromosomes during zygonema or in certain dipteran tissues that undergo endomitosis.

synaptonemal complex A proteinaceous complex that apparently mediates synapsis during the zygotene stage and then disintegrates.

syncytium A cell that has many nuclei not separated by cell membranes.

synexpression group A group of eukaryotic genes that are involved in the same function or pathway and are induced together.

synteny test A test that determines whether two loci belong to the same linkage group by observing concordance in hybrid cell lines.

synthetic medium A chemically defined substrate that microorganisms are grown upon.

TACTAAC box A consensus sequence surrounding the lariat branch point of eukaryotic messenger RNA introns.

TATA-binding protein A protein, part of TFIID, that binds the TATA consensus sequence in eukaryotic promoters.

TATA box An invariant DNA sequence at about -25 in the promoter region of eukaryotic genes; analogous to the Pribnow box in prokaryotes.

tautomeric shift Reversible shifts of proton positions in a molecule. The bases in nucleic acids shift between the keto and enol forms or between the amino and imino forms.

TBP-associated factors (TAFs) Proteins that bind with the TATA-binding protein to form TFIID. They aid in the selectivity of TFIID.

T-cell receptors Surface proteins of T cells that allow the T cells to recognize host cells that have been infected.

telocentric chromosome A chromosome whose centromere lies at one of its ends.

telomerase An enzyme that adds telomeric sequences to the ends of eukaryotic chromosomes.

telomeres The ends of linear chromosomes.

telophase The terminal stage of mitosis or meiosis in which chromosomes uncoil, the spindle breaks down, and cytokinesis usually occurs.

telson The posterior end of the arthropod embryo, where the end of the alimentary canal is located.

temperate phage A phage that can enter into lysogeny with its host.

temperature-sensitive mutant An organism with an allele that is normal at a permissive temperature but mutant at a restrictive temperature.

template A pattern serving as a mechanical guide. In DNA replication, each strand of the duplex acts as a template for the synthesis of a new double helix.

template strand See anticoding strand.

terminator sequence A sequence in DNA that signals the termination of transcription to RNA polymerase.

terminator stem A configuration of the leader transcript that signals transcriptional termination in attenuator-controlled amino acid operons.

tertiary structure The further folding of a protein, bringing α helices and β sheets into three-dimensional arrangements.

testcross The cross of an organism with a homozygous recessive organism.

testing of hypotheses The determination of whether to reject or fail to reject a proposed hypothesis based on the likelihood of the experimental results.

testis-determining factor (TDF) General term for the gene determining maleness in human beings (*Tdf* in mice).

tetrads The meiotic configuration of four chromatids first seen in pachynema. There is one tetrad (bivalent) per homologous pair of chromosomes.

tetranucleotide hypothesis

The hypothesis, based on incorrect information, that DNA could not be the genetic material because its structure was too simple—that is, that repeating subunits contain one copy each of the four DNA nucleotides.

tetraploids Organisms with four whole sets of chromosomes.

tetratype (TT) A spore arrangement in Ascomycete fungi that consists of the two parental and two recombinant spores.

theta structure An intermediate structure formed during the replication of a circular DNA molecule.

three-point cross A cross involving three loci.

thymine See pyrimidines.

t-loop A loop that forms at the end of mammalian telomeres by the interdigitation of the 3' free end into the DNA double helix.

topoisomerase An enzyme that can relieve (or create) supercoiling in DNA by creating transitory breaks in one (type I) or both (type II) strands of the helical backbone.

topoisomers Forms of DNA with the same sequence but differing in their linkage number (coiling).

totipotent The state of a cell that can give rise to any and all adult cell types, as compared with a differentiated cell whose fate is determined.

trailer The length of messenger RNA from the nonsense codon to the 3' end or, in polycistronic messenger RNAs, from a nonsense codon to the next gene's leader.

trans Meaning "across" and referring usually to the geometric configuration of mutant alleles across from each other on a homologous pair of chromosomes.

trans-acting Referring to mutations of, for example, a repressor gene, that act through a diffusible protein product; the normal mode of action of most recessive mutations.

transcription The process whereby RNA is synthesized from a DNA template.

transcription factors Eukaryotic proteins that aid RNA polymerase in recognizing promoters. See general transcription factors and specific transcription factors.

transducing particle A defective phage, carrying part of the host's genome.

transduction A process whereby a cell can gain access to and incorporate foreign DNA brought in by a viral particle.

transfection The introduction of foreign DNA into eukaryotic cells.

transfer operon (*tra*) Sequence of loci that impart the male (F-pili-producing) phenotype on a bacterium. The male cell can transfer the F plasmid to an F⁻ cell.

transfer RNA (tRNA) Small RNA molecules that carry amino acids to the ribosome for polymerization.

G-16

Glossary

transformation A process whereby prokaryotes take up DNA from the environment and incorporate it into their genomes, or the conversion of a eukaryotic cell into a cancerous one.

transgenic Eukaryotic organisms that have taken up foreign DNA.

transition mutation A mutation in which a purine-pyrimidine base pair replaces a base pair in the same purine-pyrimidine relationship.

translation The process of protein synthesis wherein the nucleotide sequence in messenger RNA determines the primary structure of the protein.

translocase (EF-G) Elongation factor in prokaryotes necessary for proper translocation at the ribosome during the translation process. Replaced by eEF2 in eukaryotes.

translocation A chromosomal configuration in which part of a chromosome becomes attached to a different chromosome. Also a part of the translation process in which the messenger RNA is shifted one codon in relation to the ribosome.

translocation channel (translocon) A protein-lined pore or channel in a membrane through which nascent proteins are transported during translation.

transposable genetic element A region of the genome, flanked by inverted repeats, a copy of which can be inserted at another place; also called a transposon or a jumping gene.

transposon See transposable genetic element.

transversion mutation A mutation in which a purine replaces a pyrimidine, or vice versa.

trihybrid An organism heterozygous at three loci.

triploids Organisms with three whole sets of chromosomes.

trisomic A diploid cell with an extra chromosome.

tRNA See transfer RNA (tRNA).

trp RNA-binding attenuation protein (TRAP) Protein that can bind to the attenuation region of the messenger RNA of the tryptophan operon in *Bacillus subtilis*, causing a terminator stem to form and halting further transcription.

true heritability See heritability.

tumor Abnormal growth of tissue.

tumor-suppressor genes Genes that normally prevent unlimited cellular growth. When both copies of the gene are mutated, cellular transformation follows. Examples are the *p53* gene and the genes for retinoblastoma and Wilm's tumor.

two-point cross A cross involving two loci.

type I error In statistics, the rejection of a true hypothesis.

type II error In statistics, the accepting of a false hypothesis.

typological thinking The concept that organisms of a species conform to a specific norm. In this view, variation is considered abnormal.

ubiquitin A peptide of twenty-six amino acid residues that enzymes attach to proteins that the proteasome will degrade.

unequal crossing over Nonreciprocal crossing over caused by the mismatching of homologous chromosomes. Usually occurs in regions of tandem repeats.

uninemic chromosome A chromosome consisting of one DNA double helix.

unique DNA A length of DNA with no repetitive nucleotide sequences.

unmixed families Groups of four codons sharing their first two bases and coding for the same amino acid.

unusual bases Other bases, in addition to adenine, cytosine, guanine, and uracil, found primarily in transfer RNAs.

UP element See upstream element.

upstream A convention on DNA related to the position and direction of transcription by RNA polymerase (5'→3'). Downstream (or 3' to) is in the direction of transcription whereas upstream (5' to) is in the direction from which the polymerase has come.

upstream element A sequence of about twenty AT-rich bases centered at -50 in promoters of prokaryotic genes that are expressed strongly.

uracil See pyrimidines.

variable-number-of-tandem-repeats (VNTR) loci Loci that are hypervariable because of tandem repeats. Presumably, variability is generated by unequal crossing over.

variance The average squared deviation about the mean of a set of data.

variegation Patchiness; a type of position effect that results when particular loci are contiguous with heterochromatin.

virion A virus particle.

viroids Bare RNA particles that are plant pathogens.

V(D)J joining The process of joining variable, diversity, and joining gene segments (V-J, V-D, or D-J joining) in the formation of a functioning immunoglobulin gene.

Wahlund effect The effect of subdivision on a population, causing it to contain fewer heterozygotes than predicted despite the fact that all subdivisions are in Hardy-Weinberg proportions.

western blotting A technique for probing for a particular protein using antibodies. See Southern blotting.

whole-genome shotgun method

A method of sequencing entire genomes by breaking up the genomes into small pieces, sequencing the pieces, and then using computers to establish order by overlapping the sequences.

wild-type The phenotype of a particular organism when it is first seen in nature.

Wilm's tumor A childhood kidney cancer caused by the inactivation of an anti-oncogene.

wobble Referring to the reduced constraint over the third base of an anticodon as compared with the other bases, thus allowing additional complementary base pairings.

xeroderma pigmentosum A disease in human beings caused by a defect in the UV mutation repair system.

X-inactivation center (XIC) Locus at which inactivation is initiated on the X chromosome in mammals.

X linked See sex-linked.

X-ray crystallography A technique, using X rays, to determine the atomic structure of molecules that have been crystallized.

yeast artificial chromosome (YAC) Originating from a bacterial plasmid, a YAC contains additionally a yeast centromeric region (CEN) and a yeast origin of DNA replication (ARS). YACs are capable of including large pieces of foreign DNA during cloning.

Y-junction The point of active DNA replication where the double helix opens up so that each strand can serve as a template.

Y linked Inheritance pattern of loci located on the Y chromosome only. Also refers to the loci themselves.

Z DNA A left-handed form of DNA found under physiological conditions in short GC segments that are methylated. It may be involved in regulating gene expression in eukaryotes.

ZFY gene Originally believed to be the human male sex-switch gene, located on the short arm of the Y chromosome. *ZFY* stands for zinc finger on the Y chromosome.

zinc finger Configuration of a DNA-binding protein that resembles a finger with a base, usually cysteines and histidines, binding a zinc ion. Discovered in a transcription factor in *Xenopus*.

zygonema (zygotene stage) The stage of prophase I of meiosis in which synapsis occurs.

zygotic induction The beginning of vegetative growth when a prophage is passed into an F⁺ cell during conjugation.

zygotic selection The forces acting to cause differential mortality of an organism at any stage (other than gametes) in its life cycle.

INDEX

A

ABC excinuclease, 342
Abelson, J., 268
ABO blood groups, 25–26, 32, 559
Ac-Ds system, 468
Acentric fragment, 178, 179
Acetabularia (green alga), 515
Acinonyx jubatus. *See* Cheetah
Acquired immunodeficiency syndrome.
 See AIDS
Acridine dyes, 337
Acrocentric chromosome, 48, 49
Acron, 470–71
Activators, for transcription, 263
Active site, of enzyme, 206, 208
Acute lymphocytic leukemia, 485
Adaptation and Natural Selection: A Critique of Some Current Evolutionary Thought (Williams, 1966), 604–5
Adaptive mutations, 339
Adaptive value, 577
 Additive models, of variation, 533, 538, 548
Adenines
 chemistry of nucleic acids, 213, 215
 dideoxy method of DNA sequencing, 385
 tautomeric shifts, 329, 330
Adenosine deaminase, 397
Adjacent-1 and adjacent-2 types, of segregation, 184–85
A DNA, 219
Affected individuals, and pedigree analysis, 98
Affymetrix, Inc., 396
Africa, and AIDS, 502
African sleeping sickness, 275
AGAMOUS gene, 483
Agriculture
 genetically modified crops, 371, 398
 life cycle of plants, 66
 polyploidy in plants and, 199
 recombinant DNA techniques and, 13
Agrobacterium tumefaciens, 371–72, 376–77
AIDS, 13, 276, 502–4
Alberts, B., 234, 252
Albinism, 33, 34, 37, 97–98
Alcoholism, 103–4
Alkaptonuria, 37, 38, 582
Allele. *See also* Allelic frequency; Allelism
 classical genetics and, 9
 dominant and recessive traits, 18
 Hardy-Weinberg equilibrium, 558–59
 lethal-equivalent, 561
 multiple, 25–26, 558–59
Allelic exclusion, 495
Allelic frequency. *See also* Allele
 Hardy-Weinberg equilibrium, 553–54, 555, 561–62, 566–67

 heterozygote advantage, 584
 mutational equilibrium, 572–73
 selection-mutation equilibrium, 581
Allelism, 319, 354. *See also* Allele
Allium cepa. *See* Onion
Allo lactose, 407, 408, 409
Allopatric speciation, 592, 592, 594
Allopolyploidy, 198
Allosteric protein, 409
Allotypes, 493
Allozygosity, 561
Alternate segregation, 184
Alternative energy, and biotechnology, 398
Alternative splicing, 271–72
Altman, Sidney, 265, 268
Altruism, and sociobiology, 14, 603–5, 606, 607–8
Alu family, 459
American Red Cross, 26
Ames, Bruce, 332
Ames test, for carcinogens, 332
Amino acids
 experimental methods and sequencing of, 284–86
 mutations, 313
 proteins and structure of, 282
 rate of evolutionary change and sequences of, 601–602, 603
 RNA and, 12, 281
 sequence of dihydrofolate reductase, 513
Aminoacyl site (A site), 292
Aminoacyl-tRNA synthetases, 257, 286, 287
ampC gene, 389
Amphidiploids, 198
Ampicillin, 427
Anagenesis, 590
Anaphase, of mitosis, 52, 54, 56, 57
Anaphase A and anaphase B, 54
Anaphase I, and meiosis, 59, 60, 62
Anaphase-promoting complex (APC), 51
Anemia. *See* Thalassemia
Aneuploids and aneuploidy, 83, 190, 192–97
Anfinsen, Christian, 303
Angelman syndrome, 524
Angiogenesis, 492
Angiosperms, 64
Animal Dispersion in Relation to Social Behavior (Wynne-Edwards, 1962), 603–4
Animals. *See also* Cattle; Mouse; Sheep
 life cycle, 47
 meiosis, 63–64
 polyploidy, 198–99
 vectors and cloning, 371
Antennapedia complex (ANT-C), 478

Antibiotics and antibiotic resistance. *See also*
 Drug resistance
 antibacterial mechanisms, 153
 Chlamydomonas, 517–18
 composite transposons, 427
 mitochondrial protein synthesis, 515
 resistance plasmids, 523
 translation and, 294–95
Antibodies
 ABO blood group system, 25
 definition of, 492
 diversity of, 494–97
 molecular genetics and, 13
Anticoding strand, of DNA, 249
Anticodons, 256, 258, 309. *See also* Codons
Antigens, 25, 492
Antimutator mutations, 338
Anti-oncogenes, 13, 485
Antiparallel strands, of DNA, 218
Antirepressor, 421
Antirrhinum majus (Snapdragon), 35–36, 484
Antisense RNA, 431
Anti-sigma factor (AsiA), 430
Antiterminator protein, 420
AP endonuclease, 341, 354
Apolipoprotein-B (*apoB*) gene, 275
Apoptosis, 484
Apple maggot fly (*Rhagoletis pomonella*), 593
Arabidopsis thaliana (Meadow weed)
 dwarf form of, 373
 flower formation, 482–83
 methylation of DNA, 466–67
 study of plant genetics and, 372
Arabinose, 435
Arberg, W., 4, 359
Archaea, 149
Armillaria bulbosa (fungus), 441
Aromatic amines, 492
Arsenic, 492
Asbestos, 492
Ascaris spp., 50, 87
Ascospores, 123
Ascus, 123
AsiA (anti-sigma factor), 430
Aspartate transcarbamylase, 433, 434
Aspergillus nidulans (fungus), 132
Assignment test, 136–37
Assortative mating, 554, 560
Aster, 53
Asthma, 545
Ataxia-telangiectasia, 484
Attenuator-controlled system, of gene expression, 415–18
Attenuator region, 415

I-2 Index

- Attenuator stem, 416
Autogamy, 519, 520, 521
Autonomously replicating sequences (ARS), 239
Autopolyploidy, 198
Autoradiography, and DNA replication, 222–24
Autosomal dominant inheritance, 102, 104
Autosomal linkage, 134
Autosomal recessive inheritance, 102
Autosomal set, 84
Autosomes, 83
Autotrophs, 150
Autozygosity, 561, 564
Auxiliary factors, and introns, 271
Auxotrophs, 150
Avery, Oswald, 4, 154, 209
AZT, 504
- B**
- Bacillus, 149
Bacillus subtilis (soil bacterium)
DNA-RNA complementarity, 247
gene transcription, 253
transformation and transformation mapping, 155, 156–57, 172
trp operon, 417, 418
Bacillus thuringiensis (Bt), 398
Bacitracin, 153
Backcross, 22
Back mutations, 327
Bacteria. *See also Escherichia coli*;
Prokaryotes
antibiotic resistance, 294–95
cultivation of, 150–51
genetic research and, 149–50
life cycle, 64
phenotypes of, 151–54
recombinant DNA technology, 349–51
sexual processes in, 154–62
size of cells, 441
Bacterial artificial chromosomes (BACs), 393, 395
Bacterial lawn, 151
Bacterial viruses, 66, 149–50. *See also* Viruses
Bacteriophages
genetic research and, 8, 149–50
life cycle of, 163–65
sexual processes in, 154–62
transduction, 165–68
Balaenoptera musculus (Blue whale), 441
Balbiani, E. G., 449
Balbiani rings, 449
Baltimore, David, 276
*Bam*HI restriction endonuclease, 360
Barr, M., 90
Barr body, 90
Bar system, 185–86, 188
Basal bodies, 52
Base excision repair, 340–41
Base flipping, 341
Basic/helix-loop-helix/leucine zipper, 480, 481
Bates, H. W., 604
Batesian mimicry, 604, 605
Bateson, William, 21
B-cell chronic lymphocytic leukemia, 485
B DNA, 219, 220
Beadle, George, 10, 38–39
Beagle (ship), 589
Bean (*Phaseolus vulgaris*), 541. *See also*
Broad bean
Behavioral genetics, human, 547. *See also*
Sociobiology
Bennett, J. C., 494
Benzer, Seymour, 318, 320, 321, 322–23, 324
Benzene, 492
Berg, Paul, 4, 370, 383
Bertram, E., 90
*Bgl*I restriction enzyme, 360
Bicoid gene, 472, 474, 475
Binary fission, 519
Binomial expansion, 73
Binomial theorem, 72
Biochemical genetics, 37–39
Bioinformatics, 396
Biological transfer, 374, 375
Biological containment, 370–71
Biological species concept, 590
Biomedical applications
AIDS and retroviruses, 502–4
Ames test for carcinogens, 332
antibiotics and antibiotic resistance, 294–95
of multiple-stranded DNA, 221, 222
of prions, 213
Biometrics, 534–35
*Bit*borax gene, 478, 479
Bivalents, and meiosis, 58
Blackburn, Elizabeth H., 454, 455
Blastoderm, 470
Blobel, Gunter, 301
Blood coagulating factor III, 136
Blood pressure, and concordance, 546
Blood transfusions, 26
Blood types
Hardy-Weinberg equilibrium, 557, 558, 559
migration, 573–74
phenotypes, 25–26, 32
Blunt-end ligation, 363–64
Boa constrictor (*Constrictor constrictor*), 50
Bobbed gene, 87, 96
Bodmer, W., 562
Body plan, 471–75
Bolivar, F., 363
Bos taurus. *See* Cattle
Bottlenecks, genetic, 576–77
Bounty (ship), 576
Bouquet stage, 56
Boveri, Theodor, 3
Boyer, H., 360
Branch migration, 348, 350
Brassica oleracea (cabbage), 198
Breakage-fusion-bridge cycle, 178
Breakage-and-reunion process, 347
Breast cancer, 390, 391, 392, 546
Brenner, Sydney, 325, 483
Brewer, B., 252
Bridges, Calvin B., 84, 121, 199
Britten, Roy J., 457–58
Broad bean (*Vicia faba*), 50
Brody, E., 268
Broker, Thomas, 265, 266
Brower, L., 604
Brown, Robert, 3
Buoyant density, of DNA, 451, 452, 514
Burkitt's lymphoma, 485
Burt, Cyril, 546
Bush, Guy, 592
Butterflies, and mimicry, 604, 605
- C**
- Cabbage. *See Brassica oleracea*
Cactus ground-finch (*Geospiza scandens*), 588, 595
Cactus protein, 467
Caedobacter conjugatus (kappa particle), 521
Caedobacter taeniospiralis (kappa particle), 520, 522
Caenorhabditis elegans (Roundworm)
genetic control of development, 483–84
genome sequencing, 396
Cairns, John, 222–24, 339
Calculus of genes, 606
Cameleo pardalis. *See* Chameleon
Cancer. *See also* Breast cancer
Ames test for carcinogens, 332
environmental causes of, 492
gene expression, 484
gene mapping, 390, 391, 392
mutational nature of, 484–87, 506
recombinant DNA technology, 13
viral nature of, 487–92, 506
Cancer-family syndromes, 484–85
Cannibalism, and kuru, 213
Cap, and eukaryotic transcription, 265
Capsids, 149
Capsomeres, 149
Carcinogens
Ames test, 332
environmental causes of cancer, 492
Carcinomas, 484
Carpel whorl, 479, 482
Cassette mechanism, 469
Cat, tortoiseshell, 91, 95
Catabolite activator protein (CAP), 412–13
Catabolite repression, 412–13
Catostomus clarki (fish), 599, 600
Cat's eye syndrome, 195
Cattle (*Bos taurus*), 66, 543
Cavalli-Sforza, L., 562
C-bands, 451, 452
CCR5 protein, 502
Cdc2 gene, 50–51
CD4 protein, 502
Cech, Thomas, 265–66, 268
Celera Genomics, 396
Cell. *See also* Meiosis; Mitosis
chromosomes and complements of, 48–49
daughter cells, 49, 57
diploid cells, 9, 48
disomic cells, 197
eukaryotic compared to prokaryotic, 440
expression of foreign DNA and eukaryotic, 372–75

- F cells, 160, 161
 follicle cells, 470, 474
 haploid cells, 9, 48
 helper T cells, 502
 mitosis and lymphocytic cells, 15
 monosomic cells, 190
 nullisomic cells, 190
 nurse cells, 470
 pole cells, 470
 size of, 441
 sperm cells, 63
 totipotent cells, 469
 trisomic cells, 190
 upregulation of growth, 487
- Cell cycle, 50–51
 Cell-free system, 306
 Cellular immunity, 492
 Centimorgan, 111
 Central dogma, of genetic information, 244, 275–76
 Centric fragment, 178
 Centrioles, 52–53
 Centromeres, 48, 453, 454
 Centromeric breaks, 185
 Centromeric fission, 185
 Centrosome, 52, 53
 Chain-terminating dideoxy nucleotides, 383–84
 Chameleon (*Cameleo pardalis*), 2
 Chaperone proteins, 303
 Chaperonins, 303
 Chapeville, E., 287
 Chargaff, Erwin, 216
 Chargaff's ratios, 216
 Charon phages, 360, 363
 Chase, Martha, 209–10
 Checkpoints, and cell cycle, 51
 Cheetah (*Acinonyx jubatus*), 552
 Chemical method, of DNA sequencing, 383, 390
 Chemical mutagenesis, 330–31, 336–37
 Chemiluminescent techniques, 367
 Chiasmata, 59
 Chicken. *See also* Poultry
 epistatic interactions and phenotype of, 37
 genotypic interactions and combs of, 30–32, 34
 sex-linked traits, 96, 98
 Chimeras, 190
 Chimeric plasmid, 360
 Chirality, on amino acid, 281
Cbironomus pallidivittatus (midge), 449
Cbironomus tentans (midge)
 chromosome puffs, 449, 450
 messenger RNA, 300
 Chi site, 350
 Chi square analysis, 76–77, 79, 557, 558, 567
Cblamydomonas reinbardi (green alga), 517–18
 Chloramphenicol, 294, 515
 Chloroplasts, and cytoplasmic inheritance, 515–18
 Chondrodystrophy, 582
Cbortbippus parallelus. *See* Grasshopper
 Chow, Louise T., 265, 266
 Chromatids, 52, 67
 Chromatin
 definition of, 48
 higher-order structure of, 448
 nucleosome structure, 443–44
 types of in eukaryotic chromosomes, 453
 Chromatin assembly factors, 444
 Chromatin remodeling, 446, 465
 Chromatosome, 443
 Chromium, 492
 Chromomeres, 48
 Chromosomal banding, 134, 451–53
 Chromosomal breaks, 178–85, 200
 Chromosomal maps. *See also* Mapping
 of *Drosophila melanogaster*, 121, 122, 169–71
 of humans, 132–40
 Chromosomal painting, 486
 Chromosomal rearrangements, in humans, 186–90
 Chromosomal theory, 3–4, 66
 Chromosome
 attachment point, 48
 cancer and defects in, 484, 485
 chromatids compared to, 67
 combinations of maternal and paternal in gametes, 67
 complements of cell, 48–49
 cytogenetics and variations in structure of, 178–90
 cytological crossing over, 120–22
 definition of, 154
 Down syndrome, 177
 eukaryotic, 440–61, 462
 sex determination, 83
 species and numbers of, 49–50
 three-point cross, 119
 two-point cross, 111, 114
 Chromosome jumping, 390–91
 Chromosome number, 190–99
 Chromosome puffs, 449–50, 462
 Chromosome walking, 390–91, 392
 Chronic myelogenous leukemia, 485
 Chronic wasting disease, 213
 Chymotrypsin, 284
 Cigarettes, and cancer, 492
Cis-acting mutants, 411
Cis configuration, 111
Cis-dominant mutation, 411, 412
Cis-trans arrangement, and X linkage, 133
Cis-trans complementation test, 318
 Cistron, 318
 Cladogenesis, 590, 592–94, 595
 Clamp-loader complex, 232
 Classical genetics, 4–5, 9–10, 66
 Classical linkage maps, 393
 Cleft lip, 545
 Clinal selection, 600
 Clonal cancers, 484
 Clonal evolution theory, of cancer, 484
 Clones and cloning. *See also* Recombinant DNA technology
 assignment test, 137
 benefits from, 397–98
 chromosome location of gene, 141
 ethical debate on, 374–75
 particular genes, 364–66
 probing for cloned gene, 368–77
 restriction enzymes, 360–61
 of sheep, 1
 Clubfoot, and concordance, 546
C-myc gene, 490
 Coal products, as carcinogens, 492
 Coccus, 149
 Coding strand, of DNA, 249
Codium (green alga), 515
 Codominance, 23
 Codon library, 602–3
 Codon preference, 431
 Codons. *See also* Anticodons
 common and alternative meanings of, 311
 synthetic, 306–7
 transcription, 11–12
 transfer RNA, 256
 wobble hypothesis, 307–8, 309
 Coefficient of coincidence, 118
 Coefficient of relationship, 605–6
 Cohen, S., 360
 Cohesin, 53
 Coiling, direction of in snails, 509–10, 525
 Cointegrate state, of transposition, 428
 Colicins, 522–23
 Colinearity, and mutations, 324–25
 Collins, Francis S., 358
 Colony morphology
 bacteria, 151
 fungi, 124
 Color. *See also* Pigment and pigmentation;
 Skin color
 flowers and inheritance of, 23, 37
 metabolic pathways, 36
 mouse and genotypes, 33–34
 multilocus control of in wheat, 531–33
 proplastids and variegation in plants, 516
 Color blindness, 133
 Col plasmids, 522–23
Columba livia. *See* Pigeon
 Combinatorial control, 264–65
 Comings, David E., 452
 Common ancestry, 560–61
 Comparative studies, 8
 Compensosome, 94
 Competence factor, 155
 Complement, 492
 Complementarity
 DNA-RNA hybridization, 246–47
 double helix of DNA, 11
 Complementary DNA (cDNA), 365–66
 Complementation, and mutations, 318–19
 Complementation groups, 321
 Complete linkage, 115
 Complete medium, 150
 Complementarity, and structure of DNA, 218, 219
 Complementation matrix, 320, 323
 Component of fitness, 577
 Component numbers, and transfer RNA, 281, 286
 Composite transposons, 427
 Computer programs, and population genetics, 583–84

I-4 Index

- Concordance, among twins, 546
 Condensin, 444
 Conditional-lethal mutant, 150, 327–28
 Confidence limits, 74
 Congenital malformations, and inbreeding, 562
 Congenital heart disease, 545
 Conjugation, 154, 157, 158, 161
 Consanguineous degrees, of relationship, 99
 Consensus sequence, 248
 Conservative replication, of DNA, 220
 Conserved sequence, 248, 277–78
 Constitutive heterochromatin, 452–53
 Constitutive mutants, 410–12
 Constitutive puffs, 450
Constrictor constrictor. See *Boa constrictor*
 Containment, and recombinant DNA technology, 370–71
 Contigs, 393, 394, 395
 Continuous replication, of DNA, 225
 Continuous variation, 17, 531, 533
 Control, and experimental design, 6
 Controlling elements, and transposons, 468
 Corepressor, 414
 Corey, R. B., 206
 Corn. See *Zea mays*
 Correlation and correlation coefficient, 539–40
 Correns, Carl, 4, 17
 Cotransduction, 166, 167
 Coulson, Alan, 383
 Coupling, of alleles, 111
 Covariance, 539–40
 CpG islands, 524
 Creationism, 591
 Creighton, Harriet, 120–22, 123
 Crick, Francis, 4, 10, 11, 205, 215, 216, 244, 272, 276, 304, 328, 330
 Cricket (*Gryllus domesticus*), 50
 Cri du Chat syndrome, 188–90
 Crisscross pattern of inheritance, 96
 Critical chi-square, 76
Cro-gene product, 423
 Crossbreeding, 17
 Cross-fertilization, 17–18
 Crossing over, and meiosis, 58–59
 Crossover suppression, 179–80
 Crow, James, 88, 533, 534, 562
 Crown gall tumor, 371, 372, 377
Csis oncogene, 491
Cucurbita pepo. See *Summer squash*
 Cultivation
 of bacteria, 150–51
 of viruses, 151
 CURLY LEAF gene, 483
 CUX family, of codons, 309
 C-value paradox, 457
 Cyanobacteria, and introns-early hypothesis of exon shuffling, 274
 Cyclic AMP (cAMP), 412–13
 Cyclin, 50–51
 Cyclin-dependent kinase (CDK), 50
 Cyclosome, 51
 Cysteines, 152, 283
 Cystine, 283
 Cystic fibrosis, 13
 Cytogenetics. See also Molecular genetics
 chromosomal structure, 178–90
 chromosome number, 190–99
 definition of, 178
 Cytokinesis, 47, 49–50
 Cytological crossing over, 120–22, 123
 Cytoplasmic inheritance. See also Inheritance
 chloroplasts, 515–18
 definition of, 509
 infective particles, 518–22
 mitochondria, 511–15
 prokaryotic plasmids, 522–24
 Cytosine, 213, 215
 Cytotoxic T lymphocytes, 492
- D**
 Damage reversal, and DNA repair, 340
Danaus plexippus. See *Monarch butterfly*
 Danna, K., 359
 Darnell, J., 273
 Darwin, Charles, 3, 4, 5, 13, 17, 589, 594
 Dauermodification, 509
 Daughter cells, 49, 57
 Davies, David, 221
 DDT, resistance to in *Drosophila*, 533–34, 535, 540
 Dean, D., 252
 Deer mouse (*Peromyscus maniculatus*), 92
 Degenerate code, 307
 Degrees of freedom, 76, 79
 Delbrück, Max, 4, 163, 164, 316, 317
Deleted in colorectal cancer (DCC) gene, 491
 Deletion chromosome, 178, 179
 Deletion mapping, 321–23
 Deletions, and mutations, 336–37
 Demes, 553
 Denaturation studies, of DNA, 218
 Denominator elements, 85
 Density-gradient centrifugation, 221
 Depauperate fauna, 594
 Depression, and heritability, 545
 Derepressed operon, 414
 Dervan, P., 221
 Deterministic population size, 574
 Development
 definition of, 469
 gene expression and patterns, 469–84
 hemoglobins and human, 459–61
 Diabetes mellitus, 545
 Diakinesis, 56
 Dicentric chromosome, 178, 179
 Dictyotene stage, of meiosis, 64
 Dideoxyinosine, 504
 Dideoxy method, of DNA sequencing, 383–86, 390
 Dideoxynucleotides (dd), 384–85
 Diener, Theodor, 272
 Dihybrids, 26, 30, 33, 36
 Dihydrofolate reductase, 513
 Dimerization, 340
 Dioecious flowers, 87
 Diploid cells, 9, 48
 Diploid mapping, 110–11, 114–22
 Diplonema, 56
 Directed mutation, 339
 Directional selection, 577–78
 Disassortative mating, 560
 Discontinuous DNA replication, 225, 227
 Discontinuous variation, 17, 531
 Discrete generations, 556–57
 Disomic cells, 197
 Dispersive replication, of DNA, 220
 Disruptive selection, 578, 579
 Dissortative mating, 554
 Disulfide bridge, and cysteines, 283
 Dizygotic twins, 546
 D-loop model, of DNA replication, 238, 239
 DNA. See also Complementary DNA; Mitochondrial DNA; Mutation; Repetitive DNA; Satellite DNA; Transcription; Translation; Triplex DNA; Z DNA
 alternative forms of, 219
 biologically active structure, 214–15
 chemistry of, 211–19
 cloning and creation of, 364–66
 computer-generated images, 204, 315
 control of proteins, 205–7
 denaturation studies, 218
 double helix structure, 10–11, 206–7, 216–18
 enzymes and, 218–19
 eukaryotic cells and expression of foreign, 372–75
 eukaryotic chromosome and arrangement of, 440–42
 genomic library, 366
 location of, 208–9, 219
 methylation of, 466–67
 misalignment mutagenesis, 337
 normal and tautomeric forms of bases, 328, 329
 partitioning of, 238
 phage labeling, 209–10
 plasmids, 357
 protein motifs of recognition, 480–81
 RecA protein, 345
 recombinant DNA technology, 12–13, 347–51, 352
 repair, 339–46, 348
 replication of, 207, 219, 220–39, 382
 translation, 276
 transcription, 246–55, 260–75
 transformation, 155, 156, 209
 variation, 602–3
 X-ray crystallography, 215–16
 DNA-DNA hybridization, 457–58
 DNA fingerprints, 381, 382. See also Fingerprints
 DNA glycosylases, 340–41
 DNA gyrase, 236
 DNA helicase, 230
 DNA ligase, 227–29
 DNA polymerase, 225, 227, 229, 230
 DNA-RNA hybridization, 246–47
 DNA sequencing, 383–90
 DNA tumor viruses, 371
 Docking protein (DP), 301
 Dominance and dominant traits
 heterozygotes and homozygotes, 22–23

- Mendel's experiments, 18, 19
pedigree analysis, 99, 102
Doolittle, W. E., 273
Dorsal gene, 466, 467
Dosage compensation, 90–95
Dot blotting, 368
Doublebar gene, 185–86, 188
Double breaks, in chromosome, 179
Double crossovers, 117, 118
Double digests, and restriction mapping, 379–80
Double helix, structure of DNA, 10–11, 206–7, 216–18
Double-strand break model, of recombination, 347–49, 365
Double-strand break repair, 344
Double-stranded DNA, 221, 225
Double transformation, 157
Douglas, L., 30
Down, John Langdon, 192
Downstream direction, of DNA transcription, 249
Downstream promoter element (DPE), 262
Down syndrome, 177, 192–94, 546
Dreyer, W. J., 494
Drosophila ananassae, 182–84
Drosophila melanogaster (fruit fly)
 allelism, 354
 chromosomal breaks, 200
 chromosomal map, 121, 122, 169–71
 chromosome number, 50
 chromosome puffs, 450
 complementational analysis, 319, 320
 DDT resistance, 533–34, 535, 540
 development, 469–79
 diagram of chromosome 2, 9
 disruptive selection, 578, 579
 dosage compensation, 94–95
 duplications of chromosomal segments, 185–86
 epistasis, 33
 gene mapping, 396
 generation interval, 66
 genic balance, 84–85
 geotactic response, 541, 542
 giant banded chromosomes, 449
 giant salivary gland chromosome, 119–20, 179, 439, 455
 gynandromorph, 192
 heterozygosity, 599
 homeobox genes, 482
 infective particles, 522
 inversion, 180, 181
 linkage groups of, 110
 as model organism, 8, 65
 mutants and mutations, 24
 nomenclature, 23–24
 nondisjunction, 190, 191
 nuclease-hypersensitive sites of DNA, 446
 nucleolus, 260
 population genetics, 566–67
 realized heritability, 543
 satellite DNA, 452
 scanning electron micrograph of, 109
 sex determination, 87
 sex-linked inheritance, 95–96, 103, 104
 size of chromosome, 441–42
 testcrossing of, 110–11, 114–16, 118–20, 140–41
 twin spots, 132
 unequal crossing over, 188
Drosophila pseudoobscura, 596, 599
Drug resistance. *See also* Antibiotics and antibiotic resistance
 bacteria and, 153
 fungi and, 124
Drug sensitivity, 154
Duffy blood group, 134
Duplications, of chromosomal segments, 185–86
Dusts, as carcinogens, 492
Dyad chromatid pair, 59
Dynein, 52
Dysplasia, 491
- E**
Ectoderm, 470
Edman method, 284–85
Edward syndrome, 194, 195
Ehrman, Lee, 597
Eicher, Eva, 86
Eichwald, Ernst, 85
Either-or rule, 72
Electrophoresis
 evolution and empirical data, 14, 553
 experimental methods, 92–94
 glucose-6-phosphate dehydrogenase system, 91
 polymorphisms, 14, 598
Electroporation, 374
Ellis, J., 303
Elongation, and translation, 288, 292–93, 296
Elongation complex, 262–63, 264
Elongation factors (EF-Ts and EF-Tu), 292, 293
Eldredge, N., 594
Encephalopathies, and animal diseases, 213
Endoderm, 470
Endogenote, 159
Endomitosis, 119–20
Endonucleases, 226, 334
Endosomic vesicles, 500
Endotoxins, 398
Energy cost
 of protein biosynthesis, 313
 of translation, 297
Energy industry, and biotechnology, 398
Enhancers, and transcription, 263
Enhancer of zeste gene, 483
Enriched medium, 150
Env (envelope) gene, 503
Environment
 carcinogens, 492
 evolutionary role of sexual reproduction, 88
 induced versus inherited traits, 509
 phenotypic distributions, 533, 534
Environmentally induced puffs, 450
Enzymes
 active site, 206, 208
 classical genetics, 10
 DNA and control of, 218–19
 DNA repair in *E. coli*, 348
 DNA replication in *E. coli*, 231
 Tay-Sachs disease, 23
 three-dimensional structure, 206–7
Enzymology, of DNA replication, 225–38
Eosinea gene, 35–36
Ephestia kühniella (flour moth), 510, 511
Epidemiology, of AIDS, 502–3
Epistasis, 32–37, 40–41
Epstein, R. H., 296
Epuliscium fishelsoni (bacteria), 441
Equational division, 60
Erythromycin, 153, 515
Escherich, Theodor, 149
Escherichia coli. *See also* Bacteria
 autoradiography of DNA, 223
 conditional-lethal mutants, 328
 conjugation, 157, 158
 DNA repair, 348
 DNA replication in, 231, 233, 238
 DNA-RNA hybridization, 246
 F factor, 161
 fine-structure mapping, 321
 gene expression, 406–12, 435–36
 generation interval, 66, 149
 genetic variation, 316
 heat shock proteins, 430
 host-range mutations, 163
 lacZ gene and directed mutations, 339
 medium for cultivation, 150
 as model organism, 8
 nucleotide excision repair, 341
 N-formyl methionine and protein synthesis, 288
 nucleolus, 278
 phage λ , 420
 pili, 159
 plasmids and cloning, 357, 361
 promoter of ribosomal RNA gene, *rrnB*, 249
 recombinant DNA techniques, 13
 ribosome, 257
 RNA polymerase, 250
 scanning electron micrograph, 148
 T4 bacteriophage, 405
 transduction experiments, 168, 172
 transformation, 155
 tRNAs for methionine, 289
Essay on the Principle of Population, An (Malthus, 1798), 589
Ethics, and genetics
 cloning, 374–75
 evolution, 590–91
 human behavioral genetics, 547
 human genome sequencing, 397
 Lysenko affair, 6–7
 recombinant DNA technology, 370–71
 sociobiology, 606
Euchromatin, 48, 452
Eugenics, 547
Eukaryotes
 chromosome, 440–61, 462
 clones and cloning, 369–70
 control of transcription, 465–69
 daughter cells, 49

I-6 *Index*Eukaryotes—*Cont.*

- DNA replication, 231–32, 238–39
- DNA transcription, 260–75, 278
- foreign DNA, 372–75
- gene expression, 12
- initiation factor, 290
- life cycle, 64
- nucleotide excision repair, 343
- prokaryotes compared to, 47, 261–62, 298, 440
- size of cells, 441

Eukaryotic vectors, 369–70

Euploidy, 197–98

Eusocial hymenoptera, 606

Even-skipped gene, 476, 477

Evolution

- Darwinian, 5, 589, 590–91
 - DNA synthesis, 228–29
 - genetic code, 311–12
 - hemoglobin genes, 460
 - homeotic control, 479
 - imprinting, 524
 - intron function, 272–74
 - inversions, 182–84
 - mutations, 339
 - population genetics, 553
 - proto-oncogenes, 491
 - reproductive isolating mechanisms, 607
 - sexual reproduction, 88–89
 - sociobiology, 603–6
 - speciation, 589–94, 595
 - supergenes, 181
 - transposons, 429–30
- Evolutionary genetics, 4, 5, 13–14
- Evolutionary rates, 601, 603
- Ewing's sarcoma, 485
- Excisionase, 418
- Excision repair, 340–46
- Exonote, 159
- Exons, 265
- Exon shuffling, 272–74
- Exonucleases, 226–27
- Experimental design, 74
- Experimental methods
- adaptive mutations, 339
 - amino acid sequencing, 284–86
 - chromosomal painting, 486
 - computer program for allelic equilibrium under heterozygote advantage, 583–84
 - gene sequencing, 388–89
 - high-speed chromosomal sorting, 444–45
 - inversions and evolutionary sequences, 182–84
 - in vitro* site-directed mutagenesis, 333–35
 - lethal equivalents, 562
 - mapping of quantitative trait loci, 537–38
 - mimicry, 604
 - polymerase collisions, 252
 - protein motifs of DNA recognition, 480–81
 - size of cells, 441
 - transcription in real time, 250–51
 - viroids and introns, 272–73
- Expression vectors, 369
- Expressivity, of phenotype, 97, 98
- Extrachromosomal inheritance, 509
- Eye color, and concordance, 546

F

- Facioscapular muscular dystrophy, 582
- Factorials, 72
- Familial Down syndrome, 193–94
- Familial insomnia, 213
- Family tree, 98–99
- Fate map, 471
- F cells, 160, 161
- F-duction, 161
- Fecundity selection, 577
- Feedback inhibition, and posttranslational control, 433
- Felsenfeld, Gary, 221
- Female-lethal gene, 85
- Fertility factor (F factor), 158–59
- Fetus, human
 - hemoglobin, 460
 - inbreeding and death rates, 562
 - Rh locus and maternal incompatibility, 582
- F factor, 161, 162
- Field mouse. *See Peromyscus polionotus*
- Filial generation, 18
- Fine-structure mapping, 320–23
- Fingerprints, heritability of human, 544, 545.
 - See also* DNA fingerprints
- First-division segregation (FDS), 128–29, 130
- Fisher, Ronald A., 4, 13, 553, 604
- Fitness, and natural selection, 577, 585, 605–6. *See also* Survival of the fittest
- 5-bromouracil (5BU), 331
- 5' untranslated region (5' UTR), 291
- Flemming, W., 3
- Floral induction, 479, 482
- Floral meristem, 479
- Floral-meristem identity genes, 483
- Flow cytometry, 445
- Flowers. *See also* Plants
 - genetic control of development, 479, 482–83
 - inheritance of color, 23, 37
 - sex determination, 87, 90
- Flow karyotype, 444
- Fluctuation test, 316–17, 318
- Fluorescent in situ hybridization (FISH), 486
- EMR-I* gene, 187–88
- Fokker-Planck equation, 575
- Follicle cells, 470, 474
- Follicular lymphoma, 485
- Food processing, and biotechnology, 398
- Footprinting, 248
- Ford, Edmund B., 596
- Forensics, and DNA fingerprint, 381
- Four o'clock plants (*Mirabilis jalapa*), 23, 517
- Founder effect, 576
- Four-stranded DNA, 221
- F-pili, 159
- Fraenkel-Conrat, H., 210–11
- Fragile site, of chromosome, 490
- Fragile-X syndrome, 186–88
- Fragment length polymorphisms (RFLPs), 537
- Frameshift, and genetic code, 304–5
- Frameshift mutation, 326–27, 338
- Franklin, Rosalind, 205, 215
- FrdC* gene, 389
- Free energy of activation, 205

Frequency-dependent selection, 597

Fruit fly. *See Drosophila melanogaster*

Functional alleles, 318, 320

Fundamental number (NF), 185

Fungi

- haploid mapping, 123
- heterokaryon test, 509
- phenotypes, 124
- size of cells, 441

Furberg, S., 206

Fusbi tarazu gene, 477

Futch, David, 182–84

G

- Gag* (group antigen gene), 503
- β -Galactosidase, 406, 407
- β -Galactoside acetyltransferase, 406
- β -Galactoside permease, 406
- Galápagos Islands, 594
- Gallo, Robert C., 502
- Galton, Francis, 535, 547
- Gametes
 - combinations of chromosomes in, 67
 - meiosis, 55
 - rule of independent assortment, 27
 - rule of segregation, 18
- Gametic selection, 577
- Gametophyte, 64
- Gancyclovir, 376
- Gap* genes, 475
- Garden pea (*Pisum sativum*)
 - chromosome number, 50
 - classical genetics, 9
 - independent assortment, 27
 - Mendel's experiments, 17–18, 65
 - phenotypic and genotypic ratios of dihybrid, 32
- Garrod, A. E., 37
- Gastrulation, 470
- G-banded chromosomes, 134, 138–39, 451
- Gehring, Walter J., 478
- Gene. *See also* Gene expression; Gene mapping; Gene order
 - classical genetics, 9
 - HIV and, 503–4
 - segregation, 18
- Gene amplification, 459
- Gene cloning. *See* Clones and cloning; Recombinant DNA technology
- Gene conversion, 351
- Gene expression. *See also* Transcription; Translation
 - cancer, 484–92
 - catabolite repression, 412–13
 - immunogenetics, 492–501
 - inducible system, 406–12, 435
 - lytic and lysogenic cycles in phage λ , 418–24
 - operon model, 12, 406, 408, 418
 - patterns in development, 469–84
 - posttranslational control, 433
 - transcription factors, 430
 - translational control, 430–33
 - transposable genetic elements, 425–30
 - trp* operon, 413–18

- Gene family, 458–59
- Gene flow, 592
- Gene mapping. *See also* Mapping
- human mitochondrial chromosome, 512
 - phage λ , 419
 - sequencing of human genome, 390–97
- Gene order
- haploid mapping, 129–32
 - Hfr strains, 162
 - three-point cross, 118
 - two-factor cotransduction, 167
- Gene overlap, 388–89
- Gene pool, 555
- General interval, 66
- Generalized transduction, 166
- General transcription factors, 262, 263
- Generation time, 556–57
- Gene therapy, 397
- Genetically modified crops (GM crops), 371
- Genetic code
- breaking of, 305
 - DNA control of enzymes, 218–19
 - evolution and, 311–12
 - overlap and punctuation of, 305, 306
 - synthetic codons, 306–7
 - synthetic messenger RNAs, 306
 - triplet nature of, 304–5
 - universality of, 308–11
 - wobble hypothesis, 307–8
- Genetic diseases
- gene therapy, 397
 - hemoglobins, 460–61
 - imprinting, 524
 - inbreeding, 562
 - mitochondrial inheritance, 514–15
- Genetic drift, 555, 575–76, 585
- Genetic engineering. *See* Recombinant DNA technology
- Genetic fine structure, 317–24
- Genetic load, 597
- Genetic polymorphism, 596
- Genetics. *See also* Biomedical applications;
- Cytogenetics; Ethics; Experimental methods; Heredity; Inheritance;
 - Molecular genetics
- bacteria and bacterial viruses and research in, 149–50
 - classical, evolutionary, and molecular, 4–5, 9–14
 - definition of, 3
 - fruit flies and colon bacteria as subjects of experiments, 7–8
 - history of, 3–4
 - Nobel laureates, 113–14
 - scientific method, 5–7
 - techniques of study, 8
- Genetic variation, 316, 596–603
- Genic balance, in *Drosophila*, 84–85
- Genome
- electrophoresis and sampling of, 598
 - HIV-1, 504
 - mapping, 110
 - mitochondrial, 512
 - sequencing of human, 390–97
- Genomic equivalence, 469
- Genomic library, 366, 367
- Genomics. *See also* Recombinant DNA technology
- benefits of cloning, 397–98
 - DNA sequencing, 383–90
 - polymerase chain reaction, 381–83
 - probing for cloned gene, 368–77
 - prokaryotic vectors, 360–66
 - restriction endonucleases, 359–60
 - restriction mapping, 377–81
 - southern blotting, 366–68
- Genophore, 154
- Genotypes
- evolutionary genetics, 14
 - genotypic interactions, 30–37
 - independent assortment, 27
 - pedigree analysis, 97–98
 - segregation, 18, 21
 - transposition, 429–30
- Genotypic interactions, 30–37
- Geochelone elephantopus vanderburgh*. *See* Giant land tortoise
- Geospiza fortis* (Ground finch), 590–91, 595
- Geotactic response, 541, 542
- Gerald, Park, 197
- Gerstmann-Straussler-Scheinker syndrome, 213
- Giant* gene, 476
- Giant land tortoise (*Geochelone elephantopus vanderburgh*), 529
- Giant salivary gland chromosomes, 119–20, 179, 439, 455
- Giemsa stain, 451
- Gilbert, Walter, 272–73, 274, 383, 390, 412
- β -Globin gene, 367–68
- Globin gene family, 459–61
- Glucose-6-phosphate dehydrogenase (G-6-PD), 90–91
- Glycolytic pathway, 10
- Goodfellow, Peter, 86
- Gooseberry* gene, 477
- Gould, Stephen J., 594
- Grandfather method, and X linkage, 132–33
- Grant, B. & P., 590
- Grasshopper (*Chorthippus parallelus*), 59
- Green fluorescent protein, 377
- Griffith, F., 154, 209
- Ground finches, of Galápagos Islands, 588, 590–91, 594, 595
- Ground squirrels, 606
- Group I and Group II introns, 266
- Group selection, 603–4, 606
- Grunberg-Manago, M., 306
- Gryllus domesticus*. *See* Cricket
- G-tetraplex, 456, 457
- Guanine, 213, 215
- Guide RNA (gRNA), 275
- Gurken* gene, 474–75
- Gynandromorphs, 191, 192
- H**
- Haemantibus katherinae*, 53, 54
- Haemophilus influenzae*, 396
- Hair color, and concordance, 546
- Hairless mutation, of *Drosophila*, 24
- Haldane, J. B. S., 4, 13, 553
- Hall, B., 246
- Hamilton, W. D., 605–6
- Hammerhead ribozyme, 266, 270
- Hamster, chromosome of, 448
- Handedness (left or right), and concordance, 546
- Haploid cells, 9, 48
- Haploidioidy, 606
- Haploid mapping, 122–32
- Hard selection, 597
- Hardy, G. H., 13, 553
- Hardy-Weinberg equilibrium
- allelic frequencies, 553–54
 - definition of, 553
 - extensions of, 558–59
 - generation time, 556–57
 - migration, 573
 - multiple loci, 559–60
 - mutation, 572, 573
 - natural selection, 555
 - proof of, 555–56
 - random mating, 554–55, 560
 - selection models, 578
 - small population size, 574
 - testing for fit of, 557–58
- HAT medium, 135
- Heart disease, congenital, 545
- Heat, and transversions, 336
- Heat shock proteins, 249–50, 430
- Helicase II, 342
- Helix-turn-helix motif, 480
- Helper T cells, 502
- Hemizygous genes, 96
- Hemoglobin
- human development, 459–61
 - sickle-cell anemia, 39
- Hemophilia, 95, 100–102
- Heredity, chromosomal theory of, 66. *See also* Genetics; Inheritance
- Heritability. *See also* Inheritance
- measurement of, 544–45, 548
 - partitioning of variance, 543–44
 - realized, 542–43
- Hermaphrodites, 191
- Hershey, A. D., 209–10
- Hertwig, O., 3
- Heterochromatin, 48
- Heteroduplex DNA, 351, 352
- Heterogametic chromosomes, 83
- Heterogeneous nuclear mRNAs (hnRNAs), 265
- Heterokaryon test, 134, 509
- Heteromorphic chromosome pair, 49, 83
- Heteroplasmy, 511
- Heterothallic mating type, 469
- Heterotrophs, 150
- Heterozygote advantage, 582, 583–84, 596–97
- Heterozygotes, 21, 22–23
- Heterozygosity, 564–65, 599
- Heterozygous DNA, 351
- Hexosaminidase-A, 23
- Hfr strains, 158–59, 160, 161, 162
- High-speed chromosomal sorting, 444–45
- HindII* restriction enzyme, 360
- Histocompatibility Y antigen (H-Y antigen), 85

I-8 *Index*

- Histone acetyl transferases (HATs), 446
 Histone genes, 459
 Histone proteins, 443
 Histones, composition of, 447
 History, of genetics, 3–4, 17, 28–29, 30–31, 47
 HIV, and genes, 503–4. *See also* AIDS
 Ho, David, 504
 Hogness box, 262
 Holandric traits, 95
 Holland, J. J., 276
 Holliday, R., 347
 Holliday junction, 347, 350
 Holoenzyme, 231
 Homeo box, 478–79, 482, 505
 Homeo domain, 478, 479, 505
 Homeotic genes, 477–78
 Homogametic chromosomes, 83
 Homogentisic acid, 38
 Homologous chromosomes, 48
 Homologous recombination, 347, 375
 Homology-directed recombination, 344
 Homomorphic chromosome pairs, 48, 83
 Homoplasmy, 511
Homo sapiens. *See* Humans
 Homothallic mating type, 469
 Homozygosity, and inbreeding, 561–62, 562, 564
 Homozygotes, 21, 22–23
 Horseshoe crab. *See* *Limulus polyphemus*
 Host-range mutations, 163
 Hot spots, and mutations, 323
 Hox clusters, 479
 Hubby, J. L., 596, 598
 Human Genome Project, 6, 390–97, 537
 Humans (*Homo sapiens*). *See also* Fetus
 aneuploidy, 192–97
 behavioral genetics, 547
 chromosomal maps, 132–40
 chromosomal rearrangements, 186–90
 chromosome number, 50
 composition of DNA, 216
 fingerprints and heritability, 544, 545
 flow karyotype of chromosomes, 444
 gene mapping and sequencing of genome, 390–97
 generation interval, 66
 hemoglobins, 459–61
 heterozygosity, 599
 karyotype, 49
 linkage groups of, 110
 melanin synthesis, 36
 metacentric, submetacentric, and acrocentric chromosomes, 49
 mitochondria and mitochondrial inheritance, 512, 514–15
 monosomy, 200
 nondisjunction of sex chromosomes, 191
 nuclease-hypersensitive sites of DNA, 446
 quantitative inheritance, 545–46
 Rh locus and maternal-fetal incompatibility, 582
 sex determination, 83, 84, 85–86, 87
 telomeric sequence, 455
 xeroderma pigmentosum, 343
 Humoral immunity, 492
Hunchback gene, 475, 476
 Hungry codons, 432
 Huntington disease, 188, 582
 Hybrid DNA, and recombinant DNA technology, 351
 Hybridomas, 494
 Hybrid plasmid, 360
 Hybrids and hybridization, 18, 199
 Hybrid vectors, 361, 363
 Hybrid vehicle, 360
 Hybrid zones, 592
 Hydrogen bonding, in DNA, 218
 Hyperplasia, 491
 Hypervariable loci, 380–81
 Hypothesis testing, 74–76
 Hypotrichosis, 100
 Hypoxanthine phosphoribosyl transferase (HPRT), 134–35
- ## I
- Identity by descent, 561
 Idiogram, 48, 49
 Idiotypic variation, 493
 Idling reaction, 432
Iij genotype, 517–18
 Imino acid, 281
 Immune system, and adenosine deaminase, 397
 Immunity, definition of, 492
 Immunity reactions, of ABO blood types, 25
 Immunogenetics, 492–501
 Immunoglobulins
 definition of, 492
 formation of, 498
 genes and antibodies, 497
 hybridomas, 494
 variability of, 493, 506
 Imprinting, 509, 524
 Imprinting center (IC), 524
Inborn Errors of Metabolism (Garrod 1909), 37
 Inbreeding
 definition of, 554
 genetic diseases, 562
 genotypic proportions in population with, 563
 homozygosity from, 561–62
 nonrandom mating, 560
 Inbreeding coefficient, 561, 567
 Inbreeding depression, 542
 Incestuous unions, 99
 Inclusive fitness, 605–6
 Incomplete dominance, 23
 Independent assortment
 classical genetics, 10
 example of, 34
 meiosis and mitosis, 63, 67
 rule of, 26–30
 Independent events, 71–72
 Indian fern (*Ophioglossum reticulatum*), 50
 Induced mutation, 325–26
 Inducer, and gene expression, 408
 Inducible system, of gene expression, 406–12, 435
 Induction, and gene expression, 165, 409, 489–91
 Industry, and genetic engineering, 398
 Infective particles, and cytoplasmic inheritance, 518–22
 Information transfer, and translation, 281–303
 Inheritance. *See also* Cytoplasmic inheritance; Genetics; Heredity; Heritability; Quantitative inheritance
 determination of non-mendelian, 509
 heritability, 542–45
 polygenic inheritance, 533–35, 541
 population statistics, 535–40
 selection experiments, 541–42
 traits controlled by multiple loci, 531–35
 Initiation codon, 288
 Initiation complex, and translation, 288–91
 Initiation factors (IF1, IF2, IF3), 289
 Initiation signals, for transcription, 248
 Initiator Element (Inr), 262
 Initiator proteins, 230
 Insecticides, 398
 Insertion mutagenesis, 488
 Insertion sequences (IS elements), 425, 426
 Insertions, and mutations, 336–37
 Insomnia, familial, 213
 Instantaneous speciation, 593
 Interbreeding, 591
 Intercalary heterochromatin, 453
 Intergenic suppression, and mutations, 337–38
 Integrase, 418
 Intelligence quotient (IQ), and heritability, 546
 Interkinesis, 59
 Internal ribosome entry site, 291
 Interphase, of mitosis, 48, 53, 56, 62
 Interpolar microtubules, 53
 Interrupted mating, 159–61, 162
 Intersex, 85
 Intervening sequences. *See* Introns
 Intestinal bacterium. *See* *Escherichia coli*
 Intra-allelic complementation, 323–24
 Intragenic suppression, 327
 Introns
 eukaryotic transcription, 265, 266
 evolution, 272–74
 self-splicing of, 271
 single-stranded DNA loops, 268
 Tetrahymena ribozyme, 269
 viroids, 272–73
 Introns-early and introns-late views, of exon shuffling, 273, 274
 Inversion, and chromosome breaks, 179–81, 200
 Inverted-repeat sequence, 251
In vitro research, 209
In vitro site-directed mutagenesis, 333–35
Io locus, 516
 Ionizing radiation, 492
 Iron oxide, 492
 IS elements, 425, 426, 427
 Isochromosome, 194, 195
 Isopropyl oil, 492
 Isozymes, 94

J

Jacob, François, 159–60, 161, 409
 Jeffreys, A., 381
 Johannsen, Wilhelm, 21, 541
 Junctional diversity, 495
 Junk DNA, 458–59

K

Kaposi's sarcoma, 502
 Kappa particles, 520–21, 522
 Karotype, 48, 49
 Karpechenko, G. D., 198
 Karyokinesis, 47
 Kavenoff, Ruth, 440–41
 Khrushchev, Nikita, 7
 Killer paramecium, 520–21, 525
 Kimura, Motoo, 562, 599
 Kinesin, 52
 Kinetochore, 48, 55, 453
 Kinetochore microtubules, 53
 Kinetoplasts, 275
 King, Mary-Claire, 391
 Kin selection, 605–6
 Kleckner, Nancy, 429, 430
 Klenow fragment, 232
 Klinefelter syndrome, 85, 197
 Klotz, L., 440–41
 Klug, A., 480
Knirps gene, 476
 Knockout mice, 375–76
 Koehn, R., 599
 Kornberg, Arthur, 225
Krüppel gene, 476
 Kuhn, Thomas, 591
 Kuru, 213
 Kynurenin, 39, 510, 511

L

Lac operon, and gene expression, 406–12, 436
 Lactate dehydrogenase (LDH), 93, 94
 Lactose metabolism, 406, 407, 436
LacZ gene, 339
 Ladder gels, 386, 399
 Lagging strand, 225
 Lahn, Bruce, 89
 Lamarck, Jean-Baptiste, 5
 Lampbrush chromosomes, 450–51
 Landsteiner, Karl, 25
Lathyrus odorata. *See* Sweet pea
 Leader, and RNA transcription, 255
 Leader peptide gene, 416–19
 Leader transcript, 415–16
 Leading strand, 225
 Leber optic atrophy, 514–15
 Leder, Phillip, 307
 Lederberg, Joshua, 157, 158, 161, 165, 166
Leishmania tarentolae, 275
 Lejeune, Jérôme, 192–93
 Lemmings, and population density, 597
 Leptonema, 56
 Lethal-equivalent alleles, 561, 562
 Leucine zipper, 480, 481
 Leukemia, 484, 485
 Level of significance, 77–78
 Lewin, B., 523
 Lewis, Edward B., 478

Lewontin, Richard C., 596, 598
 LexA protein, 346, 347
 Life cycle
 of bacteriophages, 163–65
 of eukaryotes, 64
 generalized for animals and plants, 47
 of *Paramecium*, 518–19
 of plants, 64–66
 of yeast, 126
 Ligation, and DNA replication, 227–29
Liguus fasciatus (Tree snail), 570
 Lily (*Lilium longiflorum*), 50
Limenitis archippus. *See* Viceroy butterfly
Limnea peregra (Pond snail), 510
Limulus polyphemus (Horseshoe crab), 599
 Linear measurement, metric units of, 48
 Linkage
 classical studies of, 66
 equilibrium and disequilibrium, 559–60
 Linkage groups, 110
 Linkage number, 236
 Linkers, and blunt-end ligation, 363–64, 365
 Liposarcoma, 485
 Liposome-mediated transfection, 374
 Liposomes, 374
 Literature, scientific, 6–7
 Littlefield, J. W., 134
 Liver, and sickle-cell anemia, 39
 Location, of DNA, 208–9, 219
 Lock-and-key model, of enzyme functioning, 206–7
Lod score method, 135
 Long interspersed elements (LINES), 458
 Lovell-Badge, Robin, 86
 Luciferase gene, 377
 Luciferin, 377
 Luft disease, 514
 Luria, Salvador E., 4, 316, 317
 Lwoff, André, 423
 Lymphocytic cell, and mitosis, 15
 Lymphomas, 484
 Lyon, Mary, 90
 Lyon hypothesis, 90–91
 Lysate, 163, 164
 Lysenko, T. D., 6–7
 Lysogenic cycles, in phage λ , 418–24
 Lysogeny, 165
 Lytic cycles, in phage λ , 418–24

M

MacLeod, C., 209
 Major histocompatibility complex (MHC), 492
 Malaria, 596
 Male-specific lethal (*msl*) gene, 94
 Malthus, Thomas, 589
 Maniatis, Tom, 461
 Manic-depression, and concordance, 546
 Map distances, 116–18, 131
 Mapping. *See also* Chromosomal maps; Gene mapping; Restriction mapping
 classical studies of, 66
 conjugation, 161
 diploid, 110–11, 114–22
 haploid, 122–32
 human chromosomal maps, 132–40

quantitative trait loci, 537–38
 sequencing of human genome, 390–97
 transduction, 166–68
 transformation, 155–57
 Mapping function, 117–18
 Map units, 10, 111
 Margulis, Lynn, 513
 Mariculture, and biotechnology, 398
 Mass production methods, and Human Genome Project, 396
Master-switch genes, 478, 505
 Mate-killer infection, 521–22
 Maternal effects
 definition of, 509
 maternal-effect genes, 472, 473, 475
 moth pigmentation, 510, 511
 snail coiling, 509–10
 Maternal-effect genes, 472, 473, 475
 Mating types, 124, 469
 Matthei, J. H., 306
 Maturation-promoting factor (MPF), 50
 Maxam, Allan, 383
 McCarthy, B. J., 276
 McCarty, M., 209
 McClintock, Barbara, 120–22, 123, 425, 468
 McKusick, Victor A., 99, 137
 Meadow weed. *See Arabidopsis thaliana*
 Mean, statistical, 74, 535, 536, 538–39
 Mean fitness, of population, 579–80
 Measles, and concordance, 546
 Measurement
 heritability, 544–45, 548
 metric units of linear, 48
 Medicine, and genetic engineering, 397. *See also* Biomedical applications
 Meiosis, 62
 animals and, 63–64
 definition of, 47
 origins of term, 49
 process of, 55–61
 rules of segregation and independent assortment, 67
 significance of, 61, 63
 triploids, 197
 Meiosis II, 60–61
 Meiotic drive, 577
 Melanin, 33, 34, 36
 Melanoma, 485
 Mendel, Gregor
 classical genetics, 9–10
 dominance, 22–23
 experiments in genetics, 17–18, 97
 genotypic interactions, 30–37
 history of genetics, 3, 4, 65
 independent assortment, 26–30
 multiple alleles, 25–26
 nomenclature, 23–24
 quantitative traits, 534
 rediscovery of, 47
 rule of segregation, 18–22, 63
 statistics, 71, 74
 techniques of study, 8
 Merozygote, 159, 409–10
 Meselson, Matthew, 220–21
 Mesoderm, 470

I-10 *Index*

- Messenger RNA (mRNA)
 cloning, 364–65, 366
 function of, 245
 polycistronic and monocistronic, 298, 299, 300
 prokaryotes, 277
 synthetic, 306, 307
- Messing, J., 386
- Metabolic pathways, of color production in dihybrids, 36. *See also* Metabolism
- Metabolism, and biochemical genetics, 37–38
- Metacentric chromosome, 48, 49
- Metafemales, 85
- Metagon, 521
- Metamales, 85
- Metaphase, of mitosis, 52, 54, 57
- Metaphase I, and meiosis, 59, 60, 62
- Metaphase plate, 54
- Metastasis, 484
- Methionine, 152, 288, 289
- Methylation, of DNA, 466–67
- Metrical variation, 533
- Metric measurements, linear, 48
- MHC proteins, 498–500, 501
- Michigan Technological University, 441
- Microsatellite DNA, 382
- Microscope, history of, 3
- Microtubule organizing centers, 52
- Microtubules, 48, 52
- Midparent, and wing length, 539
- Migration
 Hardy-Weinberg equilibrium, 555
 population genetics, 573–74
- Mimicry, 604, 605
- Minimal medium, 124, 150
- Mirabilis jalapa*. *See* Four o'clock plants
- Misalignment mutagenesis, 337
- Mismatch repair, 343–44, 354
- Missense mutations, 338
- Mitochondrial DNA (mtDNA), 600
- Mitochondrial transfer RNAs, 309
- Mitochondrion
 cytoplasmic inheritance, 511–15
 genetic code of yeast, 311
 scanning electron microscope image of, 508
 signal hypothesis, 303
- Mitosis
 definition of, 47
 late anaphase, 56
 lymphocytic cell, 15
 origins of term, 49
 process of, 52–55
 rules of segregation and independent assortment, 67
 significance of, 55
 stages of, 46, 57
- Mitosis-promoting factor, 50
- Mitotic crossing over, 132
- Mitotic spindle, 52–53, 55
- Mixed families, of codons, 307
- Model organisms, 8
- Modern linkage map, 393, 395
- Molecular chaperones, 303
- Molecular evolutionary clock, 600–602
- Molecular genetics. *See also* Cytogenetics
 Down syndrome, 193
 evolutionary processes, 553
 as general area of study in genetics, 4, 5, 10–13
 growth in field of, 358
 search for genetic material, 205–11
- Molecular imprinting, 524
- Molecular mimicry, 297
- Moloney murine leukemia virus, 374
- Monarch butterfly (*Danaus plexippus*), 604, 605
- Monocistronic messenger RNAs, 299
- Monoclonal antibody, 493
- Monod, Jacques, 409
- Monoecious flowers, 87
- Monohybrids, 18
- Monosomic cells, 190
- Monosomy, 195, 200
- Monovalent chromosome, 59
- Monozygotic twins, 546
- Montagnier, Luc, 502
- Morgan, Thomas Hunt, 83, 95, 110, 111
- Morphogen, 472
- Morphological species concept, 591
- Morton, Newton E., 135
- Mosaicism, 90, 190–92, 194
- Moths, and pigmentation, 510, 511
- Mouse (*Mus musculus*). *See also* Deer mouse
 allelic frequency, 600
 c-banding of chromosomes, 452
 chromosome number, 50
 epistasis and color of, 33–34, 35, 37
 generation interval, 66
 genetic bottlenecks, 577
 heterozygosity, 599
 knockout, 375–76
 nuclease-hypersensitive sites of DNA, 446
 population density, 597
 transgenic, 373
 Turner syndrome, 196
- Müller, E., 604
- Muller, H. J., 88, 121, 325–26, 330
- Müller-Hill, B., 412
- Müllerian mimicry, 604, 605
- Muller's ratchet, 88
- Multihybrids, 30, 31
- Multilocus control, of inheritance, 532–33
- Multilocus selection models, 598–99
- Multinomial expansion, 73, 558
- Multiple alleles, 25–26
- Multiple loci, 532–33, 559–60
- Mu particles, 521–22
- Muscular dystrophy, 13, 188
- Mustard gas, 492
- Mutability, of DNA, 207
- Mutational equilibrium, 571–73
- Mutation rates, 326, 572–73
- Mutations and mutants
 adaptive, 339
 AIDS virus, 504
 amino acids, 313
 bicoid gene, 472
 cancer, 484–87, 506
 chemical mutagenesis, 330–31, 336–37
 classical genetics, 10
 colinearity, 324–25
 definition of, 316
 evolutionary theory, 5, 88, 312
 fluctuation test, 316–17
 genetic fine structure, 317–24
 Hardy-Weinberg equilibrium, 555
 homeotic genes, 477–78
 intergenic suppression, 337–38
 in vitro site-directed mutagenesis, 333–35
 lac operon, 409–12
 misalignment mutagenesis, 337
 mutator and antimutator, 338
 nomenclature, 24
 point mutations, 326–28
 population genetics, 571–73
 selection-mutation equilibrium, 581
 sexual reproduction, 88
 spontaneous mutagenesis, 329–30
 spontaneous versus induced, 325–26
- Mutator mutations, 338
- Muton, 320
- Mutually exclusive events, 71
- Mycobacterium tuberculosis*, 216, 546
- Mycoplasmas, 441
- ## N
- Nail-patella syndrome, 134, 135
- Nanos* gene, 472–73, 474
- Nathans, D., 4, 359
- National Academy of Sciences, 370
- National Institutes of Health
 gene therapy, 397
 guidelines on recombinant DNA technology, 370
 Human Genome Research Institute, 358
- Natural selection
 effects of, 577–78
 evolution, 5, 14
 fitness, 577
 Hardy-Weinberg equilibrium, 555
 models of, 581–82, 584
 recessive homozygote, 578–81
 selection-mutation equilibrium, 581
- Nef* gene, 504
- Negative assortative mating, 554
- Negative interference, 118
- N-end rule, 433
- Neo-Darwinism, 13, 589
- Neomycin, 375, 376
- Neoplasms, 484
- Neurospora crassa* (Bread mold)
 biosynthesis of niacin, 38–39
 chromosome number, 49
 generation interval, 66
 haploid mapping and, 123, 141
 mutations of, 124
 ordered spores, 125–27
 phenylalanine synthesis, 41
- Neutral alleles, 599
- Neutral gene hypothesis, 598
- Neutral petite, 514
- Newt. *See* *Notophthalmus viridescens*; *Triturus*
- N-formyl methionine, 288
- Niacin, 38, 39
- Nickel, as carcinogen, 492
- Nicotinamide adenine dinucleotide (NADH), 93
- Nilsson-Ehle, H., 532

Nirenberg, Marshall W., 305, 306
 Nitrous acid, 331
Nivea gene, 35–36
 Nobel, Alfred, 112
 Nobel Prize, in genetics, 112–14
 Nomenclature, 23–24
 Noncoding strand, of DNA, 249
 Nondisjunction
 mosaicism, 190–92
 sex chromosomes, 83
 Nonhistone proteins, 448
 Non-Hodgkin's lymphoma, 485
 Nonhomologous chromosomes, 182–85
 Nonhomologous end joining, 344
 Non-Mendelian inheritance, 509
 Nonparental ditype (NPD), 125, 127
 Nonparental phenotypes, 111
 Nonrandom mating, 560–65
 Nonreciprocity, and sex linkage, 96
 Nonrecombinant phenotypes, 111, 118
 Nonsense codons, 296–97
 Nonsense mutations, 337–38
 Normal distributions, 535, 536, 538
 Northern blotting, 368
Notophthalmus viridescens (newt), 451
 Novitski, E., 30
 N segments, 496–97
 Nuclease-hypersensitive sites, 445–46
 Nucleic acids, chemistry of, 211–19
 Nucleolar organizer, 53
 Nucleolus, 53, 260–61, 278
 Nucleoprotein, 47, 440, 442–61
 Nucleoside, 212, 214
 Nucleosome structure, 442–47
 Nucleotides, 8
 biomedical applications of triple-stranded
 chains, 221
 conserved sequence, 277–78
 DNA replication, 226
 excision repair, 341–44
 nomenclature, 215
 nucleic acids, 211–12, 213–14
 techniques of genetic research, 8
 Null hypothesis, 77–78
 Nullisomic cells, 190
 Numerator elements, 85
 Nurse cells, 470
 Nüsslein-Volhard, Christiane, 471
 Nutritional-requirement mutants, 327
 Nutritional requirements
 of bacteria, 151, 153
 of fungi, 124

O

Ochoa, Severo, 305, 306
 Okazaki fragments, 225, 228, 229, 230
 Oligonucleotide primer, 387
 Oncogenes, 13, 484, 488–92
One-gene-one-enzyme rule, 10, 38–39
 Onion (*Allium cepa*), 46
 Oocyte, nucleus and follicle cells, 474
 Oogenesis, 64
 Oogonia, 64
 Open reading frames (ORFs), 291
 Operator, and gene expression, 408–9, 414
 Operon model, 12, 406, 408, 418

Opbioglossum reticulatum. *See* Indian fern
 Ordered spores, 125–27
 Original literature, 6–7
 Origin recognition complex (ORC), 239
Origin of Species, The (Darwin), 3, 589
Ostreococcus tauri (green alga), 441
 Outbreeding, 554, 560
 Ovary determining (Od) gene, 86
 Overlap, of genetic code, 305, 306
 Ovum, 64
 Oxidative phosphorylation, 512
Oxytricha nova, 456, 457

P

Pachynema, 56
 Page, David, 85, 89
Pair-rule genes, 475
 Palindrome, 359, 361
 Palmer, J., 274
 Panmictic population, 574
 Paracentric inversion, 180
 Paramecin, 520
Paramecium, 518–19, 525
 Parameters, statistical, 538
 Parapatric speciation, 592, 593
 Parasegments, 470, 471
 Parental ditype (PD), 125, 127
 Parental imprinting, 524
 Parental phenotypes, 111
 Parthenogenesis, 64
 Partial digest, 379
 Partial dominance, 23
 Partitioning, of variance, 543–44
 Pascal's triangle, 73
 Patau syndrome, 194–95
 Path diagram, 562–64, 565
 Pattern formation, and flower development,
 479, 482
 Pauling, Linus, 206, 214–15, 602
 pBR322 cloning plasmid, 363, 364, 372
 Pea. *See* Garden pea; Sweet pea
 Pearson, K., 535
 Pedigree analysis, 97–102, 562–64, 565, 567
 Peer review, and scientific journals, 7
 Penetrance, of genetic trait, 97–98, 103, 104
 Penicillin, 153, 154, 172, 294
 Pepsin, 284
 Peptic ulcer, 545
 Peptide bond formation, 292
 Peptide fingerprint, 284
 Peptide map, 284
 Peptidyl site (P site), 292
 Peptidyl transferase, 292
 Pericentric inversion, 180
 Permissive temperature, and mutation, 327
Peromyscus maniculatus. *See* Deer mouse
Peromyscus polionotus (Field mouse), 599
 PEST hypothesis, 433
 Petal whorl, 479, 482
 Petite mutations, 514, 525
 Petroleum, as carcinogen, 492
 P53 gene, 487, 488
 Phage, and cloning with restriction en-
 zymes, 361. *See also* Bacteriophages
 Phage labeling, 209–10
 Phage M13, 390

Phage λ , lytic and lysogenic cycles in, 418–24,
 435–36
 Phage resistance, 163
 Phage T4, 430
Phaseolus vulgaris. *See* Bean
 Phenocopy, 98
 Phenotypes
 of bacteria, 151–54
 blood groups, 25–26
 Drosophila melanogaster, 8
 of fungi, 124
 independent assortment, 29
 lac operon, 436
 segregation, 21
 transposition, 429–30
 of viruses, 154
 Phenylalanine, 41
 Phenylisothiocyanate (PITC), 285
 Phenylketonuria (PKU), 558, 582
 Pheromones, 469
 Phosphodiester bonding, 214
 Phosphoglycerate kinase, 283
 Photocrosslinking, 250
 Photoreactivation, 340
 Phyletic evolution, 590
 Phyletic gradualism, 594, 595
 Phylogenetic tree, 601
 Physical crossover, 120–22
 Physical map, and Human Genome Project,
 393, 395
 Pigeon (*Columba livia*), 50, 59
 Pigment and pigmentation. *See also* Color;
 Skin color
 epistasis in mice, 33–34
 maternal effects in moths, 510, 511
 Pili (fimbriae), 159
 Pink bread mold. *See Neurospora crassa*
 Pistil, 87
Pisum sativum. *See* Garden pea
 Plants. *See also Arabidopsis thaliana*;
 Flowers; Garden pea; *Zea mays*
 genetically altered crops, 398
 genetic control of development, 479, 482–83
 life cycle, 47, 64–66
 polyploidy, 198–99
 proplastid formation and variegation, 516
 sex determination, 87, 90
 vectors and cloning of, 371–72
 Plaques, bacterial, 151
 Plasmids
 bacteria in genetic research, 149
 E. coli and, 357
 F factor, 158
 prokaryotic and cytoplasmic inheritance,
 522–24
 recombinant DNA technology, 13
 Plastids, 515
 Pleiotropy, 39
 Plus-and-minus method, of DNA sequencing,
 383, 388
Pneumocystis carinii, 502
 Pneumonia, and AIDS, 502
 Point centromere, 453
 Point mutations, 326–28
 Polar body, 64
 Polarity, of DNA structure, 218

I-12 Index

- Polar mutants, 425
 Pole cells, 470
Pol (polymerase) gene, 503
 Politics, and ethics in genetics, 6–7
 Pollen grain, 65
 Poly-A tail, 265, 267
 Polycistronic messenger RNAs, 297, 300
 Polydactyly, 99, 100
 Polygenes and polygenic inheritance, 533–35, 541
 Polymerase chain reaction (PCR), 381–83
 Polymerase collisions, 252
 Polymerase cycling, 231
 Polymerization, of nucleotides, 214, 215
 Polymorphisms, 14, 596–99. *See also* Variation
 Polynucleotide phosphorylase, 306
 Polyploids, 83, 197–98
 Polyribosome, 297
 Polysome, 297, 300
 Polytene chromosomes, 439, 449, 455
 Pond snail. *See* *Limnea peregra*
 Population, definition of, 553
 Population analysis, 564–65
 Population density, 597
 Population genetics
 computer programs to simulate, 583–84
 evolution, 553, 589
 Hardy-Weinberg equilibrium, 553–60
 migration, 573–74
 models for, 571
 mutation, 571–73
 natural selection, 577–84
 nonrandom mating, 560–65
 small population size, 574–77
 Population statistics, 535–40
Populus tremuloides (Quaking aspen), 441
 Position effect, 179
 Positive assortative mating, 554
 Positive interference, 118
 Postreplicative repair, of DNA, 344–46
 Posttranscriptional modifications, 261–62
 Posttranslational control, 433
 Posttranslocational state, of ribosome, 293, 298
 Postzygotic mechanisms, 592
 Potato spindle tuber viroid (PSTV), 272–73
 Poultry, and realized heritability, 543. *See also* Chicken
 PPL Therapeutics, 374
 Prader-Willi syndrome, 524
 Preemptor stem, 417
 Preer, J., 520, 521
 Preinitiation complex (PIC), 262
 Pretranslocational state, of ribosome, 293, 298
 Prezygotic mechanisms, 592
 Pribnow box, 248
 Primary oocytes, 64
 Primary spermatocytes, 63
 Primary structure, of protein, 281
 Primary transcript, 265
 Primase, 225–26
 Primers
 creation of general-purpose, 386–87, 390
 DNA replication, 225–27, 228, 232
 Prions, 213
 Probability, 71–74, 78
 Probability theory, 71
 Proband, 98
 Product rule, 72
 Progeny testing, 21–22
 Prokaryotes. *See also* Bacteria;
 Escherichia coli
 DNA transcription, 246–55, 277, 465
 eukaryotes compared to, 47, 261–62, 298, 440
 genomics, 360–66
 mitochondrial ribosomal RNA, 512–13
 operon model and, 12
 origins of term, 149
 plasmids, 522–24
 ribosomes, 301
 Prokaryotic vectors, 360–66
 Prolactin, and signal peptide, 302
 Promoters
 efficiency of, 430
 lac operon, 408
 repressor operon of phage λ , 435–36
 transcription in eukaryotes, 262–65
 transcription in prokaryotes, 248–51
 Proofreading, and DNA polymerase, 227
 Prophage, 165
 Prophase, of mitosis, 52, 53, 57
 Prophase I and II, and meiosis, 56–60, 62
 Proplastids, 515, 516
 Proposita and propositus, 98
 Proteasome, 499
 Protein degradation, and posttranslational control, 433
 Protein electrophoresis, 92–94
 Protein-folding problem, 303
 Protein-mediated splicing, 268–72
 Proteins
 DNA and control of, 205–7
 DNA repair in *E. coli*, 348
 energy requirements of biosynthesis, 313
 methods of depiction, 283
 motifs of DNA recognition, 480–81
 primary, secondary, tertiary, and quaternary structures, 281
 synthesis of, 245, 283, 287
 Proteomics, 396–97
 Proto-oncogenes, 13, 489, 491
 Prototrophs, 150
 Prusiner, Stanley, 213
 Pseudoalleles, 319
 Pseudoautosomal regions, 87, 95
 Pseudodominance, 96, 178
 Ptashne, Mark, 412
 Punctuated equilibrium, 14, 594, 595
 Punctuation, of genetic code, 305, 306
 Punnett, Reginald C., 21, 26
 Punnett square diagram, 21, 26, 28, 30, 532
 Purine nucleotides, 213, 214
 Puromycin, 294, 295
 Pyrimidine nucleotides, 213, 214
- ## Q
- Quantitative inheritance, 533, 534, 537–38, 545–46. *See also* Inheritance
 Quaternary structure, of proteins, 281
- ## R
- Rabl, K., 3
 Racism, and sociobiology, 606
 RAG recombinase, 495, 496
 Random genetic drift, 555, 575–76, 585
 Random mating, 554–55, 556
 Random strand analysis, 122
Rapbanobrassica (cabbage-radish cross), 198
Rapbanus sativus (radish), 198
Ras oncogene, 491
 R-bands, 452
 Read-through process, and terminators, 251
 Realized heritability, 542–43
 Real time, and observation of transcription, 250–51
 RecA protein, 345–46, 435
 RecBCD protein, 349–51, 354
 Recessive homozygote, and natural selection, 578–81
 Recessive inheritance and recessive traits, 18, 99–100, 102
 Reciprocal cross, 18, 115
 Reciprocal translocation, 182, 186, 187
 Recombinant DNA technology. *See also* Genomics
 bacteria, 349–51
 bacteriophages, 163–64
 double-strand break model of, 347–49
 ethical debate on, 370–71, 374–75
 genome sequencing, 110
 heteroduplex DNA, 351, 352
 hybrid DNA, 351
 meiosis, 61, 63
 overview of techniques, 358
 restriction endonucleases, 12–13
 Recombinant phenotypes, 111, 116
 Recombinant plasmids, 360, 362
 Recombination nodules, 59
 Recombination signal sequences, 494
 Recon, 320
 Red clover, 26
 Red eye, mutation of *Drosophila*, 24
 Reductional division, 59
 Redundant controls, and amino acid operons, 418
 Regional centromeres, 453
 Regression and regression analysis, 539–40
 Regulator gene, 406
 Reinitiation, of translation, 291
 Relaxes mutant, 432
 Release factors (RF), 296
 Repair, of DNA
 categories of systems, 339–40
 damage reversal, 340
 in *E. coli*, 348
 excision repair, 340–46
 Repetitive DNA, 458
 Replication, of DNA
 control of enzymes, 219
 enzymology of, 225–38
 in eukaryotes, 231–32, 238–39
 mutability, 207
 process of, 220–24
 structures and, 238
 Replication-coupling assembly factor, 444

- Replicons, of DNA, 224
 Replisome model, of DNA replication, 234, 235
 Reporter systems, 376–77, 399
 Repressible system, 406
 Repression, gene expression and maintenance of, 421–23
 Repressor, and gene expression, 406, 408, 409
 Repressor transcription, 421
 Reproductive isolating mechanisms, 592, 607
 Reproductive success, 577
 Repulsion, of alleles, 111
 Rescue experiment, 472
 Residues, of amino acids, 281
 Resistance transfer factor (RTF), 523
 Resolution, of transposition, 428
 Resolvase, 428
 Restricted transduction, 165
 Restriction digest, 378
 Restriction endonucleases, 12–13, 359–60
 Restriction enzymes, 360–61, 400
 Restriction fragment length polymorphisms (RFLPs), 378, 380–81
 Restriction mapping, 377–81
 Restriction sites, 359
 Restrictive temperature, and mutation, 327
 Retinoblastoma, 486–87
 Retrotransposons, 458
 Retroviruses
 AIDS, 502–4
 cancer, 487–88
 Reverse bands, 452
 Reverse transcription, 276
 Reverse transcriptionase, 276
 Reversion, and mutation rates, 326
Rev (regulation of expression of virion proteins) gene, 504
 Rhesus system, and blood types, 32
 Rh locus, and maternal-fetal incompatibility, 582
 Rhoades, M., 516
 Rho-dependent terminators, 251, 254, 255
 Rho-independent terminators, 251, 255
Rhomboid gene, 466
 Rho protein, 251
 Ribosomal RNA (rRNA), 245, 246
 Ribosome recycling factor (RRF), 297
 Ribosomes
 messenger RNA, 245
 transcription, 256
 translation, 290–91, 292, 299, 301
 Ribozyme, 266
 Rice, and genetic modified varieties, 398
 Rice, William, 89
 Rich, Alexander, 219, 221
 Rickets, vitamin-D-resistant, 97–98, 102, 509
 Ritland, D., 604
 RNA. *See also* Messenger RNA; Ribosomal RNA; Transfer RNA
 antisense, 431
 computer model of serine transfer, 243
 editing, 275
 genetic code dictionary of, 12
 genetic material and, 210–11
 guide, 275
 priming of DNA synthesis, 228, 230, 232
 self-replication, 276
 types of, 245–46
 viruses, 213
 RNA phages, 276
 RNA polymerase
 observation of, 250
 prokaryotic and eukaryotic, 247, 260
 transcription, 11, 12
 RNA replicase, 276
 RNA tumor viruses, 276
 Roberts, Richard J., 265, 266
 Robertson, E., 544
 Robertson, W., 185
 Robertsonian fusion, 185
 Rodriguez, R., 363
 Rolling-circle model, of DNA replication, 238, 239
 Roslin Institute, 374
 Roundworm. *See Ascaris* spp.
 Rous, Peyton, 487
 R plasmids, 522–23
 RRE (rev response element) gene, 504
R II screening techniques, 321, 322, 323
 R222 plasmid, 523
 Rule of independent assortment, 27–29, 67
 Rule of segregation, 18, 21–22, 63, 67
Rumex acetosa. *See* Sorrel
S
Saccharomyces cerevisiae (Baker's yeast)
 centromeres, 453
 genetically modified, 398
 haploid mapping, 123
 telomeres, 456
 unordered spores, 124–25
 Sager, Ruth, 517
Salmonella typhimurium
 Ames test for carcinogens, 332
 generalized transduction, 166
 transposon orientation, 429, 468
 Sampling distribution, 74
 Sampling error, 574–75
 Sanger, Frederick, 284, 383
 Sarcomas, 484
 Satellite DNA, 451, 452, 458
 Scaffold structure, of chromatin, 448, 449
 Scanning hypothesis, 291
 Schizophrenia, 545, 546
Schizosaccharomyces pombe (fission yeast), 111
 Scientific journals, 7
 Scientific method, 5–7, 71
 Scrapie, 213
 Screening, for drug resistance, 154
 Sea urchin, 216
 Secondary oocyte, 64
 Secondary sources, and scientific literature, 7
 Secondary spermatocytes, 64
 Secondary structure, of proteins, 281
 Second-division segregation (SDS), 128–29, 130, 131
 Second messenger, 412
 Securin, 54
 Segmentation genes, 475–77
Segment-polarity genes, 475
 Segregation
 classical genetics, 10
 meiosis and mitosis, 67
 rule of, 18–22, 63, 67
 translocation, 183–85
 Segregational load, 597
 Segregational petite, 514
 Segregation distortion, 577
 Selection coefficient, 577
 Selection experiments, 541–42
 Selection model, 578–79
 Selection-mutation equilibrium, 581
 Selective medium, 150
 Selenocysteine, 310–11
 Selfed. *See* Self-fertilization
 Self-fertilization, 17, 21–22, 31
 Selfish DNA, 430
 Self-splicing, 265–66, 268
 Semiconservative replication, of DNA, 220
 Semisterility, 181
 Sendai virus, 134
 Sepal whorl, 479, 482
 Separin, 54
 Sequence-tagged sites (STSs), 393
Sequoiadendron giganteum (giant redwood), 441
 Seryl tRNA synthetase, 280
Serratia marcescens (bacteria), 151
 Sex chromosomes, 83, 191, 197
 Sex-conditioned traits, 96
 Sex determination, 83–90
 Sex-determining region Y (SRY), 86
 Sexduction, 161
 Sex-influenced traits, 96
 Sexism, and sociobiology, 606
 Sex-lethal (*Sxl*) gene, 85, 95
 Sex-limited traits, 96
 Sex linkage, 95–96, 104
 Sex-linked inheritance, 100–102, 103, 104
 Sex-ratio phenotype, 522
 Sex-reversed individuals, 85–86
 Sex-switch gene, 85
 Sexual reproduction, and evolution, 88–89
 Sexual selection, 577
 Shapiro, J. A., 427–28
 Sharp, Philip A., 265, 266
 Sheep (*Ovis aries*)
 cloning of, 1, 374–75
 generation interval, 66
 realized heritability, 543
 Shepherd's purse (*Capsella bursa-pastoris*), 37
 Sheppard, P. M., 604
 Sherman, P., 606
 Sherman Paradox, 187
Shigella, and resistance plasmids, 523
 Shine-Dalgarno hypothesis, 290–91, 300
 Shoot apical meristem, 479
 Short interspersed elements (SINES), 458–59
 Shotgun cloning, 364, 367
 Shub, D., 274
 Shunting, and translation initiation, 291
 Sickle-cell anemia, 39, 460–61, 596, 597
 Sigma factor, 247

I-14 Index

- Signal hypothesis, 301–3
Signal peptidase, 301
Signal peptide, 301, 302
Signal recognition particle (SRP), 301
Signal-sequence receptors, 302
Signal sequences, and mitochondrial genomes, 512
Signal transduction, 467–69
Signal transduction pathway, 467
Silene latifolia. *See* White campion
Simian immune deficient viruses (SIVs), 502
Singer, B., 211
Single breaks, 178
Single-nucleotide polymorphisms (SNPs), 393
Single-strand binding proteins, 232, 234
Sister chromatids, 52
Site-specific recombination, 418
Site-specific variation, in codon reading, 309–10
Size
 of cells and chromosomes, 441–42
 of population, 555, 574–77
Skin color, and quantitative inheritance in humans, 545, 546
Skolnick, M., 391
Sleeping sickness, 275
Small nuclear ribonucleoprotein particles (snRNPs), 261, 268, 270
Small nucleolar RNAs (snoRNAs), 261
Small population size, 574–77
SMC proteins, 444
Smith, H., 4, 359
Smoking, and cancer, 492
Snails, and direction of coiling, 509–10, 525.
 See also Pond snail; Tree snails
Snapdragon. *See* *Antirrhinum majus*
Sobrero, Ascanio, 112
Sociobiology, 14, 603–6
Sociobiology: The New Synthesis (Wilson, 1975), 603
Soft selection, 597
Solenoid model, for chromatin fiber, 448
Somatic-cell hybridization, 134–40
Somatic crossing over, 132
Somatic doubling, 198
Somatic hypermutation, 497
Sonneborn, Tracy M., 519, 520
Sorrel (*Rumex acetosa*), 90
SOS box, 346
SOS response, 346
Southern, Edwin, 368
Southern blotting, 366–68
Soviet Union, and Lysenko affair, 6–7
Spätzle protein, 467
Specialized transduction, 165
Speciation, and evolution, 589–94, 595
Species
 allopolyploidy and cross-fertilization between, 198
 definition of, 553
 inversion process and evolutionary history of groups, 181
 speciation, 589–94, 595
Specific transcription factors, 263
Speckles, and nuclear messenger RNAs, 268
Spermatids, 64
Spermatogenesis, 63–64
Spermatogonium, 63
Sperm cells, 63
Spiegelman, S., 246
Spina bifida, and concordance, 546
Spindle pole body, 52
Spiral cleavage, 510
Spirillum, 149
Spliceosome, 268–72
Splicing, of exons, 265
Splicing factors, and introns, 271
Spontaneous mutagenesis, 328–30
Spontaneous mutation, 325–26
Sporophyte, 64
Stability, of mutational equilibrium, 571–72
Stabilizing selection, 578
Stadler, L. J., 325–26
Stage-specific puffs, 449–50
Stahl, Franklin W., 220–21
Stalin, Joseph, 6
Stamen, 87
Stamen whorl, 479, 482
Stanbridge, E., 487
Standard deviation, 74, 536, 538–39
Standard error of the mean (SE), 539
Standard method, and Human Genome Project, 392
Stanford University, 371
Starvation, and attenuator regulation, 417
Statistics and statistical methods, 74–78
Stature, and heritability of human traits, 545
Steiz, Joan A., 268, 269
Stem-loop structure, 252
Stepladder gels, 386, 387
Stern, Curt, 132
Stochastic events, 71
Streptococcus pneumoniae, 209
Streptomyces antibioticus, 151
Streptomyces griseus, 151
Streptomycin, 153, 294, 517–18
Stringent factor, 432
Stringent response, 432
Strobel, S., 221
Strong inference, and scientific method, 5
Structural alleles, 318–19, 320
Sturtevant, Alfred H., 4, 121, 186
Submetacentric chromosome, 48, 49
Substrate transition, and DNA replication, 329
Subtelocentric chromosome, 48
Sucrose density-gradient centrifugation, 256
Sulfur bacteria, 441
Summer squash (*Cucurbita pepo*), 37, 40–41
Sum rule, 72
Supercoiling, 234, 236, 237
Supergenes, 181
Superinfection, and repression, 422
Superrepressed mutation, 412
Suppressive petite, 514
Suppressor gene, 337
Suppressor of Stellae (*Su* or *Ste*) gene, 87
Surveillance mechanisms, and cell cycle, 51
Survival of the fittest, 589. *See also* Fitness
Sutton, Walter, 4, 66, 83, 110
SV40 virus, 371, 372, 378
Svedberg, T., 256
Svedberg unit (S), 256
Sweet pea (*Lathyrus odorata*), 37
Swine, and realized heritability, 543
SWI/SNF complex, 446–47
Sympatric speciation, 592–94
Synapsis, and meiosis, 56
Synaptonemal complex, 58
Syncytium, 470
Syn-expression group, 465
Synteny test, 136
Synthetic codons, 306–7
Synthetic medium, 150
Synthetic messenger RNAs, 306, 307
Szostak, J., 347
- T**
TACTAAC box, 268
Tanksley, Steven, 538
Tat (trans-activating transcription factor), 503–4
TATA-binding protein (TBP), 262
TATA box, 262, 263
Talmud, 95
TAR (trans-activating response element), 504
Tatum, Edward L., 10, 38–39, 157, 161
Tautomeric shifts, 328, 329
Taylor, J., 440
Tay-Sachs disease, 23, 582
T/B-cell lymphoma, 485
TBP-associated factors (TAFs), 262
T-cell receptors, 492, 498, 499
Telocentric chromosome, 48
Telomerase, 454–55, 456–57
Telomeres, 178, 454–57
Telophase, of mitosis, 52, 54–55, 56, 57
Telophase I, of meiosis, 59–60, 62
Telson, 471
Temin, Howard, 276
Temperate phages, 165
Temperature-sensitive mutants, 327
Template, and DNA replication, 220
Template strand, of DNA, 249
Template transition, and DNA replication, 329
Termination
 of DNA replication, 236, 238
 of translation, 296–301
Termination signals, for transcription, 248
Terminators, and transcription, 248, 251–55
Terminator sequence, 248
Terminator stem, 416
Tetramycin, 153
Tertiary structure, of proteins, 281
Testcross and testcrossing. *See also* Three-point cross; Two-point cross
 dihybrids, 33
 examples of, 140–41
 multihybrids, 30
 segregation rule, 22
Testing, for AIDS, 504
Testis-determining factor (TDF), 85
Tetracycline, 153, 294
Tetrad analysis, and haploid mapping, 122–32
Tetrads, 59
Tetrabyomena thermophila, 269, 272–73

- Tetranucleotide hypothesis, 216
 Tetraploids, 83, 197
 Tetratype (TT), 125, 127
 Thalassemia, 460, 461, 582
Thermus aquaticus (hot-spring bacteria), 382
 Theta structure, of DNA, 223–24
Tbimargarita namibiensis (sulfur bacteria), 441
 3-hydroxyanthranilic acid, 38, 39
 Three-locus control, 532
 Three-point cross, 114–16, 118–20
 Thymidine kinase (TK), 134–35
 Thymine, 213, 215
 Time frame, for population equilibrium, 580–81
 Tissue-specific puffs, 450
 T-loop, and telomeres, 456, 458
 T-lymphocytes, 464
 Tobacco mosaic virus, 210–11, 212
 Tobacco plants, and recombinant DNA technology, 377
 Toll gene, 473
 Topoisomers and topoisomerases, 236, 237
 Torpedo gene, 474–75
 Torso gene, 473
 Tortoise. *See Geochelone elephantopus vanderburgh*
 Totipotent cells, 469
 Touchette, Nancy, 213
 Trailer, and RNA transcription, 255
 Trans-acting mutations, 411
 Trans configuration, 111, 133
 Transcription. *See also* Gene expression
 classical genetics, 11–12
 control of in eukaryotes, 465–69
 definition of, 244
 early and late in phage infection of *E. coli*, 420–21
 eukaryotic DNA, 260–75
 flow of genetic information and, 244, 275–76
 observation of in real time, 250–51
 prokaryotic DNA, 246–55
 prokaryotic initiation and termination signals for, 248
 transfer RNA, 256–60
 translation, 297, 299
 types of RNA, 245–46
 Transcription factors, 430, 466–69
 Transducing particle, 166
 Transduction, 154, 165–68, 172
 Transfection, 372
 Transfer operon (tra), 523
 Transfer RNA (tRNA)
 function of, 245
 transcription, 256–60
 translation, 281–83, 286–88, 292
 Transfer RNA loops, 258, 260
 Transformation
 bacteria and bacteriophages, 154–57
 DNA and, 155, 156, 209
 in eukaryotes, 371
 Transformation mapping, 155–57, 172
 Transformed cancers, 484
 Transgenic organisms, 372, 397
 Transient polymorphism, 597
 Transition mutation, 329, 331, 336
 Translation. *See also* Gene expression
 classical genetics, 12
 control of, 430–33
 definition of, 244
 DNA involvement, 276
 energy cost of, 297
 genetic code, 304–12
 information transfer, 281–303
 initiation of, 288–91
 start and stop signals of, 313
 transcription, 297, 299
 Translational control, 430–33
 Translocation
 segregation, 183–85
 translation and elongation, 292–93, 296
 Translocation channel (translocon), 301
 Transposable genetic elements, 425–30
 Transposase, 428
 Transposition, mechanism of, 427–28
 Transposons, 425–30, 468, 469
 Transversion mutation, 329–30, 336
 Tree snail. *See Liguis fasciatus*
 Trihybrids, 30, 33, 40
 Triplet nature, of genetic code, 304–5
 Triplex DNA, 221, 222
 Triple-X female, 197
 Triploids, 83, 197
 Trisomic cells, 190
 Trisomy 8, 195
 Trisomy 13, 194–95
 Trisomy 18, 194, 195
 Trisomy 21, 192–94
Triticum wheat, 198
Triturus (newt), 260
Trp operon, 413–18
Trp RNA-binding attenuation protein (TRAP), 417–18
 True heritability, 543
 Trypanosomes, and RNA editing, 275
 Trypsin, 284
 Tryptophan, 39, 413–14, 416–17
 Tuberculosis, 216, 546
 Tumor-inducing (*Ti*) plasmid, 371
 Tumors, and cancer, 484
 Tumor-suppressor genes, 485–87
 Turner syndrome, 85, 195–96
 Twin spots, and mitotic recombination, 132
 Twin studies, of concordance, 546
Twist gene, 466
 2-aminopurine (2AP), 331
 Two-dimensional chromatography, 284–86
 Two-locus control, 531–32
 Two-point cross, 110–11, 114
 Type I and type II errors, 75
 Typological thinking, 590
 Tyrosine, 36
 Tyrosine kinases, 491
- U**
 UAA, UAG, and UGA stop codons, 309
 Ubiquitin, 51
 Ultraviolet (UV) light
 as carcinogen, 492
 dimerization, 340
 Unequal crossing over, 186
 Uninemic chromosome, 440, 442
 Unique DNA, 458
 Universality, of genetic code, 308–11
 University of California, 371
 University of Toronto, 441
 Unmixed families, of codons, 307
 Unordered events, 72
 Unordered spores, and yeast, 124–25
 Unusual bases, of transfer RNA, 257–58
 Upregulation, of cell growth, 487
 Upstream direction, of DNA transcription, 249
 Upstream element (UP element), 249
 Uracil, 213, 215, 344
 Uracil-DNA glycosylase, 342
 U-tube experiment, 158
UvrC and *uvrD* genes, 341, 342
- V**
 Van Breden, E., 3
 Vancomycin, 294
 Van Leeuwenhoek, Anton, 3
 Variable-number-of-tandem-repeats (VNTR), 381
 Variance
 partitioning of, 543–44
 statistical methods, 536, 538–39
 Variation. *See also* Polymorphisms
 additive models, 533, 538, 548
 causes of, 316
 chromosome number, 190–99
 codon-reading and site-specific, 309–10
 continuous, 17, 531, 533
 Darwinian evolution, 313, 589, 596
 discontinuous, 17, 531
 frequency-dependent selection, 597
 heterozygote advantage, 596–97
 multilocus selection models, 598–600
 patterns of, 600–603
 Variegation, of color, 179
 Vavilov, Nikolai, 6
 V(D)J joining, 496
 Venter, J. Craig, 358, 396
 Viceroy butterfly (*Limenitis archippus*), 604, 605
Vicia faba. *See* Broad bean
 Victoria (Queen of England), 100–102
Vif gene, 504
 Vinyl chloride, 492
 Virchow, Rudolf, 3
 Virion, 154
 Viroids, and introns, 272–73
 Viruses. *See also* Bacterial viruses; Retroviruses
 cancer, 13, 487–92
 cultivation of, 151
 life cycle, 64
 operon model, 12
 phenotypes of, 154
 recombinant DNA technology, 371
 RNA, 213
 Vitamin-D-resistant rickets, 97–98, 102
 V-J joining, 494, 495, 497
V-myc gene product, 491

I-16

Index

Von Mohl, Hugo, 3
Von Nageli, C., 3, 48
Vonnegut, Kurt, 213
Von Tschermak, Erich, 4, 17
Vpr gene, 504
Vpu gene, 504
Vries, Hugo de, 4, 17

W

Wahlund effect, 574
Waldeyer, W., 3, 48
Wallace, Douglas, 514, 597
Walzer, Stanley, 197
Washburn, Linda, 86
Watson, James, 4, 10, 11, 205, 215, 216,
328, 330
Weiling, F., 31
Weinberg, W., 13, 553
Weismann, August, 3
Western blotting, 368, 369
Wheat, and grain color, 531–33. *See also*
Triticum wheat
White campion (*Silene latifolia*), 90
Whole-genome shotgun method, 392
Wieschaus, Eric F., 471
Wild-type
 metabolic pathways in *Neurospora*, 41
 phenotypes of *Drosophila*, 23–24
Wilkins, Maurice, 205
Williams, G., 604–5
Williams, R., 210–11
Wilm's tumor, 487

Wilson, Edward O., 603
Wobble hypothesis, 307–8, 309, 310, 312
Wollman, E., 159–60, 161
Wood and leather dust, 492
Wright, Sewall, 4, 13, 553
Wynne-Edwards, V. C., 603–4

X

X chromosome, 90–95, 100
Xeroderma pigmentosum, 343, 484
X-gal, 386–87
X inactivation center (XIC), 91
X linkage, 95–96, 132–33
XO-XX system, of sex determination, 86
X-ray crystallography, of DNA, 215–16
XY system, of sex chromosomes, 83, 87, 90
XYY karyotype, 196–97

Y

Yanofsky, Charles, 324, 415
Y chromosome, 87, 88–89
Yeast. *See also Saccharomyces cerevisiae*;
Schizosaccharomyces pombe
 antibiotic resistance, 515
 centromeres, 454
 computer model of transcriptional fac-
 tor, 244
 genetic code and mitochondria, 311
 mating type, 469
 mitochondrial DNA, 513, 515
 nucleotide excision repair, 343
 petite mutations, 514, 525

testcrossing, 111, 141
transfer RNA, 259
unordered spores, 124–25
vectors and cloning, 370
Yeast artificial chromosomes (YACs), 370
Yjunctions, of DNA, 223, 232, 234, 236
Ylinked chromosomes, 95
Yunis, J., 486–87

Z

Z DNA, 219, 220, 467
Zea mays (corn)
 Ac-Ds system, 468
 cytoplasmic inheritance, 516–17
 epistasis, 34–35, 37
 generation interval, 66
 genotypic interactions and phenotypes, 32
 life cycle of, 65, 66
 meiosis, 60, 62
 testcross of trihybrid, 40
Zeigler, D., 252
ZFY gene, 86
Zimm, B., 440
Zinc finger, 480, 481
Zinder, N. D., 166
Zuckerkindl, E., 602
ZW system, 87
Zygonema, 56
Zygotes, and gene pool concept of forma-
 tion, 555
Zygotic induction, 165
Zygotic selection, 577